

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T08:06:39

Source: data/json/noise_experiments/v2/noise8i_fixed_memprimary_game/claude-sonnet-4.5/game/noisy8_bidirectional_informed_memprimary_fixed/noise8i_fixed_memprimary_game_001_neutral_rep01.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 8
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy8_bidirectional_informed_memprimary_fixed
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 1
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 1
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard",
  "Agent_3": "standard",
  "Agent_4": "standard",
  "Agent_5": "standard",
  "Agent_6": "standard",
  "Agent_7": "standard",
  "Agent_8": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

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- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will

have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_3; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_4; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_5; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

Agent_6; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_7; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Agent_8; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

ROUND 1

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$7; payoffs Agent_1=\$9, Agent_2=\$2;
noise/visible: sent visible=\$3.52, received visible=\$10.56, returned visible=\$6.1
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 9.0, "min": 0, "raw": 7.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_2: Agent_3 sent \$2.5; received \$7.5; Agent_4 returned \$3; payoffs Agent_3=\$5.5, Agent_4=\$4.5;
noise/visible: sent visible=\$2.22, received visible=\$6.66
Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 7.5, "min": 0, "raw": 3.0}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
dyad_3: Agent_5 sent \$3; received \$9; Agent_6 returned \$4; payoffs Agent_5=\$6, Agent_6=\$5;
noise/visible: sent visible=\$2.78, received visible=\$8.34, returned visible=\$4.96
Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
dyad_4: Agent_7 sent \$3; received \$9; Agent_8 returned \$7; payoffs Agent_7=\$9, Agent_8=\$2;
noise/visible: sent visible=\$3.65, received visible=\$10.95, returned visible=\$6.92
Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 9.0, "min": 0, "raw": 7.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 9.0,
"Agent_2": 2.0,
"Agent_3": 5.5,
"Agent_4": 4.5,
"Agent_5": 6.0,
"Agent_6": 5.0,
"Agent_7": 9.0,
"Agent_8": 2.0

}

Visible balances after round: {

"Agent_1": 8.1,
"Agent_2": 3.5600000000000005,
"Agent_3": 5.5,
"Agent_4": 3.66,
"Agent_5": 6.96,
"Agent_6": 4.34,
"Agent_7": 8.92,
"Agent_8": 3.9499999999999993

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:51:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
"agents": [
"Agent_1",
"Agent_2"
],
"investor": "Agent_1",
"roles": {
"Agent_1": "investor",
"Agent_2": "trustee"
},
"trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

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This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.


```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

```
Usage: {
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:51:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
```

```
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
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- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: \$1.48 + \$7.00 = \$8.48 (profit of \$3.48)
- I keep: \$10.56 - \$7.00 = \$3.56 (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

Reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment.
This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value ($\$10.56 - \$3.52 = \7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:51:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    }
  }
}
```

```

    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocity from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

```
{'send': 2.5}
```

Reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 705,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:52:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {
```

```

    "Agent_3": "investor",
    "Agent_4": "trustee"
  },
  "trustee": "Agent_4"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal

approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

```
{"return": 3.00}
```

Reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have \$7.22, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split -giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

```
Usage: {  
  "input_tokens": 529,
```

```
"output_tokens": 2312,
"reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:52:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3x3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better.

```

    r than the initial endowment.
    ...
Usage: {
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}

```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:53:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Assistant output:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

Reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84-a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself....

```
Usage: {  
  "input_tokens": 529,  
  "output_tokens": 1197,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:53:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

Reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness,

I have several options:

Option A: Send \$0 - safe but no potential gains

Option B: Send a moderate amount (e.g., \$2-3) - test the waters

Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with-enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

```
Usage: {  
  "input_tokens": 500,  
  "output_tokens": 952,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:54:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 1,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

Reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $5 + 3.65 = 8.65$
- They currently have \$1.35, so I should return $8.65 - 1.35 = 7.30$
- I would keep $10.95 - 7.30 = 3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times 10.95 = 7.99$
- Or split the total pool equally: $12.30 \div 2 = 6.15$ each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

```
Usage: {
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
 dyad_2: Agent_4=Sender, Agent_3=Receiver
 dyad_3: Agent_6=Sender, Agent_5=Receiver
 dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$3.5; received \$10.5; Agent_1 returned \$5; payoffs Agent_2=\$6.5, Agent_1=\$5.5; noise/visible: sent visible=\$2.61, received visible=\$7.83, returned visible=\$5.84
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 10.5, "min": 0, "raw": 5.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_2: Agent_4 sent \$2.5; received \$7.5; Agent_3 returned \$3.5; payoffs Agent_4=\$6, Agent_3=\$4; noise/visible: returned visible=\$2.75
 Action validation: {"returned": {"amount": 3.5, "clamped": false, "max": 7.5, "min": 0, "raw": 3.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

dyad_3: Agent_6 sent \$3.5; received \$10.5; Agent_5 returned \$8; payoffs Agent_6=\$9.5, Agent_5=\$2.5; noise/visible: sent visible=\$4.35, received visible=\$13.05, returned visible=\$8.76
 Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 10.5, "min": 0, "raw": 8.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

dyad_4: Agent_8 sent \$3.5; received \$10.5; Agent_7 returned \$7; payoffs Agent_8=\$8.5, Agent_7=\$3.5; noise/visible: sent visible=\$4, received visible=\$12, returned visible=\$6.52
 Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 10.5, "min": 0, "raw": 7.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

```
"Agent_1": 14.5,
"Agent_2": 8.5,
"Agent_3": 9.5,
"Agent_4": 10.5,
"Agent_5": 8.5,
"Agent_6": 14.5,
"Agent_7": 12.5,
"Agent_8": 10.5
```

}

Visible balances after round: {

```
"Agent_1": 10.93,
"Agent_2": 10.9,
"Agent_3": 9.5,
"Agent_4": 8.91,
"Agent_5": 12.009999999999998,
"Agent_6": 14.6,
"Agent_7": 13.92,
"Agent_8": 11.969999999999999
```

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:54:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: $\$1.48 + \$7.00 = \$8.48$ (profit of \$3.48)
- I keep: $\$10.56 - \$7.00 = \$3.56$ (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value (\$10.56 - \$3.52 = \$7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)

2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust
- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

```
Usage: {
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:54:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting

risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means: \$5 - \$3 + \$7 = \$9, but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: \$5 - \$3 = \$2
- Plus returned: \$7
- Total: \$2 + \$7 = \$9

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). This is quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous to
0. ...

```
Usage: {  
  "input_tokens": 803,  
  "output_tokens": 635,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:54:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

{"return": 3.00}

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have \$7.22, meaning I return: $\$7.22 - \$2.78 = \$4.44$

- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2312,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

```
{"send": 2.5}
```

Reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff f \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

```
Usage: {  
  "input_tokens": 873,  
  "output_tokens": 742,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:54:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

{'send': 2.5}

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything

- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.

```
{'return': 3.5}
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is \$5 - \$2.5 + \$3.0 = \$5.5, correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 \times \7.5 , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = \$5.0, so I return \$2.5

...

```
Usage: {  
  "input_tokens": 762,  
  "output_tokens": 625,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:55:13

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: \$2.22 + \$4.00 = \$6.22 (a gain of \$1.22)
- I keep: \$8.34 - \$4.00 = \$4.34

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $2.78 \times 3 = \$8.34$
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: $\$8.34$
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: $\$8.34$ (I received) vs $\$2.78$ (originally sent) = net gain of $\$5.56$
- If we split the gains equally, each gets $\$2.78$ extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has $\$2.22$, Receiver has $\$8.34$
- Total: $\$10.56$

If we wanted to split the total equally, each would get $\$5.28$. For the sender to reach that, I'd need to return $\$3.06$, leaving me with $\$5.28$ as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal split might be appropriate to reward that trust.

Testing different return amounts: returning $\$3.50$ would give the sender $\$5.72$ while I keep $\$4.84$ -a significant advantage for me. Returning $\$4.00$ instead would give the sender $\$6.22$ with a $\$1.22$ profit, while I'd keep $\$4.34$. If I return exactly half the total gain at $\$4.17$, we'd each end up with $\$6.39$ and $\$4.17$ respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1197,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent $\$3.0$, it became $\$9.0$, saw $\$4.96$ returned, visible payoff $\$6.96$.

Your total visible earnings across all rounds are $\$4.34$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

My co-player showed good trust in Round 1 by sending $\$3.0$ (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

Reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

```
Usage: {  
  "input_tokens": 836,  
  "output_tokens": 625,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:55:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
}
```



```

"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

Reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted $\$4.0 - \$2.78 = \$1.22$ gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is $\$5 - \$4.35 + \$6.525 = \6.175
- I would keep $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience.

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

```
Usage: {
  "input_tokens": 832,
```

```
"output_tokens": 1707,  
"reasoning_tokens": 0  
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:56:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_8",  
      "Agent_7"  
    ],  
    "investor": "Agent_8",  
    "roles": {  
      "Agent_7": "trustee",  
      "Agent_8": "investor"  
    },  
    "trustee": "Agent_7"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$
- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.949999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

Reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

```
...
Usage: {
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T07:56:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 2,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
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- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

- Option A: Send \$0 - safe but no potential gains
- Option B: Send a moderate amount (e.g., \$2-3) - test the waters
- Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

Reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 (3 × \$3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 (10.95 - 7.0)
- My payoff was \$5 - \$3 + \$7 = \$9... but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)

- I received \$12.0 (3 × \$4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6
- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

```
Usage: {
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver
 dyad_2: Agent_3=Sender, Agent_4=Receiver
 dyad_3: Agent_5=Sender, Agent_6=Receiver
 dyad_4: Agent_7=Sender, Agent_8=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$4; received \$12; Agent_2 returned \$9; payoffs Agent_1=\$10, Agent_2=\$3;
 noise/visible: sent visible=\$4.34, received visible=\$13.02, returned visible=\$8.5
 Action validation: {"returned": {"amount": 9.0, "clamped": false, "max": 12.0, "min": 0, "raw": 9.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_3 sent \$3; received \$9; Agent_4 returned \$4; payoffs Agent_3=\$6, Agent_4=\$5;
 noise/visible: sent visible=\$2.84, received visible=\$8.52, returned visible=\$3.01
 Action validation: {"returned": {"amount": 4.0, "clamped": false, "max": 9.0, "min": 0, "raw": 4.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_5 sent \$4; received \$12; Agent_6 returned \$9; payoffs Agent_5=\$10, Agent_6=\$3;
 noise/visible: sent visible=\$4.67, received visible=\$14.01, returned visible=\$8.77
 Action validation: {"returned": {"amount": 9.0, "clamped": false, "max": 12.0, "min": 0, "raw": 9.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_4: Agent_7 sent \$3.5; received \$10.5; Agent_8 returned \$8; payoffs Agent_7=\$9.5, Agent_8=\$2.5; noise/visible: sent visible=\$4.11, received visible=\$12.33, returned visible=\$8.46
 Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 10.5, "min": 0, "raw": 8.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent_1": 24.5,
 "Agent_2": 11.5,
 "Agent_3": 15.5,
 "Agent_4": 15.5,
 "Agent_5": 18.5,
 "Agent_6": 17.5,
 "Agent_7": 22.0,
 "Agent_8": 13.0

}

Visible balances after round: {

"Agent_1": 20.43,
 "Agent_2": 14.92,
 "Agent_3": 14.51,

```

"Agent_4": 13.43,
"Agent_5": 21.779999999999998,
"Agent_6": 19.61,
"Agent_7": 23.880000000000003,
"Agent_8": 16.3
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T07:56:29

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means: \$5 - \$3 + \$7 = \$9, but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: \$5 - \$3 = \$2
- Plus returned: \$7
- Total: \$2 + \$7 = \$9

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). They are quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 635,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

Reasoning:

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

```
Usage: {
  "input_tokens": 1113,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T07:56:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: $\$1.48 + \$7.00 = \$8.48$ (profit of \$3.48)
- I keep: $\$10.56 - \$7.00 = \$3.56$ (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value (\$10.56 - \$3.52 = \$7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust
- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

Reasoning:

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34.

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with: $\$0.66 + \$9.00 = \$9.66$ (profit of \$4.66)
- I keep: $\$13.02 - \$9.00 = \$4.02$ (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust. I should consider returning enough to make it very worthwhile for them.

...

```
Usage: {
  "input_tokens": 1203,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T07:56:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round

- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

```
{'send': 2.5}
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.

```
{'return': 3.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is \$5 - \$2.5 + \$3.0 = \$5.5, correct)

Now in Round 2:

- I'm the RECEIVER

- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 \times 3 = \7.5 , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = $\$5.0$, so I return \$2.5

...

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

Reasoning:

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0
- My payoff: $\$5 - \$2.5 + \$3.0 = \5.5

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75
- Their visible payoff: $\$5 - \$2.5 + \$2.75 = \5.25

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)
- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

```
Usage: {  
  "input_tokens": 1055,  
  "output_tokens": 491,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T07:57:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

```
{"return": 3.00}
```

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$

- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have $\$7.22$, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2312,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

```
{"send": 2.5}
```

reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER

- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

usage:

```
{
  "input_tokens": 873,
  "output_tokens": 742,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern:

- Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)
- Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)
- Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: $\$2.16 + \$4.00 = \$6.16$ (net gain of \$1.16)

- I keep: $\$8.52 - \$4.00 = \$4.52$

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

Reasoning:

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

```
Usage: {  
  "input_tokens": 1177,  
  "output_tokens": 777,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T07:57:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 3,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```

'''json
{"send": 3}

```

'''

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)

- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: $\$5 - \$4.35 + \$8 = \8.65 (profit of \$3.65)
- My final: $\$13.05 - \$8 = \$5.05$ (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted $\$4.0 - \$2.78 = \$1.22$ gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is $\$5 - \$4.35 + \$6.525 = \6.175
- I would keep $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience.

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

usage:

```
{
```

```
"input_tokens": 832,  
"output_tokens": 1707,  
"reasoning_tokens": 0  
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

****Round 2:****

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json  
{"send": 4}  
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

Reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

****Round 2:****

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - $\$4.35 + \$8.76 = \$9.41$ (visible to them as \$10.26)
- I ended with $\$13.05 - \$8 = \$5.05$

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

****Round 3:****

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last

t round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

```
Usage: {  
  "input_tokens": 1268,  
  "output_tokens": 684,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T07:57:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {  
      "Agent_5": "investor",  
      "Agent_6": "trustee"  
    },  
    "trustee": "Agent_6"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: \$2.22 (current) + \$2.78 (their share of gain) = \$5.00 (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84—a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1197,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited.

This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

usage:

```
{
  "input_tokens": 836,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history and current situation:

****Analysis:****

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

****Reasoning:****

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:
- Sender ends with: $\$0.33 + \$9.00 = \$9.33$ (profit of \$4.33)
- I keep: $\$14.01 - \$9.00 = \$5.01$

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

```
{'return': 9.00}
```

Reasoning:

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50, but they saw \$4.35. When tripled: $\$4.35 * 3 = \13.05 .

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So: $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \10.05 ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount: $\$13.05 - \$8.0 = \$5.05$. That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated....

```
Usage: {  
  "input_tokens": 1170,  
  "output_tokens": 1644,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T07:58:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  

```

```

    "Agent_7",
    "Agent_8"
  ],
  "investor": "Agent_7",
  "roles": {
    "Agent_7": "investor",
    "Agent_8": "trustee"
  },
  "trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

Option A: Send \$0 - safe but no potential gains

Option B: Send a moderate amount (e.g., \$2-3) - test the waters

Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with - enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ($3 \times \3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 ($10.95 - 7.0$)
- My payoff was \$5 - \$3 + \$7 = \$9... but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ($3 \times \4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6
- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

```
...
usage:
{
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with this co-player:

****Pattern Analysis:****

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

****Current Decision:****

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

Reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

****Round 2:****

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with \$1.5 + \$6.52 = \$8.02

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with \$12 - \$7 = \$5

****Pattern Analysis:****

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

****Current Decision (Round 3, I'm SENDER):****

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

```
Usage: {  
  "input_tokens": 1141,  
  "output_tokens": 570,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T07:58:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra

- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$
- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0

- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

usage:

```
{
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

Reasoning:

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

```
Usage: {
  "input_tokens": 1217,
  "output_tokens": 772,
  "reasoning_tokens": 0
}
```

ROUND 4

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
 dyad_2: Agent_4=Sender, Agent_3=Receiver
 dyad_3: Agent_6=Sender, Agent_5=Receiver
 dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4; received \$12; Agent_1 returned \$6; payoffs Agent_2=\$7, Agent_1=\$6;
 noise/visible: sent visible=\$3.14, received visible=\$9.42, returned visible=\$6.69
 Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_2: Agent_4 sent \$3; received \$9; Agent_3 returned \$5; payoffs Agent_4=\$7, Agent_3=\$4;
 noise/visible: sent visible=\$3.37, received visible=\$10.11, returned visible=\$5.04
 Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}
 dyad_3: Agent_6 sent \$4; received \$12; Agent_5 returned \$7; payoffs Agent_6=\$8, Agent_5=\$5;
 noise/visible: sent visible=\$3.5, received visible=\$10.5, returned visible=\$7.93
 Action validation: {"returned": {"amount": 7.0, "clamped": false, "max": 12.0, "min": 0, "raw": 7.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
 dyad_4: Agent_8 sent \$4; received \$12; Agent_7 returned \$8; payoffs Agent_8=\$9, Agent_7=\$4;
 noise/visible: sent visible=\$4.12, received visible=\$12.36, returned visible=\$7.68
 Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

```

Balances after round: {
  "Agent_1": 30.5,
  "Agent_2": 18.5,
  "Agent_3": 19.5,
  "Agent_4": 22.5,
  "Agent_5": 23.5,
  "Agent_6": 25.5,
  "Agent_7": 26.0,
  "Agent_8": 22.0
}
Visible balances after round: {
  "Agent_1": 23.85,
  "Agent_2": 22.61,
  "Agent_3": 19.619999999999997,
  "Agent_4": 20.47,
  "Agent_5": 25.279999999999998,
  "Agent_6": 28.54,
  "Agent_7": 28.240000000000002,
  "Agent_8": 24.98
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T07:58:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: \$1.48 + \$7.00 = \$8.48 (profit of \$3.48)
- I keep: \$10.56 - \$7.00 = \$3.56 (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.
I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value (\$10.56 - \$3.52 = \$7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)

- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total ($\$5 - \$3.0 + \$6.1 = \8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust
- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:
- Sender ends with: $\$0.66 + \$9.00 = \$9.66$ (profit of \$4.66)
- I keep: $\$13.02 - \$9.00 = \$4.02$ (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

reasoning:

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back
Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34.

The multiplication created \$8.68 in added value ($\$13.02 - \4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:
- Sender ends with: $\$0.66 + \$9.00 = \$9.66$ (profit of \$4.66)
- I keep: $\$13.02 - \$9.00 = \$4.02$ (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust. I should consider returning enough to make it very worthwhile for them.

...

usage:

```
{
  "input_tokens": 1203,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

Reasoning:

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: $\$5 - \$3.50 + \text{return} = \$1.50 + \text{return}$

Round 3: I returned \$9.00, kept \$4.02

So my earnings: $\$3.56 + (\$1.50 + \text{what I got back in round 2}) + \$4.02 = \$14.92$

This means: $\$9.08 + \text{what I got back in round 2} = \14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

```
Usage: {
  "input_tokens": 1669,
  "output_tokens": 673,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T07:58:45

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
```

```

    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1:
 - I sent \$3 (but noise made it appear as \$3.52 to the receiver)
 - The receiver got \$10.56 (3x of \$3.52)
 - They returned \$7.0
 - My visible payoff was \$3.56 (which means: $\$5 - \$3 + \$7 = \9 , but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: $\$5 - \$3 = \$2$
- Plus returned: \$7
- Total: $\$2 + \$7 = \$9$

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:
 - My current co-player (same person from Round 1) is now the SENDER
 - They sent me \$2.61 (after noise)
 - I received \$7.83
 - I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). They are quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 635,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

Reasoning:

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

```
Usage: {
  "input_tokens": 1469,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T07:59:02

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_4",
      "Agent_3"
    ],
    "investor": "Agent_4",
    "roles": {
      "Agent_3": "trustee",
```



```

    "Agent_4": "investor"
  },
  "trustee": "Agent_3"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal

ocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of $\$0.78$)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of $\$3.66$)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

```
{"return": 3.00}
```

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have \$7.22, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{
```

```
"input_tokens": 529,  
"output_tokens": 2312,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

```
{"send": 2.5}
```

reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

usage:

```
{
  "input_tokens": 873,
  "output_tokens": 742,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: \$2.16 + \$4.00 = \$6.16 (net gain of \$1.16)
- I keep: \$8.52 - \$4.00 = \$4.52

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

usage:

```
{
  "input_tokens": 1177,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

Reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0

Round 2: I sent \$2.5 (50%), co-player returned \$3.5

Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

```
Usage: {  
  "input_tokens": 1614,  
  "output_tokens": 808,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T07:59:12

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

{'send': 2.5}

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.

{'return': 3.5}

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is $\$5 - \$2.5 + \$3.0 = \5.5 , correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 + \7.5 , so $\$5.0$ extra value
- Splitting equally: each gets $\$2.5$ extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = $\$5.0$, so I return $\$2.5$

...

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw $\$2.22$ sent, it became $\$6.66$, returned $\$3.0$, visible payoff $\$3.66$.
- Round 2 against you, as SENDER: sent $\$2.5$, it became $\$7.5$, saw $\$2.75$ returned, visible payoff $\$5.25$.

Your total visible earnings across all rounds are $\$9.5$.

This round, you are the SENDER against your current co-player. You have $\$5$. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent $\$2.5$, they received and returned $\$3.0$ - showing good cooperation

Round 2: They sent me $\$2.5$, I returned $\$3.5$ (though noise showed them $\$2.75$)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

reasoning:

Let me analyze the history:

Round 1: I sent $\$2.5$ (they saw $\$2.22$ due to noise), it became $\$6.66$ for them, they returned $\$3.0$

- They kept $\$3.66$, I got back $\$3.0$
- My payoff: $\$5 - \$2.5 + \$3.0 = \5.5

Round 2: They sent me $\$2.5$, I received $\$7.5$, I returned $\$3.5$ (they saw $\$2.75$ due to noise)

- I kept $\$4.0$, they got back what they saw as $\$2.75$
- Their visible payoff: $\$5 - \$2.5 + \$2.75 = \5.25

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned $\$2.75$ when I actually returned $\$3.5$.

From their perspective:

- Round 1: They received $\$6.66$, returned $\$3.0$ (45% return rate)
- Round 2: They sent $\$2.5$, got back $\$2.75$ (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send $\$3.0$ this round.

usage:

```
{
  "input_tokens": 1055,
```

```
"output_tokens": 491,
"reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

```
{ 'return': 5.0 }
```

Reasoning:

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0

Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)
- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of \$1.63 + \$5.05 = \$6.68, which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

Usage: {

```
"input_tokens": 1407,
```

```
"output_tokens": 525,
```

```
"reasoning_tokens": 0
```

```
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T07:59:23
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
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 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
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 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 4,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal

1 split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84-a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1197,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

usage:

```
{
  "input_tokens": 836,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history and current situation:

****Analysis:****

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

****Reasoning:****

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with: \$0.33 + \$9.00 = \$9.33 (profit of \$4.33)
- I keep: \$14.01 - \$9.00 = \$5.01

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

{'return': 9.00}

reasoning:

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50,

but they saw \$4.35. When tripled: $\$4.35 \times 3 = \13.05 .

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So: $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \10.05 ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount: $\$13.05 - \$8.0 = \$5.05$. That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..
usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 1644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern with my co-player:

****Their behavior:****

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

****My decision:****

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

{ 'send': 4.00 }

Reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

```
Usage: {  
  "input_tokens": 1613,  
  "output_tokens": 523,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T07:59:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 4,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x

- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5,

which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted $\$4.0 - \$2.78 = \$1.22$ gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is $\$5 - \$4.35 + \$6.525 = \6.175
- I would keep $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 1707,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

****Round 2:****

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

****Round 2:****

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - $\$4.35 + \$8.76 = \$9.41$ (visible to them as \$10.26)
- I ended with \$13.05 - \$8 = \$5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

****Round 3:****

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our cooperative history:

****Round 1:****

- They returned \$4 on \$8.34 received (48%)

****Round 2:****

- I returned \$8 on \$13.05 received (61%)

****Round 3:****

- They returned \$9 on \$14.01 received (64%) - very generous!

****Round 4 (Current):****

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final: \$5 - \$3.5 + \$7 = \$8.5 (profit of \$3.5)
- My final: \$10.5 - \$7 = \$3.5 (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

Reasoning:

Let me review the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

****Round 2:****

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01
- They returned \$9 (about 64% of what they received)
- Very generous return!

****Round 4 (Current):****

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:

- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more ...

If I return \$7.5:

- Their final: $\$5 - \$3.5 + \$7.5 = \9 (profit of \$4)
- My final: $\$10.5 - \$7.5 = \$3$ (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance-it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

```
Usage: {
  "input_tokens": 1711,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T07:59:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
  so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
  , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

  MULTI-AGENT SETTING:
  - There are 8 agents in this run.
  - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
  , but the schedule balances sender and receiver roles across the run.
  - You play one sender-receiver game with your paired opponent each round.
  - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
  ll have acted as sender and receiver equally often when the round count allows it.
  - Agent names are hidden in this run.
  - Your paired opponent is referred to as your current co-player.
Message 2 role=user
content:
  Round 1: You are the RECEIVER. the sender sent $3.65, that is 73.0% of their total endowme
  nt. You received $10.95. How much do you return to the sender? (0-10.95)
Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed significant trust by sending $3.65 (73% of their endowment)
  - This created $10.95 for me after the 3x multiplier
  - The sender currently has $1.35 remaining

  I want to reciprocate their trust while ensuring both of us benefit from cooperating. The
  game created additional value ($12.30 total vs the original $5), and a fair approach wou
  ld be to split these gains.

  If I return $7.00:
  - Sender ends with: $1.35 + $7.00 = $8.35 (gain of $3.35)
  - I end with: $10.95 - $7.00 = $3.95 (gain of $3.95)

  This rewards the sender's trust while keeping the interaction balanced and positive for bo

```

th of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$
- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

usage:

```
{
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so y

ou received \$12.330000000000002. How much do you return? (0-12.330000000000002)
Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity.
Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 772,
  "reasoning_tokens": 0
}
```

```

    }
Message 8 role=user
content:
    Round 4

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 1 against you, as SENDER: sent $3.0, it became $9.0, saw $6.92 returned, visible p
      ayoff $8.92.
    - Round 2 against you, as RECEIVER: saw $4.0 sent, it became $12.0, returned $7.0, visible
      payoff $5.0.
    - Round 3 against you, as SENDER: sent $3.5, it became $10.5, saw $8.46 returned, visible
      payoff $9.96.
    Your total visible earnings across all rounds are $16.3.

    This round, you are the SENDER against your current co-player. You have $5. How much do yo
      u send? (0-5)
    Respond exactly as JSON: {'send': <amount>}

```

Assistant output:

```

    Looking at our history together:

    My co-player has consistently demonstrated strong trust:
    - Round 1: sent $3.0 (60%)
    - Round 3: sent $3.5 (70%)
    - Round 2 as receiver: returned $7.0, showing fair reciprocity

    They've seen good returns ($6.92 and $8.46) and have continued to cooperate strongly. This is
      clearly a cooperative partner.

    I should continue this positive relationship and send a substantial amount to show I value our
      mutual trust.

    '''json
    {"send": 4.00}
    '''

```

Reasoning:

```

    Looking at the history with my co-player:

    Round 1: They sent $3.0, received back $6.92 (visible to them), payoff $8.92
    Round 2: They saw $4.0 sent, returned $7.0, payoff $5.0
    Round 3: They sent $3.5, received back $8.46 (visible to them), payoff $9.96

    We have a strong pattern of cooperation. They've been sending high amounts ($3.0, $3.5) and wh
      en they were receiver, they returned $7.0.

    From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the return
      s they saw were $6.92 and $8.46 respectively - both quite generous.

    They've shown strong cooperation throughout. I should continue this pattern. They've sent 60%
      and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

    Given our mutual cooperation, I'll send $4.00 (80% of my endowment).
    Usage: {
      "input_tokens": 1638,
      "output_tokens": 388,
      "reasoning_tokens": 0
    }

```

GAME - Agent_7 - Receiver - dyad_4

```

    Timestamp: 2026-08-12T08:00:07
    Model: anthropic/claude-sonnet-4.5
    Temperature: 0.8
    Metadata: {
      "agent_type": "standard",
      "dyad_id": "dyad_4",
      "move_index": 7,
      "opponent_id": "Agent_8",
      "pairing": {
        "agents": [
          "Agent_8",

```

```

    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

- Option A: Send \$0 - safe but no potential gains
- Option B: Send a moderate amount (e.g., \$2-3) - test the waters
- Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with-enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.949999999999993.
Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ($3 \times \3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 ($10.95 - 7.0$)
- My payoff was \$5 - \$3 + \$7 = \$9... but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ($3 \times \4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6
- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

****Pattern Analysis:****

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

****Current Decision:****

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

****Round 2:****

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with \$1.5 + \$6.52 = \$8.02

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with \$12 - \$7 = \$5

****Pattern Analysis:****

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

****Current Decision (Round 3, I'm SENDER):****

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

usage:

```
{
  "input_tokens": 1141,
  "output_tokens": 570,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.3300000000000002, returned \$8.0, visible payoff \$4.3300000000000002.

Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with this co-player:

****Cooperation Pattern:****

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

****Current Situation:****

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

****My Decision:****

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

Reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

****Round 2:****

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)


```

**Round 3:**
- They were RECEIVER, saw $4.11 sent (I sent $3.5), got $12.33, returned $8.0 (~65%)

**Round 4 (current):**
- They are SENDER, sent $4.12, I received $12.36
- They kept $0.88

**Pattern:**
This co-player has been very cooperative:
- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts ($3.5 in R2, $4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were re
ceiver.

If I return around 65% like they did: 0.65 * $12.36 = $8.03

Alternatively, I could look at making the payoffs fair:
- They kept $0.88
- If I return $8, they get $8.88 total, I keep $4.36
- This is a good balanced outcome

Let me return $8 to match their generosity pattern.
Usage: {
  "input_tokens": 1540,
  "output_tokens": 565,
  "reasoning_tokens": 0
}

```

===== **ROUND 5** =====

```

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender,
Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver
Pairings:
  dyad_1: Agent_1=Sender, Agent_2=Receiver
  dyad_2: Agent_3=Sender, Agent_4=Receiver
  dyad_3: Agent_5=Sender, Agent_6=Receiver
  dyad_4: Agent_7=Sender, Agent_8=Receiver
Parsed game decisions:
  dyad_1: Agent_1 sent $4.5; received $13.5; Agent_2 returned $10; payoffs Agent_1=$10.5,
  Agent_2=$3.5; noise/visible: sent visible=$4.97, received visible=$14.91, returned visible=$9.44
  Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 13.5, "min": 0, "raw":
  10.0}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}
  dyad_2: Agent_3 sent $4; received $12; Agent_4 returned $6.3; payoffs Agent_3=$7.3, Agent_4=$5.7;
  noise/visible: sent visible=$4.21, received visible=$12.63, returned visible=$6.9
  Action validation: {"returned": {"amount": 6.3, "clamped": false, "max": 12.0, "min": 0, "raw":
  6.3}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
  dyad_3: Agent_5 sent $4; received $12; Agent_6 returned $6; payoffs Agent_5=$7, Agent_6=$6;
  noise/visible: sent visible=$3.06, received visible=$9.18, returned visible=$6.2
  Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw":
  6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
  dyad_4: Agent_7 sent $4; received $12; Agent_8 returned $7.5; payoffs Agent_7=$8.5, Agent_8=$4.5;
  noise/visible: sent visible=$3.62, received visible=$10.86, returned visible=$8.37
  Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw":
  7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 41.0,
  "Agent_2": 22.0,
  "Agent_3": 26.8,
  "Agent_4": 28.2,
  "Agent_5": 30.5,
  "Agent_6": 31.5,
  "Agent_7": 34.5,
  "Agent_8": 26.5
}
Visible balances after round: {
  "Agent_1": 33.79,

```

```

"Agent_2": 27.52,
"Agent_3": 27.519999999999996,
"Agent_4": 26.799999999999997,
"Agent_5": 32.48,
"Agent_6": 31.72,
"Agent_7": 37.61,
"Agent_8": 28.34
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T08:00:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 5,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means: $\$5 - \$3 + \$7 = \9 , but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: $\$5 - \$3 = \$2$
- Plus returned: \$7
- Total: $\$2 + \$7 = \$9$

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). They are quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70

- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 635,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

Reasoning:

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)

Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous

Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

Usage: {

```
"input_tokens": 1825,
"output_tokens": 315,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:00:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: $\$1.48 + \$7.00 = \$8.48$ (profit of \$3.48)
- I keep: $\$10.56 - \$7.00 = \$3.56$ (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value ($\$10.56 - \$3.52 = \7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
```

```
"output_tokens": 1020,
"reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust

- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

reasoning:

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust.

I should consider returning enough to make it very worthwhile for them.

...

usage:

```
{
  "input_tokens": 1203,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: \$5 - \$3.50 + return = \$1.50 + return

Round 3: I returned \$9.00, kept \$4.02

So my earnings: \$3.56 + (\$1.50 + what I got back in round 2) + \$4.02 = \$14.92

This means: \$9.08 + what I got back in round 2 = \$14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

usage:

```
{
  "input_tokens": 1669,
  "output_tokens": 673,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with: \$0.03 + \$10.00 = \$10.03 (profit of \$5.03)
- I keep: \$14.91 - \$10.00 = \$4.91 (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
```

'''

Reasoning:

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)
Round 3: They sent \$4.0, received \$8.5 back
Round 4: They returned \$6.0 from \$9.42 (generous)
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with: $\$0.03 + \$10.00 = \$10.03$ (profit of \$5.03)
- I keep: $\$14.91 - \$10.00 = \$4.91$ (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender: $\$0.03 + \$9.50 = \$9.53$ (profit of \$4.53)
- Me: $\$14.91 - \$9.50 = \$5.41$ (profit of \$5.41)

...

```
Usage: {  
  "input_tokens": 2083,  
  "output_tokens": 771,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:00:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

{'send': 2.5}

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return

3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.


```
{'return': 3.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is $\$5 - \$2.5 + \$3.0 = \5.5 , correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 \times 3 = \7.5 , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = $\$5.0$, so I return \$2.5

...

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

reasoning:

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0
- My payoff: $\$5 - \$2.5 + \$3.0 = \5.5

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75
- Their visible payoff: $\$5 - \$2.5 + \$2.75 = \5.25

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)
- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0
Round 2: They sent, I returned \$3.5
Round 3: I sent, they returned \$4.0
Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

{ 'return': 5.0 }

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0
Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)
- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of $\$1.63 + \$5.05 = \$6.68$, which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

usage:

```
{
  "input_tokens": 1407,
  "output_tokens": 525,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5
Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)
Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

```
{'send': 4.0}
```

Reasoning:

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)
Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous am

ounts:
- \$3.0 in Round 1
- \$4.0 in Round 3
- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

```
Usage: {  
  "input_tokens": 1754,  
  "output_tokens": 406,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:00:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_3",  
      "Agent_4"  
    ],  
    "investor": "Agent_3",  
    "roles": {  
      "Agent_3": "investor",  
      "Agent_4": "trustee"  
    },  
    "trustee": "Agent_4"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 5,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned

- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

{"return": 3.00}

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$

- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have $\$7.22$, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2312,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

```
{"send": 2.5}
```

reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)

- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

usage:

```
{
  "input_tokens": 873,
  "output_tokens": 742,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: $\$2.16 + \$4.00 = \$6.16$ (net gain of \$1.16)

- I keep: $\$8.52 - \$4.00 = \$4.52$

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

usage:

```
{
  "input_tokens": 1177,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0

Round 2: I sent \$2.5 (50%), co-player returned \$3.5

Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

usage:

```
{
  "input_tokens": 1614,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)
Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost their entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets: $\$0.79 + \$6.30 = \$7.09$ (net gain of \$2.09)
- I keep: $\$12.63 - \$6.30 = \$6.33$

This reciprocates their trust appropriately.

```
{"return": 6.30}
```

Reasoning:

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0
Round 2: I sent \$2.5, they returned \$3.5
Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)
Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0
Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)
- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32
- Sender would have: $\$0.79 + \$6.32 = \$7.11$ (gain of \$2.11)
- I would have: $\$12.63 - \$6.32 = \$6.31$

This seems fair and rewards their very high trust. Let me round to \$6.30 for a clean number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have: $\$0.79 + \$6.50 = \$7.29$

...

```
Usage: {  
  "input_tokens": 2041,  
  "output_tokens": 754,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:00:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_5",  
      "Agent_6"  
    ],  
    "investor": "Agent_5",  
    "roles": {
```

```

    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so y

ou received \$13.049999999999999. How much do you return? (0-13.049999999999999)
Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted $\$4.0 - \$2.78 = \$1.22$ gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is $\$5 - \$4.35 + \$6.525 = \6.175
- I would keep $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience.

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain

\$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 1707,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

****Round 2:****

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

****Round 2:****

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - \$4.35 + \$8.76 = \$9.41 (visible to them as \$10.26)
- I ended with \$13.05 - \$8 = \$5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocate

d generously by returning \$8.

****Round 3:****

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

****Round 1:****

- They returned \$4 on \$8.34 received (48%)

****Round 2:****

- I returned \$8 on \$13.05 received (61%)

****Round 3:****

- They returned \$9 on \$14.01 received (64%) - very generous!

****Round 4 (Current):****

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:
- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

reasoning:

Let me review the history with this co-player:

```
**Round 1:**
- I sent $3 (they saw $2.78 due to noise)
- They received $8.34, returned $4.0
- This was fair reciprocation (about 48%)

**Round 2:**
- They sent $4.35 (very high trust, 87% of endowment)
- I received $13.05
- I returned $8 (about 61% of what I received)
- They saw $8.76 returned due to noise

**Round 3:**
- I sent $4 (they saw $4.67 due to noise)
- They received $14.01
- They returned $9 (about 64% of what they received)
- Very generous return!

**Round 4 (Current):**
- They sent $3.5 (70% of their endowment)
- I received $10.5
- They're at $1.50 right now
```

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:
- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:
- Their final: $\$5 - \$3.5 + \$7.5 = \9 (profit of \$4)
- My final: $\$10.5 - \$7.5 = \$3$ (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance—it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

usage:

```
{
  "input_tokens": 1711,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```


Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our established cooperative pattern:

****Round 2:****

- They sent \$3.5 (70% of endowment)
- I returned generously

****Round 3:****

- I sent high (they saw \$4.67)
- They returned \$9 - very generous (64% of what they received)

****Round 4:****

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

****Round 5 (Current):****

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

Reasoning:

Looking at the history with this co-player:

****Round 2:****

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

****Round 4:****

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

****Round 5 (Current):****

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

```
Usage: {
  "input_tokens": 2169,
  "output_tokens": 521,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:01:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)

)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84-a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{  
  "input_tokens": 529,  
  "output_tokens": 1197,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

usage:

```
{
  "input_tokens": 836,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history and current situation:

****Analysis:****

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

****Reasoning:****

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with: \$0.33 + \$9.00 = \$9.33 (profit of \$4.33)
- I keep: \$14.01 - \$9.00 = \$5.01

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

{'return': 9.00}

reasoning:

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50, but they saw \$4.35. When tripled: $4.35 * 3 = \$13.05$.

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So: $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \10.05 ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount: $\$13.05 - \$8.0 = \$5.05$. That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..
usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 1644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

****Their behavior:****

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

****My decision:****

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

```
{'send': 4.00}
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

usage:

```
{
  "input_tokens": 1613,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

```

**Decision:**
Their trust level deserves fair reciprocation.

If I return $6.00:
- Sender ends with: $1.94 + $6.00 = $7.94 (profit of $2.94)
- I keep: $9.18 - $6.00 = $3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

{'return': 6.00}
Reasoning:
Let me analyze the situation:

**Current situation:**
- My co-player sent $3.06 (61.2% of their endowment)
- They have $1.94 remaining
- I received $9.18

**Their recent behavior as receiver:**
- Round 2: returned $8.0 out of $13.05 (61%)
- Round 4: returned $7.0 out of $10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

**My reciprocation pattern:**
Looking at what I've done when receiving:
- Round 1: returned $4.00 out of $8.34
- Round 3: returned $9.00 out of $14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return $5.50:
- Sender ends with: $1.94 + $5.50 = $7.44 (profit of $2.44)
- I keep: $9.18 - $5.50 = $3.68

If I return $6.00:
- Sender ends with: $1.94 + $6.00 = $7.94 (profit of $2.94)
- I keep: $9.18 - $6.00 = $3.18

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. $6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.
Usage: {
  "input_tokens": 2029,
  "output_tokens": 565,
  "reasoning_tokens": 0
}

GAME - Agent_7 - Sender - dyad_4
Timestamp: 2026-08-12T08:01:18
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
}

```



```

"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

- Option A: Send \$0 - safe but no potential gains
- Option B: Send a moderate amount (e.g., \$2-3) - test the waters
- Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ($3 \times \3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 ($10.95 - 7.0$)
- My payoff was $\$5 - \$3 + \$7 = \$9...$ but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ($3 \times \4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6
- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

****Pattern Analysis:****

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

****Current Decision:****

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

****Round 2:****

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with \$1.5 + \$6.52 = \$8.02

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with \$12 - \$7 = \$5

****Pattern Analysis:****

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

****Current Decision (Round 3, I'm SENDER):****

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal

I continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

usage:

```
{
  "input_tokens": 1141,
  "output_tokens": 570,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.949999999999993.
 - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
 - Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history with this co-player:

****Cooperation Pattern:****

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

****Current Situation:****

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

****My Decision:****

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

****Round 2:****

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)

****Round 3:****

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

****Round 4 (current):****

- They are SENDER, sent \$4.12, I received \$12.36
- They kept \$0.88

****Pattern:****

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did: $0.65 * \$12.36 = \8.03

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88
- If I return \$8, they get \$8.88 total, I keep \$4.36
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$28.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with this co-player:

****Established Pattern:****

- Round 2: They sent \$3.5, trusted me
- Round 3: They returned \$8 generously
- Round 4: They sent \$4 (high trust), I returned \$8

****Analysis:****

This co-player has been consistently cooperative:

- Always returns generously as receiver (\$7-8)
- Shows increasing trust as sender (\$3.5 ? \$4)
- We've built a strong reciprocal relationship

****Current Decision:****

They sent \$4 last round (80% of endowment), showing strong trust. I should match this level of trust to maintain our cooperative dynamic.

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3.5, I returned ~\$7 (they saw \$6.52 due to noise)

Round 3: I sent ~\$3.5 (they saw \$4.11), they returned \$8

Round 4: They sent \$4 (I saw \$4.12), I returned \$8 (they saw \$7.68)

This co-player has been:

- Consistently generous as receiver (returned \$7, then \$8)

- Trusting as sender (sent \$3.5, then \$4)
- Maintaining high cooperation levels

The pattern shows escalating trust and cooperation. They sent \$4 last time, which is 80% of their endowment.

I should continue this cooperative trajectory. Since they've been so consistently cooperative and generous, I should send a similar or slightly higher amount to maintain the positive relationship.

I'll send \$4 - matching their last send amount and showing equal trust.

```
Usage: {
  "input_tokens": 1974,
  "output_tokens": 404,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:01:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_7",
      "Agent_8"
    ],
    "investor": "Agent_7",
    "roles": {
      "Agent_7": "investor",
      "Agent_8": "trustee"
    },
    "trustee": "Agent_8"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$

- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0

- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

usage:

```
{
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 772,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This

is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92
Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0
Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the returns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

usage:

```
{
  "input_tokens": 1638,
  "output_tokens": 388,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

Reasoning:

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86

- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender: \$1.38 + \$7.50 = \$8.88
- Me: \$10.86 - \$7.50 = \$3.36

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

```
Usage: {  
  "input_tokens": 2006,  
  "output_tokens": 390,  
  "reasoning_tokens": 0  
}
```

ROUND 6

Roles: Agent_1=Receiver, Agent_2=Sender, Agent_3=Receiver, Agent_4=Sender, Agent_5=Receiver, Agent_6=Sender, Agent_7=Receiver, Agent_8=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver
dyad_2: Agent_4=Sender, Agent_3=Receiver
dyad_3: Agent_6=Sender, Agent_5=Receiver
dyad_4: Agent_8=Sender, Agent_7=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$4.5; received \$13.5; Agent_1 returned \$10; payoffs Agent_2=\$10.5, Agent_1=\$3.5; noise/visible: sent visible=\$5, received visible=\$15, returned visible=\$10.74
Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 13.5, "min": 0, "raw": 10.0}, "sent": {"amount": 4.5, "clamped": false, "max": 5, "min": 0, "raw": 4.5}}

dyad_2: Agent_4 sent \$4; received \$12; Agent_3 returned \$4.75; payoffs Agent_4=\$5.75, Agent_3=\$7.25; noise/visible: sent visible=\$3.15, received visible=\$9.45, returned visible=\$4.49
Action validation: {"returned": {"amount": 4.75, "clamped": false, "max": 12.0, "min": 0, "raw": 4.75}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_3: Agent_6 sent \$4; received \$12; Agent_5 returned \$8; payoffs Agent_6=\$9, Agent_5=\$4; noise/visible: sent visible=\$4.32, received visible=\$12.96, returned visible=\$8.83
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

dyad_4: Agent_8 sent \$4; received \$12; Agent_7 returned \$9; payoffs Agent_8=\$10, Agent_7=\$3; noise/visible: sent visible=\$4.4, received visible=\$13.2, returned visible=\$9.22
Action validation: {"returned": {"amount": 9.0, "clamped": false, "max": 12.0, "min": 0, "raw": 9.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent_1": 44.5,
"Agent_2": 32.5,
"Agent_3": 34.05,
"Agent_4": 33.95,
"Agent_5": 34.5,
"Agent_6": 40.5,
"Agent_7": 37.5,
"Agent_8": 36.5

}

Visible balances after round: {

"Agent_1": 38.79,
"Agent_2": 38.76,
"Agent_3": 32.22,
"Agent_4": 32.29,
"Agent_5": 37.44,
"Agent_6": 41.55,
"Agent_7": 41.81,
"Agent_8": 38.56

```
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T08:01:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: $\$1.48 + \$7.00 = \$8.48$ (profit of \$3.48)
- I keep: $\$10.56 - \$7.00 = \$3.56$ (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)

- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value (\$10.56 - \$3.52 = \$7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1020,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust
- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

reasoning:

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34.

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with: $\$0.66 + \$9.00 = \$9.66$ (profit of \$4.66)
- I keep: $\$13.02 - \$9.00 = \$4.02$ (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust. I should consider returning enough to make it very worthwhile for them.

...

usage:

```
{
  "input_tokens": 1203,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: $\$5 - \$3.50 + \text{return} = \$1.50 + \text{return}$

Round 3: I returned \$9.00, kept \$4.02

So my earnings: $\$3.56 + (\$1.50 + \text{what I got back in round 2}) + \$4.02 = \$14.92$

This means: $\$9.08 + \text{what I got back in round 2} = \14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

usage:

```
{
  "input_tokens": 1669,
  "output_tokens": 673,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with: $\$0.03 + \$10.00 = \$10.03$ (profit of \$5.03)
- I keep: $\$14.91 - \$10.00 = \$4.91$ (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)
Round 3: They sent \$4.0, received \$8.5 back
Round 4: They returned \$6.0 from \$9.42 (generous)
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with: $\$0.03 + \$10.00 = \$10.03$ (profit of \$5.03)
- I keep: $\$14.91 - \$10.00 = \$4.91$ (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender: $\$0.03 + \$9.50 = \$9.53$ (profit of \$4.53)
- Me: $\$14.91 - \$9.50 = \$5.41$ (profit of \$5.41)

...

usage:

```
{
  "input_tokens": 2083,
  "output_tokens": 771,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.

Your total visible earnings across all rounds are \$27.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history with my co-player:

They have been highly cooperative throughout:

- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (strong return)

This co-player has demonstrated:

1. High trust as sender (escalating from \$4.0 to \$4.5)
2. Fair/generous reciprocation as receiver
3. Consistent success in our cooperative relationship

They've been rewarded well for their trust and have reciprocated fairly. I should continue this strong cooperation by matching their recent trust level.

```
'''json
{"send": 4.50}
'''
```

Reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$4.0, received \$8.5 back (very good return)
- Round 4: They returned \$6.0 from \$9.42 (generous)
- Round 5: They sent \$4.5, received \$9.44 back (excellent return)

This co-player has been:

1. Highly trusting as sender (sent \$4.0, then \$4.5)
2. Generous as receiver (returned \$6.0 from \$9.42)
3. Consistently rewarded with good returns

The relationship has been very cooperative and mutually beneficial. They've been sending high amounts and getting good returns. As a receiver, they've been fair/generous.

Given this strong cooperative history, I should send a high amount to continue this mutually beneficial relationship. They sent \$4.5 in round 5, so I should match or exceed that level of trust.

I'll send \$4.50 to match their recent trust level.

```
Usage: {
  "input_tokens": 2556,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:01:41

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
```

```

"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```

'''json
{"send": 3}

```

'''

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

'''json

```
    {"return": 5.0}
    """
```

reasoning:

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means: $\$5 - \$3 + \$7 = \9 , but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: $\$5 - \$3 = \$2$
- Plus returned: \$7
- Total: $\$2 + \$7 = \$9$

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). They are quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 635,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what

they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction.

n. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)

Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous

Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show how strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

usage:

```
{
  "input_tokens": 1825,
  "output_tokens": 315,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$33.79.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my co-player's consistent pattern:

Round 3: Returned \$9.0 out of \$13.02 (69%)

Round 5: Returned \$10.0 out of \$14.91 (67%)

Now in Round 6, they've sent me the maximum amount possible (\$5), showing complete trust in me. This is the strongest signal of cooperation they could give.

I should reciprocate this maximum trust with a generous return. Returning about 67% of the \$15 I received (matching their typical return rate) would be appropriate.

```
'''json
{"return": 10.0}
'''
```

Reasoning:

Looking at the pattern with my co-player:

Round 3: They returned \$9.0 out of \$13.02 (69%)

Round 5: They returned \$10.0 out of \$14.91 (67%)

And now in Round 6, they sent me the maximum amount (\$5), which became \$15. This is showing ma

ximum trust in me.

They've been extremely generous throughout our interactions:

- Always sending substantial amounts when they're the sender
- Always returning generous amounts (around 67-69%) when they're the receiver

They just sent me the maximum possible amount (\$5), which shows complete trust in me. I should reciprocate this trust with a generous return.

Given their consistent pattern of returning around 67-69% and their maximum trust in sending \$5, I should return around 65-70% of \$15, which would be about \$9.75-\$10.50.

I'll return \$10, which is 67% of what I received, matching their typical return rate and showing I value our cooperative relationship.

```
Usage: {  
  "input_tokens": 2188,  
  "output_tokens": 385,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:01:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

```
{"return": 3.00}
```

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$

- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have $\$7.22$, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22-that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could stick with the proportional approach and return 44.4% of the \$6.66 I received. more than initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44 and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'm reconsidering whether equal splits of the joint gain actually make sense here, or if I should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multiple rounds. Returning nothing is purely selfish, but returning everything sent treats it like a simple reciprocal exchange. A middle ground-returning around \$3-would let the sender gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself and signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might discourage them from trusting again. The proportion approach-returning 44.4% of what I received-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. This balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what was sent. Considering everything, I'll return \$3.00 as a round number that rewards their trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 2312,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

```
{"send": 2.5}
```

reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

usage:

```
{
  "input_tokens": 873,
  "output_tokens": 742,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: $\$2.16 + \$4.00 = \$6.16$ (net gain of \$1.16)
- I keep: $\$8.52 - \$4.00 = \$4.52$

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

usage:

```
{
  "input_tokens": 1177,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned
Round 2: I sent \$2.5 (50%), they returned \$3.5
Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0
Round 2: I sent \$2.5 (50%), co-player returned \$3.5
Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

usage:

```
{  
  "input_tokens": 1614,  
  "output_tokens": 808,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)

Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost the entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets: $\$0.79 + \$6.30 = \$7.09$ (net gain of \$2.09)
- I keep: $\$12.63 - \$6.30 = \$6.33$

This reciprocates their trust appropriately.

{"return": 6.30}

reasoning:

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0

Round 2: I sent \$2.5, they returned \$3.5

Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)

Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0

Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)
- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32
- Sender would have: $\$0.79 + \$6.32 = \$7.11$ (gain of \$2.11)
- I would have: $\$12.63 - \$6.32 = \$6.31$

This seems fair and rewards their very high trust. Let me round to \$6.30 for a cleaner number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have: $\$0.79 + \$6.50 = \$7.29$

...

usage:

```
{
  "input_tokens": 2041,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

Your total visible earnings across all rounds are \$26.799999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 4: I sent \$3.37, they returned \$5.0 (strong reciprocation)

Round 5: They sent \$4.0 (80% of endowment), saw \$6.9 returned (good outcome)

My co-player has been increasing their trust levels and has reciprocated well when in the receiver role. They're demonstrating strong cooperative behavior.

To match their high trust level and continue building on this cooperative relationship, I should send a similarly high amount.

```
{"send": 4.0}
```

Reasoning:

Looking at the visible history from my co-player's perspective:

Round 3: They sent \$3.0, saw \$3.01 returned (seemed like poor reciprocation)

Round 4: They saw I sent \$3.37, returned \$5.0 (good reciprocation - 49.5%)

Round 5: They sent \$4.0, saw \$6.9 returned (good reciprocation - 57.5%)

My co-player has been showing increasing trust (sending \$3.0, then \$4.0). They're being cooperative despite the noise in Round 3 where it appeared I only returned \$3.01.

Given their pattern of increasing trust and good returns when they're the receiver, I should continue with high cooperation. Sending around 60-80% would be appropriate to match their escalating trust levels.

I'll send \$4.0 (80%) to match their Round 5 amount and show continued high trust.

```
Usage: {  
  "input_tokens": 2474,  
  "output_tokens": 351,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:01:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 3,  
  "opponent_id": "Agent_4",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

```
{'send': 2.5}
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.

```
{'return': 3.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is $\$5 - \$2.5 + \$3.0 = \5.5 , correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 \times 3 = \7.5 , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = $\$5.0$, so I return \$2.5

...

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

reasoning:

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0

- My payoff: $\$5 - \$2.5 + \$3.0 = \5.5

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75

- Their visible payoff: $\$5 - \$2.5 + \$2.75 = \5.25

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)

- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

{'return': 5.0}

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0

Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)

- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of $\$1.63 + \$5.05 = \$6.68$, which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

usage:

```
{
  "input_tokens": 1407,
  "output_tokens": 525,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5
Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)
Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

```
{'send': 4.0}
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)
Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous amounts:

- \$3.0 in Round 1
- \$4.0 in Round 3

- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

usage:

```
{
  "input_tokens": 1754,
  "output_tokens": 406,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.

Your total visible earnings across all rounds are \$27.519999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.15, so you received \$9.45. How much do you return? (0-9.45)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued partnership:

Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0
Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$3.15, giving me \$9.45

This co-player continues to be highly cooperative and generous, returning around 50% each time. They've maintained trust throughout despite noise. I'll reciprocate with a generous return to sustain this excellent relationship.

{'return': 4.75}

Reasoning:

Looking at our history:

Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0
Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$3.15, giving me \$9.45

This co-player has consistently been generous:

- Round 3: returned \$4.0 out of \$8.52 (47%)
- Round 5: returned \$6.3 out of \$12.63 (50%)
- Round 6: sent \$3.15 (down from \$3.0 but still good)

They've maintained trust and cooperation throughout. I should continue reciprocating generously. Given they sent \$3.15 and I received \$9.45, a fair return that rewards their trust would be around 50% or so.

Let me return about \$4.5-5 to maintain our cooperative relationship.

Usage: {

```
"input_tokens": 2138,
"output_tokens": 366,
"reasoning_tokens": 0
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:02:04
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8
 Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 4,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 6,
 "task": "game",
 "task_index": 0
 }

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent wi

ll have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

{'return': 4.00}

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal

1 split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84-a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1197,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

usage:

```
{
  "input_tokens": 836,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history and current situation:

****Analysis:****

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

****Reasoning:****

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with: \$0.33 + \$9.00 = \$9.33 (profit of \$4.33)
- I keep: \$14.01 - \$9.00 = \$5.01

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

{'return': 9.00}

reasoning:

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50,

but they saw \$4.35. When tripled: $\$4.35 * 3 = \13.05 .

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So: $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \10.05 ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount: $\$13.05 - \$8.0 = \$5.05$. That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..
usage:

```
{
  "input_tokens": 1170,
  "output_tokens": 1644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

****Their behavior:****

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

****My decision:****

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

```
{'send': 4.00}
```

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

usage:

```
{
  "input_tokens": 1613,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

****Decision:****

Their trust level deserves fair reciprocation.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

```
{'return': 6.00}
```

reasoning:

Let me analyze the situation:

****Current situation:****

- My co-player sent \$3.06 (61.2% of their endowment)
- They have \$1.94 remaining
- I received \$9.18

****Their recent behavior as receiver:****

- Round 2: returned \$8.0 out of \$13.05 (61%)
- Round 4: returned \$7.0 out of \$10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

****My reciprocation pattern:****

Looking at what I've done when receiving:

- Round 1: returned \$4.00 out of \$8.34
- Round 3: returned \$9.00 out of \$14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return \$5.50:

- Sender ends with: $\$1.94 + \$5.50 = \$7.44$ (profit of \$2.44)
- I keep: $\$9.18 - \$5.50 = \$3.68$

If I return \$6.00:

- Sender ends with: $\$1.94 + \$6.00 = \$7.94$ (profit of \$2.94)
- I keep: $\$9.18 - \$6.00 = \$3.18$

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. \$6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2029,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.

Your total visible earnings across all rounds are \$31.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history:

****Analysis:****

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 3 and 5 - very high trust
- As receiver in Round 4, they returned \$7.0 out of \$10.5 (67%) - fair reciprocation
- They maintain high trust even when returns vary

****Decision:****

This co-player demonstrates strong, consistent trust. I should match their commitment level to maintain our cooperative relationship.

Sending \$4.00 (80%) matches their trust pattern and maximizes our joint potential gains.


```
{'send': 4.00}
```

Reasoning:

Looking at my co-player's history:

Round 3: As sender, sent \$4.0 (80%), saw \$8.77 returned, payoff \$9.77

Round 4: As receiver, saw \$3.5 sent, returned \$7.0 out of \$10.5 (67%), payoff \$3.5

Round 5: As sender, sent \$4.0 (80%), saw \$6.2 returned, payoff \$7.2

Pattern analysis:

- As sender: Consistently sending \$4.0 (80%) - very high trust
- As receiver: Returning 67% - fair reciprocation
- They're maintaining high trust despite varying returns

This co-player has been consistently trusting. They've sent 80% twice in a row. As receiver, they returned a decent amount (67%).

I should match their high trust level. They're showing strong cooperative behavior, and I should reciprocate that trust.

Sending \$4.00 (80%) would match their pattern and continue our cooperation.

```
Usage: {  
  "input_tokens": 2424,  
  "output_tokens": 394,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:02:14

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 5,  
  "opponent_id": "Agent_6",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted $\$4.0 - \$2.78 = \$1.22$ gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is $\$5 - \$4.35 + \$6.525 = \6.175
- I would keep $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 1707,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

****Round 2:****

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

****Round 2:****

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - $\$4.35 + \$8.76 = \$9.41$ (visible to them as \$10.26)
- I ended with \$13.05 - $\$8 = \5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

****Round 3:****

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

****Round 1:****

- They returned \$4 on \$8.34 received (48%)

****Round 2:****

- I returned \$8 on \$13.05 received (61%)

****Round 3:****

- They returned \$9 on \$14.01 received (64%) - very generous!

****Round 4 (Current):****

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final: \$5 - \$3.5 + \$7 = \$8.5 (profit of \$3.5)
- My final: \$10.5 - \$7 = \$3.5 (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

reasoning:

Let me review the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

****Round 2:****

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01

- They returned \$9 (about 64% of what they received)
- Very generous return!

****Round 4 (Current):****

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:

- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:

- Their final: $\$5 - \$3.5 + \$7.5 = \9 (profit of \$4)
- My final: $\$10.5 - \$7.5 = \$3$ (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance—it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

usage:

```
{
  "input_tokens": 1711,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established cooperative pattern:

****Round 2:****

- They sent \$3.5 (70% of endowment)
- I returned generously

****Round 3:****

- I sent high (they saw \$4.67)

- They returned \$9 - very generous (64% of what they received)

****Round 4:****

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

****Round 5 (Current):****

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

reasoning:

Looking at the history with this co-player:

****Round 2:****

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

****Round 4:****

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

****Round 5 (Current):****

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

usage:

```
{
  "input_tokens": 2169,
  "output_tokens": 521,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.

Your total visible earnings across all rounds are \$32.48.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so y

ou received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our cooperative history:

****Round 3:****

- They returned \$9 on \$14.01 (64% return rate) - generous

****Round 4:****

- I returned fairly (they saw \$7.93)

****Round 5:****

- They saw only \$3.06 sent (due to noise), received \$9.18

- They still returned \$6 (65%) - very generous despite the low apparent send

****Round 6 (Current):****

- They sent \$4.32 (86% of their endowment) - extremely high trust!

- I received \$12.96

- They're left with only \$0.68

They've consistently shown generosity and trust, especially in Round 5 where noise made it look like I sent less, yet they still returned generously. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.32 + \$8 = \$8.68 (profit of \$3.68)

- My final: \$12.96 - \$8 = \$4.96 (profit of \$4.96)

```
'''json
{"return": 8}
'''
```

Reasoning:

Looking at the history:

****Round 3:****

- I sent (they saw \$4.67)

- They returned \$9 on \$14.01 (64%)

****Round 4:****

- They sent \$4.0

- I returned (they saw \$7.93)

****Round 5:****

- I sent \$4 (they saw \$3.06 due to noise)

- They received \$9.18 and returned \$6 (65%)

- This is very generous given they only saw \$3.06 sent!

****Round 6 (Current):****

- They sent \$4.32 (86% of endowment) - very high trust!

- I received \$12.96

- They're at \$0.68

They've been consistently generous and trusting. In Round 5, despite the noise making it look like I only sent \$3.06, they still returned \$6 which is generous. Now they're showing even more trust by sending \$4.32.

Given their consistent generosity and trust:

- If I return \$8:

- Their final: \$5 - \$4.32 + \$8 = \$8.68 (profit of \$3.68)

- My final: \$12.96 - \$8 = \$4.96 (profit of \$4.96)

This rewards their trust appropriately and continues our cooperative relationship.

```
Usage: {
  "input_tokens": 2590,
  "output_tokens": 603,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:02:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.

- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$
- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment
- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

usage:

```
{
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 772,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92

Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0

Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the re

turns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

usage:

```
{
  "input_tokens": 1638,
  "output_tokens": 388,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

reasoning:

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86
- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender: \$1.38 + \$7.50 = \$8.88
- Me: \$10.86 - \$7.50 = \$3.36

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

usage:

```
{
  "input_tokens": 2006,
  "output_tokens": 390,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.

Your total visible earnings across all rounds are \$28.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperative relationship:

My co-player has been consistently trustworthy:

- Round 3: sent \$3.5 (70%)
- Round 5: sent \$4.0 (80%) - showing increasing trust
- Round 4 as receiver: returned \$8.0 from \$12.36

They've demonstrated strong reciprocity and increased their trust level over time. I should match this escalating cooperation.

```
'''json
{"send": 4.00}
'''
```

Reasoning:

Looking at the history:

My co-player has consistently shown strong trust:

- Round 3 as sender: sent \$3.5 (70%)
- Round 5 as sender: sent \$4.0 (80%)
- Round 4 as receiver: returned \$8.0 when they received \$12.36

They've received good returns (\$8.46, \$8.37) and continue to maintain high cooperation levels.

This is clearly a very cooperative partner. I should continue sending substantial amounts to maintain this mutually beneficial relationship.

I've sent:

- Round 2: \$3.50
- Round 4: \$4.00

I should maintain or even slightly increase my sending to show continued trust, especially since they increased from \$3.5 to \$4.0.

I'll send \$4.00 to maintain strong cooperation.

```
Usage: {
  "input_tokens": 2357,
  "output_tokens": 313,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:02:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",


```

"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
  "agents": [
    "Agent_8",
    "Agent_7"
  ],
  "investor": "Agent_8",
  "roles": {
    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 6,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

- Option A: Send \$0 - safe but no potential gains
- Option B: Send a moderate amount (e.g., \$2-3) - test the waters
- Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with - enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ($3 \times \3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 ($10.95 - 7.0$)
- My payoff was \$5 - \$3 + \$7 = \$9... but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ($3 \times \4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12

- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6
- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

****Pattern Analysis:****

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

****Current Decision:****

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

****Round 2:****

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with \$1.5 + \$6.52 = \$8.02

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with \$12 - \$7 = \$5

****Pattern Analysis:****

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

****Current Decision (Round 3, I'm SENDER):****

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

usage:

```
{
  "input_tokens": 1141,
  "output_tokens": 570,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
 - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
 - Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.3300000000000002, returned \$8.0, visible payoff \$4.3300000000000002.
- Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history with this co-player:

****Cooperation Pattern:****

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

****Current Situation:****

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

****My Decision:****

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

****Round 2:****

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)

****Round 3:****

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

****Round 4 (current):****

- They are SENDER, sent \$4.12, I received \$12.36
- They kept \$0.88

****Pattern:****

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did: $0.65 * \$12.36 = \8.03

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88
- If I return \$8, they get \$8.88 total, I keep \$4.36
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.33, returned \$8.0, visible payoff \$4.36.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$28.24.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the history with this co-player:

****Established Pattern:****

- Round 2: They sent \$3.5, trusted me
- Round 3: They returned \$8 generously
- Round 4: They sent \$4 (high trust), I returned \$8

****Analysis:****

This co-player has been consistently cooperative:

- Always returns generously as receiver (\$7-8)
- Shows increasing trust as sender (\$3.5 ? \$4)
- We've built a strong reciprocal relationship

****Current Decision:****

They sent \$4 last round (80% of endowment), showing strong trust. I should match this level of trust to maintain our cooperative dynamic.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3.5, I returned ~\$7 (they saw \$6.52 due to noise)

Round 3: I sent ~\$3.5 (they saw \$4.11), they returned \$8

Round 4: They sent \$4 (I saw \$4.12), I returned \$8 (they saw \$7.68)

This co-player has been:

- Consistently generous as receiver (returned \$7, then \$8)
- Trusting as sender (sent \$3.5, then \$4)
- Maintaining high cooperation levels

The pattern shows escalating trust and cooperation. They sent \$4 last time, which is 80% of their endowment.

I should continue this cooperative trajectory. Since they've been so consistently cooperative and generous, I should send a similar or slightly higher amount to maintain the positive relationship.

I'll send \$4 - matching their last send amount and showing equal trust.

usage:

```
{
  "input_tokens": 1974,
  "output_tokens": 404,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.3300000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with this co-player:

****Their Pattern:****

- Round 3: Returned \$8/\$12.33 = 65%
- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 6 (now): Sent \$4.4 (88% of endowment!)

****Current Situation:****

- They sent \$4.4, keeping only \$0.6
- I received \$13.2
- This shows exceptional trust - their highest send yet

****My Decision:****

They've consistently returned 65-69% and are now showing maximum trust. I should match their generosity pattern and reward this trust.

Returning \$9:

- They end with \$9.6 total
- I keep \$4.2

- ~68% return rate, matching their established pattern
- Sustains our strong cooperation

```
'''json
{"return": 9}
'''
```

Reasoning:

Looking at the history with this co-player:

****Their behavior:****

- Round 3: Returned \$8 out of \$12.33 (65%)
- Round 5: Returned \$7.5 out of \$10.86 (69%)

****Current situation:****

- They sent \$4.4 (88% of their endowment!) - showing very high trust
- I received \$13.2
- They kept only \$0.6

****Analysis:****

This co-player is showing increasing trust and generosity:

- They've returned 65-69% as receiver
- As sender, they're now sending even more (\$4.4 vs \$4 previously)
- This shows they believe in our cooperation

****My decision:****

I should return proportionally to their generosity pattern. They've been returning around 65-69%.

If I return \$8.5:

- They end with \$9.1 total (\$0.6 + \$8.5)
- I keep \$4.7
- Return rate: 64% (consistent with their pattern)

If I return \$9:

- They end with \$9.6 total
- I keep \$4.2
- Return rate: 68% (matches their higher range)

Given they sent \$4.4 (very high), I should be generous back. Let me return \$8.5 to maintain good cooperation while keeping the return rate reasonable.

Actually, thinking about it more - they sent \$4.4 out of \$5 (88%). This is extremely trusting. They've been consistently generous. I should match that generosity.

Return \$9:

- They get \$9.6 total (\$0.6 kept + \$9 returned)
- I keep \$4.2
- This is about 68% return rate, matching their generosity pattern
- Rewards their exceptional trust

Usage: {

```
"input_tokens": 2394,
"output_tokens": 644,
"reasoning_tokens": 0
```

}

ROUND 7

Roles: Agent_1=Sender, Agent_2=Receiver, Agent_3=Sender, Agent_4=Receiver, Agent_5=Sender, Agent_6=Receiver, Agent_7=Sender, Agent_8=Receiver

Pairings:

```
dyad_1: Agent_1=Sender, Agent_2=Receiver
dyad_2: Agent_3=Sender, Agent_4=Receiver
dyad_3: Agent_5=Sender, Agent_6=Receiver
dyad_4: Agent_7=Sender, Agent_8=Receiver
```

Parsed game decisions:

```
dyad_1: Agent_1 sent $5; received $15; Agent_2 returned $10; payoffs Agent_1=$10, Agent_2=$5;
noise/visible: sent visible=$4.81, received visible=$14.43, returned visible=$9.6
Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 15.0, "min": 0, "raw": 10.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}
```



```

dyad_2: Agent_3 sent $4; received $12; Agent_4 returned $5; payoffs Agent_3=$6, Agent_4=$7;
noise/visible: sent visible=$3.35, received visible=$10.05, returned visible=$5.43
  Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 12.0, "min": 0, "raw":
  5.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_3: Agent_5 sent $4; received $12; Agent_6 returned $7.5; payoffs Agent_5=$8.5, Agent_6=$4.5;
noise/visible: sent visible=$3.92, received visible=$11.76, returned visible=$8.17
  Action validation: {"returned": {"amount": 7.5, "clamped": false, "max": 12.0, "min": 0, "raw":
  7.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
dyad_4: Agent_7 sent $4; received $12; Agent_8 returned $10; payoffs Agent_7=$11, Agent_8=$2;
noise/visible: sent visible=$4.84, received visible=$14.52, returned visible=$9.67
  Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 12.0, "min": 0, "raw":
  10.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}
Balances after round: {
  "Agent_1": 54.5,
  "Agent_2": 37.5,
  "Agent_3": 40.05,
  "Agent_4": 40.95,
  "Agent_5": 43.0,
  "Agent_6": 45.0,
  "Agent_7": 48.5,
  "Agent_8": 38.5
}
Visible balances after round: {
  "Agent_1": 48.39,
  "Agent_2": 43.19,
  "Agent_3": 38.65,
  "Agent_4": 37.34,
  "Agent_5": 46.61,
  "Agent_6": 45.809999999999995,
  "Agent_7": 52.480000000000004,
  "Agent_8": 43.08
}

```

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

```

Timestamp: 2026-08-12T08:02:39
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round as the sender, I need to balance trust and caution since I have no prior information about my co-player.

Key considerations:

- Sending money can benefit us both (it triples)
- But I'm vulnerable to getting nothing back
- This is a multi-round game, so establishing cooperative patterns could be valuable
- There's communication noise, which adds uncertainty

I'll send a moderate amount to signal willingness to cooperate while managing risk. If I send \$3, the receiver gets \$9. If they return even a portion (say \$4-5), we both do better than if I hadn't sent anything.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the strategic considerations:

1. I receive \$5 and can send any amount from \$0-\$5 to the receiver
2. Whatever I send gets tripled
3. The receiver then decides how much to return to me
4. My final payoff = \$5 - sent + returned
5. Receiver's final payoff = 3×sent - returned

From a game theory perspective:

- If I send nothing, I keep \$5, receiver gets \$0
- If I send everything (\$5), receiver gets \$15, and I get whatever they return
- The rational selfish strategy for the receiver would be to return \$0 (since that maximizes their payoff)
- But if both players cooperate, we can maximize total welfare

However, this is more complex because:

1. There's communication noise mentioned - amounts may differ from what was actually sent/returned
2. This is a multi-round game with randomized pairings
3. There's reputation/reciprocity considerations even though agent names are hidden

In the first round with no prior information:

- I don't know my co-player's strategy
- A moderate send shows willingness to cooperate but doesn't overcommit
- Common strategies in trust games: send about half (\$2-3) to signal trust while limiting risk

I'll send \$3 as a reasonable middle ground that:

- Shows willingness to cooperate/trust
- Gives the receiver \$9 to work with
- Limits my downside risk (I keep \$2 guaranteed)
- If they return even \$3-4, we both benefit from cooperation

This seems like a balanced approach for the first round.

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 546,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means: \$5 - \$3 + \$7 = \$9, but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept: $\$5 - \$3 = \$2$
- Plus returned: \$7
- Total: $\$2 + \$7 = \$9$

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). This is quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

usage:

```
{
  "input_tokens": 803,
  "output_tokens": 635,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)

- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

usage:

```
{
  "input_tokens": 1113,
  "output_tokens": 495,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)
Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous
Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show how strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

usage:

```
{  
  "input_tokens": 1825,  
  "output_tokens": 315,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$33.79.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at my co-player's consistent pattern:

Round 3: Returned \$9.0 out of \$13.02 (69%)

Round 5: Returned \$10.0 out of \$14.91 (67%)

Now in Round 6, they've sent me the maximum amount possible (\$5), showing complete trust in me. This is the strongest signal of cooperation they could give.

I should reciprocate this maximum trust with a generous return. Returning about 67% of the \$15 I received (matching their typical return rate) would be appropriate.

```
'''json  
{"return": 10.0}  
'''
```

reasoning:

Looking at the pattern with my co-player:

Round 3: They returned \$9.0 out of \$13.02 (69%)

Round 5: They returned \$10.0 out of \$14.91 (67%)

And now in Round 6, they sent me the maximum amount (\$5), which became \$15. This is showing maximum trust in me.

They've been extremely generous throughout our interactions:

- Always sending substantial amounts when they're the sender
- Always returning generous amounts (around 67-69%) when they're the receiver

They just sent me the maximum possible amount (\$5), which shows complete trust in me. I should reciprocate this trust with a generous return.

Given their consistent pattern of returning around 67-69% and their maximum trust in sending

ng \$5, I should return around 65-70% of \$15, which would be about \$9.75-\$10.50.

I'll return \$10, which is 67% of what I received, matching their typical return rate and showing I value our cooperative relationship.

usage:

```
{
  "input_tokens": 2188,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.
- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.

Your total visible earnings across all rounds are \$38.79.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at my co-player's excellent track record:

Round 4: Sent \$4.0 (high trust)

Round 5: Returned \$10.0 out of \$14.91 (67% - very generous)

Round 6: Sent \$4.5 (continued high trust)

This co-player has been consistently cooperative and trustworthy across all our interactions.

They've shown maximum cooperation, and our relationship has been mutually beneficial.

I should reciprocate with maximum trust by sending the full amount.

```
'''json
{"send": 5}
'''
```

Reasoning:

Looking at my co-player's history:

Round 4: Sent \$4.0 as sender (high trust)

Round 5: Returned \$10.0 out of \$14.91 (67% - generous)

Round 6: Sent \$4.5 as sender (high trust)

This co-player has consistently demonstrated:

- High trust as sender (sending \$4.0-\$4.5)
- Generous returns as receiver (67%+)
- Strong cooperative behavior throughout

We've built an excellent cooperative relationship over 6 rounds. They've been trustworthy every single time.

Given their consistent cooperation and trustworthiness, I should send the maximum amount to maximize our joint payoffs. This continues our pattern of mutual trust and cooperation.

Usage: {

```
"input_tokens": 2547,
"output_tokens": 314,
"reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T08:02:51

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {


```

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.52, that is 70.4% of their total endowment. You received \$10.56. How much do you return to the sender? (0-10.56)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.52 (70.4% of their endowment)
- The tripling effect created \$10.56 for me
- The sender is left with only \$1.48 of their original \$5

The sender took a considerable risk and showed trust. To reciprocate fairly, I should ensure we both benefit from this cooperation.

The multiplication created \$7.04 in added value (\$10.56 - \$3.52). A fair approach would be to split the gains roughly equally while ensuring the sender profits from their trust.

If I return \$7.00:

- Sender ends with: $\$1.48 + \$7.00 = \$8.48$ (profit of \$3.48)
- I keep: $\$10.56 - \$7.00 = \$3.56$ (profit of \$3.56)

This creates roughly equal gains for both of us and rewards the sender's trust.

```
'''json
{"return": 7.00}
'''
```

reasoning:

In this round, I'm the receiver. Let me analyze the situation:

- The sender sent \$3.52 out of their \$5 endowment (70.4%)
- I received $\$3.52 \times 3 = \10.56
- I need to decide how much to return (between \$0 and \$10.56)

This is a trust game. The sender showed significant trust by sending 70.4% of their endowment. This is a fairly generous amount.

Let me think about what would be fair:

- If I return nothing, I keep \$10.56 and the sender gets $\$5 - \$3.52 = \$1.48$
- If I return half (\$5.28), I keep \$5.28 and the sender gets $\$1.48 + \$5.28 = \$6.76$
- If I return \$7, I keep \$3.56 and the sender gets $\$1.48 + \$7 = \$8.48$

The sender started with \$5 and sent \$3.52, keeping \$1.48.

If I return \$7, the sender ends up with \$8.48 total, which is a gain of \$3.48 from their initial \$5.

I would keep \$3.56 from the \$10.56 I received.

Let me think about reciprocity and fairness:

- The sender was generous (70.4% of their endowment)
- A common strategy in trust games is to split the gains or ensure both parties benefit
- The total in the system increased from \$5 to \$11.56 (the sender's remaining \$1.48 + my \$10.56)

If I want to reward the sender's trust and be fair:

- The sender risked \$3.52 to potentially gain something
- A return of around \$6-7 would give them a decent profit while I also keep a reasonable amount

Given that the sender was quite generous (70.4%), I should reciprocate with a reasonable return. Let me aim for around \$7, which:

- Gives the sender \$8.48 total (good profit of \$3.48)
- Leaves me with \$3.56 (reasonable gain)
- Shows I appreciate their trust

Actually, let me reconsider the math differently. The tripling created \$7.04 in extra value ($\$10.56 - \$3.52 = \7.04). If we split this gain equally, that's \$3.52 each, which means I should return \$3.52 to match their contribution and ensure we both benefit equally from the transaction....

usage:

```
{
  "input_tokens": 529,
```

```
"output_tokens": 1020,
"reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust

- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

usage:

```
{
  "input_tokens": 902,
  "output_tokens": 597,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

reasoning:

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust.

I should consider returning enough to make it very worthwhile for them.

...

usage:

```
{
  "input_tokens": 1203,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: $\$5 - \$3.50 + \text{return} = \$1.50 + \text{return}$

Round 3: I returned \$9.00, kept \$4.02

So my earnings: $\$3.56 + (\$1.50 + \text{what I got back in round 2}) + \$4.02 = \$14.92$

This means: $\$9.08 + \text{what I got back in round 2} = \14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

usage:

```
{
  "input_tokens": 1669,
  "output_tokens": 673,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with: $\$0.03 + \$10.00 = \$10.03$ (profit of \$5.03)
- I keep: $\$14.91 - \$10.00 = \$4.91$ (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)
Round 3: They sent \$4.0, received \$8.5 back
Round 4: They returned \$6.0 from \$9.42 (generous)
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with: $\$0.03 + \$10.00 = \$10.03$ (profit of \$5.03)
- I keep: $\$14.91 - \$10.00 = \$4.91$ (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender: $\$0.03 + \$9.50 = \$9.53$ (profit of \$4.53)
- Me: $\$14.91 - \$9.50 = \$5.41$ (profit of \$5.41)

...

usage:

```
{
  "input_tokens": 2083,
  "output_tokens": 771,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.

Your total visible earnings across all rounds are \$27.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history with my co-player:

They have been highly cooperative throughout:

- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (strong return)

This co-player has demonstrated:

1. High trust as sender (escalating from \$4.0 to \$4.5)
2. Fair/generous reciprocation as receiver
3. Consistent success in our cooperative relationship

They've been rewarded well for their trust and have reciprocated fairly. I should continue this strong cooperation by matching their recent trust level.

```
'''json
{"send": 4.50}
'''
```

reasoning:

Looking at the visible history with my co-player:

- Round 3: They sent \$4.0, received \$8.5 back (very good return)
- Round 4: They returned \$6.0 from \$9.42 (generous)
- Round 5: They sent \$4.5, received \$9.44 back (excellent return)

This co-player has been:

1. Highly trusting as sender (sent \$4.0, then \$4.5)
2. Generous as receiver (returned \$6.0 from \$9.42)
3. Consistently rewarded with good returns

The relationship has been very cooperative and mutually beneficial. They've been sending high amounts and getting good returns. As a receiver, they've been fair/generous.

Given this strong cooperative history, I should send a high amount to continue this mutually beneficial relationship. They sent \$4.5 in round 5, so I should match or exceed that level of trust.

I'll send \$4.50 to match their recent trust level.

usage:

```
{
  "input_tokens": 2556,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$38.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at this situation:

My co-player has shown very high trust by sending \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19.

The visible history shows excellent cooperation:

- Round 4: Returned \$6.0 from \$9.42
- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

They were extremely generous in Round 6, returning \$10.0 from \$15. Now they're showing continued high trust as a sender.

The multiplication created \$9.62 in added value. To reciprocate their strong cooperation:

If I return \$10.00:

- Sender ends with: $\$0.19 + \$10.00 = \$10.19$ (profit of \$5.19)
- I keep: $\$14.43 - \$10.00 = \$4.43$ (profit of \$4.43)

This maintains our successful cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

Reasoning:

Looking at this situation:

My co-player sent \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19. This is very high trust.

The visible history shows strong cooperation:

- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (good return)
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

The multiplication created \$9.62 in added value (\$14.43 - \$4.81).

My co-player has been very generous as a receiver (returning \$6.0, then \$10.0) and trusting as a sender (sending \$4.5, then \$4.81).

To reciprocate this high trust and maintain the cooperative relationship:

If I return \$10.00:

- Sender ends with: $\$0.19 + \$10.00 = \$10.19$ (profit of \$5.19)
- I keep: $\$14.43 - \$10.00 = \$4.43$ (profit of \$4.43)

This is fair and rewards their very high trust. Given they returned \$10.0 from \$15 last round, returning \$10.00 from \$14.43 would be consistent with what they've shown.

Actually, let me think about this more carefully. They sent \$4.81, leaving \$0.19. The multiplication created \$14.43 for me, which is an added value of \$9.62.

If I return \$10.00, they get \$10.19 total (profit \$5.19), I keep \$4.43 (profit \$4.43). We both profit well.

Given the strong cooperative history and their very high trust (96.2% of endowment), \$10.00 seems appropriate.

```
Usage: {
  "input_tokens": 2963,
  "output_tokens": 735,
  "reasoning_tokens": 0
}
```

GAME - Agent_3 - Sender - dyad_2

Timestamp: 2026-08-12T08:02:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 2,
  "opponent_id": "Agent_4",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    }
  },
}
```

```

    "trustee": "Agent_4"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I want to balance building cooperation while being somewhat cautious since I don't know my co-player yet.

Sending money creates value through the 3x multiplier, which benefits both of us if there's reciprocity. However, since pairings are randomized each round, I can't rely on direct reciprocation from the same person.

I'll send a moderate amount to signal willingness to cooperate while not being overly risky in this initial round.

```
{'send': 2.5}
```

reasoning:

This is the first round of a multi-round trust game. I'm the sender with \$5.

Let me think about this strategically:

1. If I send money, it gets tripled, which creates value
2. The receiver then decides how much to return
3. This is a repeated game with multiple rounds
4. There's communication noise mentioned - amounts may differ from what was actually sent

Key considerations:

- Sending money creates surplus (due to 3x multiplier)
- But I'm vulnerable to the receiver keeping everything
- This is round 1, so I should consider signaling cooperation
- There are multiple rounds, so reputation/reciprocity could matter
- However, pairings are randomized each round, so I won't necessarily play with the same person again
- Since pairings are randomized and agent names are hidden, direct reciprocity is limited

The standard game theory approach:

- In a one-shot game, the subgame perfect equilibrium is to send \$0 (the receiver has no incentive to return anything)
- However, behavioral economics shows people often cooperate, especially when:
 - * They can build reputation
 - * There are repeated interactions
 - * There's uncertainty

Given that:

- This is a repeated game but with random rematching
- Names are hidden, so limited ability to track specific partners
- There's noise in communication

A reasonable first move would be to send a moderate amount to:

1. Test the waters
2. Signal willingness to cooperate
3. Not risk everything at once

Common strategies in trust games:

- Send \$0 (safe but no surplus creation)
- Send half (\$2.50) - moderate risk, moderate potential
- Send everything (\$5) - maximal trust, maximal potential surplus

I'll send \$2.50 (or \$3 since we likely need whole numbers). This shows willingness to cooperate while not being overly risky in round 1.

Actually, let me reconsider - the format asks for an amount between 0-5, and looking at the response format, it should be a number. ...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 705,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

Your total visible earnings across all rounds are \$5.5.

This round, you are the RECEIVER against your current co-player. They sent you \$2.5, so you received \$7.5. How much do you return? (0-7.5)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

- In Round 1, I sent \$2.5 (they saw \$2.22 due to noise)
- They returned \$3.0 out of \$6.66 received, showing cooperative behavior (~45% return rate)
- Now they've sent me \$2.5, showing continued trust

I want to reciprocate their cooperative behavior. They were generous in returning \$3.0 last round. I'll return a fair amount that rewards their trust.

```
{'return': 3.5}
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is $\$5 - \$2.5 + \$3.0 = \5.5 , correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of $\$2.5 + \$3.5 = \$6.0$ (net gain of \$1.0)
- I would keep $\$7.5 - \$3.5 = \$4.0$ (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created: $\$2.5 \times 3 = \7.5 , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have: $\$2.5$ (kept) + $\$2.5$ (share) = $\$5.0$, so I return \$2.5

...

usage:

```
{
  "input_tokens": 762,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise ma

y have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

reasoning:

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0
- My payoff: $\$5 - \$2.5 + \$3.0 = \5.5

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75
- Their visible payoff: $\$5 - \$2.5 + \$2.75 = \5.25

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)
- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

usage:

```
{
  "input_tokens": 1055,
  "output_tokens": 491,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

```
{'return': 5.0}
```

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0
Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)
- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of $\$1.63 + \$5.05 = \$6.68$, which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

usage:

```
{  
  "input_tokens": 1407,  
  "output_tokens": 525,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5
Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)
Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

```
{'send': 4.0}
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)
Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous amounts:

- \$3.0 in Round 1
- \$4.0 in Round 3
- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

usage:

```
{  
  "input_tokens": 1754,  
  "output_tokens": 406,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.

Your total visible earnings across all rounds are \$27.519999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.15, so you received \$9.45. How much do you return? (0-9.45)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our continued partnership:

Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0
Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$3.15, giving me \$9.45

This co-player continues to be highly cooperative and generous, returning around 50% each time. They've maintained trust throughout despite noise. I'll reciprocate with a generous return to sustain this excellent relationship.

```
{'return': 4.75}
```

reasoning:

Looking at our history:

Round 3: I sent \$3.0, they returned \$4.0
Round 4: They sent \$3.0, I returned \$5.0
Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$3.15, giving me \$9.45

This co-player has consistently been generous:

- Round 3: returned \$4.0 out of \$8.52 (47%)
- Round 5: returned \$6.3 out of \$12.63 (50%)
- Round 6: sent \$3.15 (down from \$3.0 but still good)

They've maintained trust and cooperation throughout. I should continue reciprocating generously. Given they sent \$3.15 and I received \$9.45, a fair return that rewards their trust would be around 50% or so.

Let me return about \$4.5-5 to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 2138,
  "output_tokens": 366,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.

Your total visible earnings across all rounds are \$32.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our partnership:

Round 4: They sent \$3.0, I returned \$5.0

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0, I returned \$4.75 (they saw \$4.49 due to noise)

Round 7: My turn to send

This co-player has remained cooperative throughout, continuing to send good amounts despite noise making my returns appear slightly lower. We've established a strong pattern of mutual cooperation.

I'll continue sending a high amount to maximize our joint earnings.

```
{'send': 4.0}
```

Reasoning:

Looking at the history:

Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0 (wait, actually they sent \$3.15 according to round 6), I returned \$4.75 (they saw \$4.49)

Round 7: My turn to send

Wait, let me recheck the history. In Round 6, it says "They sent you \$3.15" but now in Round 7 it says "Round 6 against you, as SENDER: sent \$4.0". This is the noise at play.

So from their perspective:

- Round 6: They sent \$4.0, it became \$12.0, they saw that I returned \$4.49

From my perspective:

- Round 6: They sent \$3.15, I received \$9.45, I returned \$4.75

The noise made them see \$4.49 instead of \$4.75. That's actually not too far off.

This co-player has been consistently cooperative. They continue to send good amounts. I should continue to cooperate strongly. I'll send another high amount.

I've been sending increasing amounts: \$2.5, \$3.0, \$4.0. I should continue at a high level to maximize our joint payoffs.

Usage: {


```
"input_tokens": 2496,
"output_tokens": 473,
"reasoning_tokens": 0
}
```

GAME - Agent_4 - Receiver - dyad_2

Timestamp: 2026-08-12T08:03:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_2",
  "move_index": 3,
  "opponent_id": "Agent_3",
  "pairing": {
    "agents": [
      "Agent_3",
      "Agent_4"
    ],
    "investor": "Agent_3",
    "roles": {
      "Agent_3": "investor",
      "Agent_4": "trustee"
    },
    "trustee": "Agent_4"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.22, that is 44.4% of their total endowment. You received \$6.66. How much do you return to the sender? (0-6.66)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.22 (44.4% of their endowment)
- This created \$6.66 for me after the 3x multiplier
- The sender currently has \$2.78 remaining

I want to reward their trust while also benefiting from the cooperative exchange. A reciprocal approach would be to return a similar proportion to what they sent.

If I return around 45% of what I received (~\$3.00):

- The sender would end with: $\$2.78 + \$3.00 = \$5.78$ (a gain of \$0.78)
- I would end with: $\$6.66 - \$3.00 = \$3.66$ (a gain of \$3.66)

This rewards the sender's trust, encourages future cooperation, and gives us both a benefit from the exchange.

{"return": 3.00}

reasoning:

Let me think about this carefully.

I'm the receiver in Round 1.

- The sender had \$5 and sent \$2.22 (which is 44.4% of their endowment)
- This was multiplied by 3x, so I received \$6.66
- I need to decide how much to return (between \$0 and \$6.66)

Key considerations:

1. The sender showed trust by sending nearly half their endowment
2. If I return nothing, I keep \$6.66 and the sender keeps \$2.78 ($\$5 - \2.22)
3. If I return some amount, it builds reciprocity for future rounds
4. This is a multi-round game with 8 agents, and pairings are randomized

Let me think about what would be fair and promote cooperation:

- The sender sent 44.4% of their endowment, showing significant trust
- A common reciprocal approach would be to return enough so both players gain equally from the cooperation
- The sender currently has: $\$5 - \$2.22 = \$2.78$
- I currently have: \$6.66

If I want to split the gains equally:

- Total is now: $\$2.78$ (sender) + $\$6.66$ (me) = $\$9.44$
- Original total was: $\$5 + \$0 = \$5$
- Gain from cooperation: $\$9.44 - \$5 = \$4.44$
- Equal split would give each person: $\$5 + \$2.22 = \$7.22$
- So sender should have \$7.22, meaning I return: $\$7.22 - \$2.78 = \$4.44$
- I would keep: $\$6.66 - \$4.44 = \$2.22$

Hmm, that seems high. Let me think about this differently.

I could match their proportion and return 44.4% of what I received, which would be about \$2.96. Or I could view the \$4.44 in new value created by the multiplication as something to split-giving each of us a \$2.22 gain. Since the sender already gave up \$2.22, they'd need to end with \$7.22 total, meaning I'd return \$4.44 to achieve equal outcomes.

But checking the math: if I return \$4.44, they'd have \$7.22 while I'd have \$2.22—that's not actually equal. The total pool is \$9.44, so for truly equal splits we'd each need \$4.72. That means I should return \$1.94, leaving me with \$4.72 as well. Alternatively, I could

ld stick with the proportional approach and return 44.4% of the \$6.66 I received. more t
han initial. The sender should end with $\$5 + \$2.22 = \$7.22$, which means I'd return \$4.44
and keep \$2.22. But this heavily favors the sender since they get most of the gains. I'
m reconsidering whether equal splits of the joint gain actually make sense here, or if I
should weight things differently based on who took the initial risk.

From a game theory angle, I need to think about what encourages cooperation across multipl
e rounds. Returning nothing is purely selfish, but returning everything sent treats it l
ike a simple reciprocal exchange. A middle ground-returning around \$3-would let the send
er gain \$0.78 while I keep \$3.66, rewarding their trust while still benefiting myself an
d signaling I'm willing to cooperate in future interactions.

I could also match what they sent (\$2.22), but that leaves them with no gain, which might
discourage them from trusting again. The proportion approach-returning 44.4% of what I r
eceived-lands at roughly \$2.96, giving the sender a modest gain of \$0.74 and me \$3.70. T
his balances mutual benefit with a clear signal that I reciprocate trust proportionally.

There's also the communication noise to account for-what I see might not be exactly what w
as sent. Considering everything, I'll return \$3.00 as a round number that rewards their
trust, gives me a solid gain, and demonstrates reciprocity without overthinking it....

usage:

```
{  
  "input_tokens": 529,  
  "output_tokens": 2312,  
  "reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible pa
yoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do yo
u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative rela
tionship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of
trust.

```
{"send": 2.5}
```

reasoning:

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: $\$6.66 - \$3.00 = \$3.66$

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible p
ayoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) mig
ht be different from what they actually sent (\$2.5). And what they saw returned might al
so differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was $\$5.5 = \$5 - \$2.5 + \3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be $\$5 - \$2.5 + \$3.0 = \5.5 (gain of \$0.5)
- Their...

usage:

```
{
  "input_tokens": 873,
  "output_tokens": 742,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: $\$2.16 + \$4.00 = \$6.16$ (net gain of \$1.16)
- I keep: $\$8.52 - \$4.00 = \$4.52$

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

reasoning:

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

usage:

```
{
  "input_tokens": 1177,
  "output_tokens": 777,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0
Round 2: I sent \$2.5 (50%), co-player returned \$3.5
Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

usage:

```
{  
  "input_tokens": 1614,  
  "output_tokens": 808,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)

Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost their entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets: \$0.79 + \$6.30 = \$7.09 (net gain of \$2.09)

- I keep: $\$12.63 - \$6.30 = \$6.33$

This reciprocates their trust appropriately.

```
{"return": 6.30}
```

reasoning:

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0

Round 2: I sent \$2.5, they returned \$3.5

Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)

Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0

Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)

- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32

- Sender would have: $\$0.79 + \$6.32 = \$7.11$ (gain of \$2.11)

- I would have: $\$12.63 - \$6.32 = \$6.31$

This seems fair and rewards their very high trust. Let me round to \$6.30 for a cleaner number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have: $\$0.79 + \$6.50 = \$7.29$

...

usage:

```
{
  "input_tokens": 2041,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

Your total visible earnings across all rounds are \$26.799999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 4: I sent \$3.37, they returned \$5.0 (strong reciprocation)

Round 5: They sent \$4.0 (80% of endowment), saw \$6.9 returned (good outcome)

My co-player has been increasing their trust levels and has reciprocated well when in the receiver role. They're demonstrating strong cooperative behavior.

To match their high trust level and continue building on this cooperative relationship, I should send a similarly high amount.

```
{"send": 4.0}
```

reasoning:

Looking at the visible history from my co-player's perspective:

Round 3: They sent \$3.0, saw \$3.01 returned (seemed like poor reciprocation)

Round 4: They saw I sent \$3.37, returned \$5.0 (good reciprocation - 49.5%)

Round 5: They sent \$4.0, saw \$6.9 returned (good reciprocation - 57.5%)

My co-player has been showing increasing trust (sending \$3.0, then \$4.0). They're being cooperative despite the noise in Round 3 where it appeared I only returned \$3.01.

Given their pattern of increasing trust and good returns when they're the receiver, I should continue with high cooperation. Sending around 60-80% would be appropriate to match their escalating trust levels.

I'll send \$4.0 (80%) to match their Round 5 amount and show continued high trust.

usage:

```
{
  "input_tokens": 2474,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.

Your total visible earnings across all rounds are \$32.29.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the visible history:

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of what they received)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to demonstrate trust and has been consistently returning around 50% when in the receiver role. I should match this reciprocal behavior.

I received \$10.05. Returning approximately 50%:

- Return \$5.00
- Sender gets: \$1.65 + \$5.00 = \$6.65 (net gain of \$1.65)
- I keep: \$10.05 - \$5.00 = \$5.05

```
{"return": 5.00}
```

Reasoning:

Looking at the visible history:

Round 6: They saw I sent \$3.15, and they returned \$4.75 (50.3% of \$9.45)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to show strong trust and has been returning around 50% when in the receiver role. They sent \$3.35 this round.

I received \$10.05. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.00 (approximately 50% of \$10.05)

- Sender gets: $\$1.65 + \$5.00 = \$6.65$ (net gain of $\$1.65$)
- I keep: $\$10.05 - \$5.00 = \$5.05$

This maintains consistency with their return rates and continues our cooperative relationship.

```
Usage: {
  "input_tokens": 2838,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

GAME - Agent_5 - Sender - dyad_3

Timestamp: 2026-08-12T08:03:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 4,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_5",
      "Agent_6"
    ],
    "investor": "Agent_5",
    "roles": {
      "Agent_5": "investor",
      "Agent_6": "trustee"
    },
    "trustee": "Agent_6"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance demonstrating trust with being cautious since I have no information about my co-player's behavior yet.

The game has potential for mutual benefit - if I send money and it gets tripled, there's room for both of us to be better off than our starting positions if the receiver reciprocates fairly.

However, there's also risk - the receiver could keep everything. The mention of "communication noise" adds another layer of uncertainty.

For Round 1, I'll send a moderate amount that shows willingness to cooperate while not being overly vulnerable:

```
'''json
{"send": 3}
'''
```

This gives my co-player \$9 to work with (3×3), which should provide good opportunity for mutually beneficial reciprocation if they're inclined to cooperate.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think through the situation:

1. I start with \$5
2. I can send any amount from \$0-\$5 to the receiver
3. Whatever I send gets multiplied by 3x
4. The receiver then decides how much to return to me
5. My final payoff = \$5 - sent + returned
6. Receiver's final payoff = (sent × 3) - returned

This is a classic trust game. The socially optimal outcome is for me to send the full \$5, which becomes \$15 for the receiver. Then the receiver could split this fairly (returning \$7.50, giving us each \$7.50) or even return more to reward my trust.

However, there are a few considerations:

- The receiver might not reciprocate and could keep everything
- There's mention of "communication noise" - amounts may differ from what's actually sent/returned
- This is a multi-round game with different pairings each round
- Agents are anonymous

Since this is the first round and I have no information about how others will behave, I need to balance:

1. Building trust/cooperation (sending more)
2. Protecting against exploitation (sending less)
3. Signaling willingness to cooperate for future rounds

A reasonable first-round strategy might be to send a moderate amount to test the waters - not nothing (which shows no trust), but not everything (which could be exploited). Something like \$3 could be a good middle ground, showing willingness to cooperate while keeping some protection.

However, in repeated trust games, starting with higher trust can establish better cooperation norms. Since pairings change each round, my behavior here sets a precedent but I won't necessarily play with this same person again.

I think I'll send \$3 as a reasonable initial offer that demonstrates trust without being overly vulnerable. This gives the receiver \$9, and if they return even \$3-4, we both end up better than the initial endowment.

...

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 756,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

****Round 2 Current Situation:****

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 (2.78×3)
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be: $\$5 - \$3 + \$4 = \6 (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 (4.35×3)
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted \$4.0 - \$2.78 = \$1.22 gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is \$5 - \$4.35 + \$6.525 = \$6.175
- I would keep \$13.05 - \$6.525 = \$6.525
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience.

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

usage:

```
{
  "input_tokens": 832,
  "output_tokens": 1707,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our history together:

****Round 1:****

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

****Round 2:****

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness

- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

reasoning:

Let me analyze the history with this co-player:

Round 1:

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

Round 2:

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - \$4.35 + \$8.76 = \$9.41 (visible to them as \$10.26)
- I ended with \$13.05 - \$8 = \$5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

Round 3:

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

usage:

```
{
  "input_tokens": 1268,
  "output_tokens": 684,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$9.34.

le payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at our cooperative history:

****Round 1:****

- They returned \$4 on \$8.34 received (48%)

****Round 2:****

- I returned \$8 on \$13.05 received (61%)

****Round 3:****

- They returned \$9 on \$14.01 received (64%) - very generous!

****Round 4 (Current):****

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final: \$5 - \$3.5 + \$7 = \$8.5 (profit of \$3.5)
- My final: \$10.5 - \$7 = \$3.5 (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

reasoning:

Let me review the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

****Round 2:****

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01
- They returned \$9 (about 64% of what they received)
- Very generous return!

****Round 4 (Current):****

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:
- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:
- Their final: $\$5 - \$3.5 + \$7.5 = \9 (profit of \$4)
- My final: $\$10.5 - \$7.5 = \$3$ (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance—it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

usage:

```
{
  "input_tokens": 1711,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our established cooperative pattern:

****Round 2:****

- They sent \$3.5 (70% of endowment)
- I returned generously

****Round 3:****

- I sent high (they saw \$4.67)
- They returned \$9 - very generous (64% of what they received)

****Round 4:****

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

****Round 5 (Current):****

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

ership.

reasoning:

Looking at the history with this co-player:

****Round 2:****

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

****Round 4:****

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

****Round 5 (Current):****

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

usage:

```
{
  "input_tokens": 2169,
  "output_tokens": 521,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.

Your total visible earnings across all rounds are \$32.48.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at our cooperative history:

****Round 3:****

- They returned \$9 on \$14.01 (64% return rate) - generous

****Round 4:****

- I returned fairly (they saw \$7.93)

****Round 5:****

- They saw only \$3.06 sent (due to noise), received \$9.18
- They still returned \$6 (65%!) - very generous despite the low apparent send

****Round 6 (Current):****

- They sent \$4.32 (86% of their endowment) - extremely high trust!
- I received \$12.96
- They're left with only \$0.68

They've consistently shown generosity and trust, especially in Round 5 where noise made it look like I sent less, yet they still returned generously. Now they're trusting me with almost everything.

If I return \$8:

- Their final: $\$5 - \$4.32 + \$8 = \8.68 (profit of \$3.68)
- My final: $\$12.96 - \$8 = \$4.96$ (profit of \$4.96)

```
'''json
{"return": 8}
'''
```

reasoning:

Looking at the history:

****Round 3:****

- I sent (they saw \$4.67)
- They returned \$9 on \$14.01 (64%)

****Round 4:****

- They sent \$4.0
- I returned (they saw \$7.93)

****Round 5:****

- I sent \$4 (they saw \$3.06 due to noise)
- They received \$9.18 and returned \$6 (65%)
- This is very generous given they only saw \$3.06 sent!

****Round 6 (Current):****

- They sent \$4.32 (86% of endowment) - very high trust!
- I received \$12.96
- They're at \$0.68

They've been consistently generous and trusting. In Round 5, despite the noise making it look like I only sent \$3.06, they still returned \$6 which is generous. Now they're showing even more trust by sending \$4.32.

Given their consistent generosity and trust:

- If I return \$8:
 - Their final: $\$5 - \$4.32 + \$8 = \8.68 (profit of \$3.68)
 - My final: $\$12.96 - \$8 = \$4.96$ (profit of \$4.96)

This rewards their trust appropriately and continues our cooperative relationship.

usage:

```
{
  "input_tokens": 2590,
  "output_tokens": 603,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.

Your total visible earnings across all rounds are \$37.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our continued cooperation:

****Round 4:****

- They sent \$4.0
- I returned fairly (they saw \$7.93)

****Round 5:****

- They saw only \$3.06 sent (noise affected it)
- Still returned \$6 - very generous despite the low visible amount

****Round 6:****

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

****Round 7 (Current):****

We've maintained an excellent cooperative relationship across 6 rounds. They've consistently shown trust even when noise made things look less favorable, and I've reciprocated generously.

Continuing our pattern of high trust:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with and demonstrates my continued commitment to our mutually beneficial partnership.

Reasoning:

Looking at our history:

****Round 4:****

- They sent \$4.0
- I returned \$7 (they saw \$7.93)

****Round 5:****

- I sent \$4 (they saw \$3.06 due to noise)
- They returned \$6 - very generous despite noise

****Round 6:****

- They sent \$4.32 (I calculated they sent, but they saw they sent \$4.0)
- I returned \$8 (they saw \$8.83)

****Round 7 (Current):****

We've established a very strong cooperative relationship. They've been consistently generous:

- Round 5: Returned \$6 on \$9.18 even when it looked like I only sent \$3.06
- Round 6: Sent \$4.0 (80% of endowment) showing high trust

I've also been cooperative, returning fairly and sending high amounts.

Given our strong pattern, I should send a high amount. I'll send \$4 to maintain our mutual trust:

This gives them \$12 to work with and continues our successful cooperative pattern.

```
Usage: {
  "input_tokens": 3086,
  "output_tokens": 459,
  "reasoning_tokens": 0
}
```

GAME - Agent_6 - Receiver - dyad_3

Timestamp: 2026-08-12T08:03:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_5",
  "pairing": {
    "agents": [
```

```

    "Agent_5",
    "Agent_6"
  ],
  "investor": "Agent_5",
  "roles": {
    "Agent_5": "investor",
    "Agent_6": "trustee"
  },
  "trustee": "Agent_6"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$2.78, that is 55.6% of their total endowment. You received \$8.34. How much do you return to the sender? (0-8.34)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed moderate-to-good trust by sending \$2.78 (55.6% of their endowment)
- They currently have \$2.22 remaining
- I received \$8.34

I want to reciprocate their trust and ensure they benefit from having sent money to me, while also keeping a reasonable amount for myself.

If I return \$4.00:

- The sender ends up with: $\$2.22 + \$4.00 = \$6.22$ (a gain of \$1.22)
- I keep: $\$8.34 - \$4.00 = \$4.34$

This rewards their trust with a positive return and splits the benefit reasonably between us.

```
{'return': 4.00}
```

reasoning:

This is the first round of what appears to be a trust game experiment.

Let me understand the situation:

- I am the RECEIVER
- The sender started with \$5
- The sender sent me \$2.78 (which is 55.6% of their \$5 endowment)
- This amount was multiplied by 3, so I received $\$2.78 \times 3 = \8.34
- I need to decide how much to return (between \$0 and \$8.34)

The sender has shown a moderate level of trust by sending more than half of their endowment.

Let me think about what would be a fair return:

- The sender now has: $\$5 - \$2.78 = \$2.22$
- I currently have: \$8.34
- If I want to be reciprocal and fair, I should consider what would make this beneficial for both of us

A common approach in trust games is to:

1. Return enough that the sender benefits from having trusted me (ends up with more than their initial \$5)
2. Keep enough to make it worthwhile for me as well

If I return about half of what I received (\$4.17), the sender would end up with $\$2.22 + \$4.17 = \$6.39$, which is better than their initial \$5, and I would keep \$4.17 as well. This seems fair and reciprocal.

Alternatively, if I think about equal splitting of the total gains:

- Total money created: \$8.34 (I received) vs \$2.78 (originally sent) = net gain of \$5.56
- If we split the gains equally, each gets \$2.78 extra
- Sender would get: $\$2.22$ (current) + $\$2.78$ (their share of gain) = $\$5.00$ (back to initial)
- Wait, that doesn't seem right. Let me recalculate.

Actually, let's think about the total pool:

- Initially: Sender had \$5, Receiver had \$0
- After sending: Sender has \$2.22, Receiver has \$8.34
- Total: \$10.56

If we wanted to split the total equally, each would get \$5.28. For the sender to reach that, I'd need to return \$3.06, leaving me with \$5.28 as well. This creates an even split where both of us benefit from the trust and multiplication. Since the sender demonstrated good faith by sending over half their endowment, returning slightly more than this equal split might be appropriate to reward that trust.

Testing different return amounts: returning \$3.50 would give the sender \$5.72 while I keep \$4.84—a significant advantage for me. Returning \$4.00 instead would give the sender \$6.22 with a \$1.22 profit, while I'd keep \$4.34. If I return exactly half the total gain at \$4.17, we'd each end up with \$6.39 and \$4.17 respectively. I'm weighing whether to prioritize rewarding the sender's trust or maintaining a more balanced outcome for myself...

usage:

```
{
```

```
"input_tokens": 529,  
"output_tokens": 1197,  
"reasoning_tokens": 0  
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

```
{'send': 3.50}
```

reasoning:

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize

the total pie, trusting they'll reciprocate fairly.

...

usage:

```
{
  "input_tokens": 836,
  "output_tokens": 625,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history and current situation:

****Analysis:****

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

****Reasoning:****

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with: \$0.33 + \$9.00 = \$9.33 (profit of \$4.33)
- I keep: \$14.01 - \$9.00 = \$5.01

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

{'return': 9.00}

reasoning:

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50, but they saw \$4.35. When tripled: $4.35 * 3 = 13.05$.

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept $10.50 - 8.0 = 2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So: \$5 + received - returned = payoff
- \$5 + \$13.05 - \$8.0 = \$10.05... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount: \$13.05 - \$8.0 = \$5.05. That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..

usage:

```
{  
  "input_tokens": 1170,  
  "output_tokens": 1644,  
  "reasoning_tokens": 0  
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the pattern with my co-player:

****Their behavior:****

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

****My decision:****

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

{ 'send': 4.00 }

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)

- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

usage:

```
{
  "input_tokens": 1613,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

****Decision:****

Their trust level deserves fair reciprocation.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

{ 'return': 6.00 }

reasoning:

Let me analyze the situation:

****Current situation:****

- My co-player sent \$3.06 (61.2% of their endowment)
- They have \$1.94 remaining
- I received \$9.18

****Their recent behavior as receiver:****

- Round 2: returned \$8.0 out of \$13.05 (61%)

- Round 4: returned \$7.0 out of \$10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

****My reciprocation pattern:****

Looking at what I've done when receiving:

- Round 1: returned \$4.00 out of \$8.34

- Round 3: returned \$9.00 out of \$14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return \$5.50:

- Sender ends with: $\$1.94 + \$5.50 = \$7.44$ (profit of \$2.44)

- I keep: $\$9.18 - \$5.50 = \$3.68$

If I return \$6.00:

- Sender ends with: $\$1.94 + \$6.00 = \$7.94$ (profit of \$2.94)

- I keep: $\$9.18 - \$6.00 = \$3.18$

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. \$6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2029,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.

Your total visible earnings across all rounds are \$31.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 3 and 5 - very high trust

- As receiver in Round 4, they returned \$7.0 out of \$10.5 (67%) - fair reciprocation

- They maintain high trust even when returns vary

****Decision:****

This co-player demonstrates strong, consistent trust. I should match their commitment level to maintain our cooperative relationship.

Sending \$4.00 (80%) matches their trust pattern and maximizes our joint potential gains.

{'send': 4.00}

reasoning:

Looking at my co-player's history:

Round 3: As sender, sent \$4.0 (80%), saw \$8.77 returned, payoff \$9.77

Round 4: As receiver, saw \$3.5 sent, returned \$7.0 out of \$10.5 (67%), payoff \$3.5

Round 5: As sender, sent \$4.0 (80%), saw \$6.2 returned, payoff \$7.2

Pattern analysis:

- As sender: Consistently sending \$4.0 (80%) - very high trust
- As receiver: Returning 67% - fair reciprocation
- They're maintaining high trust despite varying returns

This co-player has been consistently trusting. They've sent 80% twice in a row. As receiver, they returned a decent amount (67%).

I should match their high trust level. They're showing strong cooperative behavior, and I should reciprocate that trust.

Sending \$4.00 (80%) would match their pattern and continue our cooperation.

usage:

```
{
  "input_tokens": 2424,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.

Your total visible earnings across all rounds are \$41.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76
- As receiver, they've returned 62-67% consistently, showing fair reciprocation

****Decision:****

Their high trust level warrants strong reciprocation.

If I return \$7.50:

- Sender ends with: \$1.08 + \$7.50 = \$8.58 (profit of \$3.58)
- I keep: \$11.76 - \$7.50 = \$4.26

This rewards their trust with a solid profit while maintaining fairness in our ongoing cooperation.

{'return': 7.50}

Reasoning:

Looking at the history with my co-player:

****Their pattern:****

- Round 4 as RECEIVER: returned \$7.0 out of \$10.5 (67%)
- Round 5 as SENDER: sent \$4.0 (80%)
- Round 6 as RECEIVER: returned \$8.0 out of \$12.96 (62%)

****Current situation:****

- They sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76

****Their behavior as receiver:****

- They've been consistently fair, returning 62-67% of what they receive

- This has been good reciprocation

****My decision:****

They showed high trust by sending \$3.92 (78.4%). I should reward this appropriately.

If I return \$7.50:

- Sender ends with: $\$1.08 + \$7.50 = \$8.58$ (profit of \$3.58)
- I keep: $\$11.76 - \$7.50 = \$4.26$

This is a fair return that rewards their trust with a good profit, similar to what they've done for me.

Actually, let me recalculate to be consistent with their pattern. They return around 62-67%. If I do similar:

- 65% of \$11.76 = \$7.64

But I also want to ensure they profit well from their trust. Let me aim for \$7.50-\$8.00.

I'll go with \$7.50 as a good balance.

```
Usage: {  
  "input_tokens": 2806,  
  "output_tokens": 558,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_7 - Sender - dyad_4

Timestamp: 2026-08-12T08:03:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 6,  
  "opponent_id": "Agent_8",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: $\$5 - \text{sent} + \text{returned}$
- Receiver keeps: $\text{received} - \text{returned}$
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

Looking at this first round, I need to balance trust with caution since I have no information about my co-player yet.

****My reasoning:****

- Sending money can create mutual benefit (it triples in value)
- But I don't know if my co-player will reciprocate
- A moderate amount signals willingness to cooperate while limiting risk
- The communication noise means my co-player might see a slightly different amount anyway

I'll send a meaningful amount to signal trust and test cooperation, but not risk everything on the first round.

```
'''json
{"send": 3}
'''
```

reasoning:

This is the first round of a trust game. I'm the sender with \$5. Let me think through the strategic considerations:

1. If I send X dollars, the receiver gets 3X dollars
2. The receiver then decides how much to return to me
3. My final payoff = 5 - X + returned amount
4. Receiver's final payoff = 3X - returned amount

This is a classic trust game. The socially optimal outcome is for me to send my full \$5 (which becomes \$15 for the receiver), and then the receiver returns enough to make us both better off than the initial endowment.

However, there are a few considerations:

- This is round 1, so I have no information about this particular receiver's behavior
- The instructions mention there may be "communication noise" - amounts I see may differ from what was actually sent/returned
- This is a multi-agent setting with randomized pairings each round
- I'll be playing multiple rounds, potentially with different partners

Strategic considerations:

1. If I send nothing, I keep \$5 for sure
2. If I send everything and the receiver returns nothing, I get \$0
3. If I send everything and receiver returns fairly (say half of what they receive = \$7.50), I'd get \$7.50 total
4. Even if receiver returns just 1/3 of what they get (\$5), we'd both end up with \$5 (status quo)

Given this is the first round and I have no information about the receiver's trustworthiness, I have several options:

Option A: Send \$0 - safe but no potential gains

Option B: Send a moderate amount (e.g., \$2-3) - test the waters

Option C: Send everything (\$5) - maximize potential social welfare but risky

In classic trust game experiments, many people start with moderate amounts to test their partner. However, some research shows that being generous can elicit reciprocity.

The note about "communication noise" is interesting - this might mean that even if I send a certain amount, the receiver might see a different amount, which could affect their perception of my trustworthiness.

Given the multi-round structure with randomized pairings and communication noise, I should send a moderate-to-generous amount to signal cooperation without risking everything. Sending \$3 (60% of my endowment) demonstrates trust while giving the receiver \$9 to work with enough for mutually beneficial returns. This approach tests the waters, establishes a cooperative stance, and protects against worst-case scenarios....

usage:

```
{
  "input_tokens": 500,
  "output_tokens": 952,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the history:

****Round 1 Analysis:****

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

****Current Situation:****

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

****My decision:****

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

reasoning:

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ($3 \times \3.65)
- They returned \$7.0
- Their visible payoff was \$3.95 ($10.95 - 7.0$)
- My payoff was $\$5 - \$3 + \$7 = \$9...$ but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ($3 \times \4)
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total ($\$1 + \6) and I'd have \$6
- If I return \$8, they'd have \$9 total ($\$1 + \8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

usage:

```
{
  "input_tokens": 781,
  "output_tokens": 752,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

****Pattern Analysis:****

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

****Current Decision:****

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver

- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

reasoning:

Let me analyze the history with this co-player:

Round 1:

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

Round 2:

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with $\$1.5 + \$6.52 = \$8.02$

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with $\$12 - \$7 = \$5$

Pattern Analysis:

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

Current Decision (Round 3, I'm SENDER):

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

usage:

```
{
  "input_tokens": 1141,
  "output_tokens": 570,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.3300000000000002, returned \$8.0, visible payoff \$4.3300000000000002.

Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so y

ou received \$12.36. How much do you return? (0-12.36)
Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the history with this co-player:

****Cooperation Pattern:****

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

****Current Situation:****

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

****My Decision:****

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

****Round 2:****

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)

****Round 3:****

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

****Round 4 (current):****

- They are SENDER, sent \$4.12, I received \$12.36
- They kept \$0.88

****Pattern:****

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did: $0.65 * \$12.36 = \8.03

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88
- If I return \$8, they get \$8.88 total, I keep \$4.36
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$28.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant**content:**

Looking at the history with this co-player:

****Established Pattern:****

- Round 2: They sent \$3.5, trusted me
- Round 3: They returned \$8 generously
- Round 4: They sent \$4 (high trust), I returned \$8

****Analysis:****

This co-player has been consistently cooperative:

- Always returns generously as receiver (\$7-8)
- Shows increasing trust as sender (\$3.5 ? \$4)
- We've built a strong reciprocal relationship

****Current Decision:****

They sent \$4 last round (80% of endowment), showing strong trust. I should match this level of trust to maintain our cooperative dynamic.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$3.5, I returned ~\$7 (they saw \$6.52 due to noise)

Round 3: I sent ~\$3.5 (they saw \$4.11), they returned \$8

Round 4: They sent \$4 (I saw \$4.12), I returned \$8 (they saw \$7.68)

This co-player has been:

- Consistently generous as receiver (returned \$7, then \$8)
- Trusting as sender (sent \$3.5, then \$4)
- Maintaining high cooperation levels

The pattern shows escalating trust and cooperation. They sent \$4 last time, which is 80% of their endowment.

I should continue this cooperative trajectory. Since they've been so consistently cooperative and generous, I should send a similar or slightly higher amount to maintain the positive relationship.

I'll send \$4 - matching their last send amount and showing equal trust.

usage:

```
{
  "input_tokens": 1974,
  "output_tokens": 404,
  "reasoning_tokens": 0
}
```

Message 12 role=user**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
 - Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
 - Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the history with this co-player:

****Their Pattern:****

- Round 3: Returned \$8/\$12.33 = 65%
- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 6 (now): Sent \$4.4 (88% of endowment!)

****Current Situation:****

- They sent \$4.4, keeping only \$0.6
- I received \$13.2
- This shows exceptional trust - their highest send yet

****My Decision:****

They've consistently returned 65-69% and are now showing maximum trust. I should match their generosity pattern and reward this trust.

Returning \$9:

- They end with \$9.6 total
- I keep \$4.2
- ~68% return rate, matching their established pattern
- Sustains our strong cooperation

```
'''json
{"return": 9}
'''
```

reasoning:

Looking at the history with this co-player:

****Their behavior:****

- Round 3: Returned \$8 out of \$12.33 (65%)
- Round 5: Returned \$7.5 out of \$10.86 (69%)

****Current situation:****

- They sent \$4.4 (88% of their endowment!) - showing very high trust
- I received \$13.2
- They kept only \$0.6

****Analysis:****

This co-player is showing increasing trust and generosity:

- They've returned 65-69% as receiver
- As sender, they're now sending even more (\$4.4 vs \$4 previously)
- This shows they believe in our cooperation

****My decision:****

I should return proportionally to their generosity pattern. They've been returning around 65-69%.

If I return \$8.5:

- They end with \$9.1 total (\$0.6 + \$8.5)
- I keep \$4.7
- Return rate: 64% (consistent with their pattern)

If I return \$9:

- They end with \$9.6 total
- I keep \$4.2
- Return rate: 68% (matches their higher range)

Given they sent \$4.4 (very high), I should be generous back. Let me return \$8.5 to maintain good cooperation while keeping the return rate reasonable.

Actually, thinking about it more - they sent \$4.4 out of \$5 (88%). This is extremely trusting. They've been consistently generous. I should match that generosity.

Return \$9:

- They get \$9.6 total (\$0.6 kept + \$9 returned)
- I keep \$4.2
- This is about 68% return rate, matching their generosity pattern
- Rewards their exceptional trust

usage:

```
{
  "input_tokens": 2394,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.

Your total visible earnings across all rounds are \$41.81.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the history with this co-player:

****Their Pattern:****

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver (generous)
- Round 6: Sent \$4 as sender again

****Analysis:****

This co-player has been exceptionally consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns generously when receiver (\$7.5-\$8)
- We've built strong mutual trust over 6 rounds

****Current Decision:****

They consistently send \$4, showing strong trust. I should match this behavior to maintain our excellent cooperative relationship.

```
'''json
{"send": 4}
'''
```

Reasoning:

Looking at the history with this co-player:

****Their Pattern:****

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver
- Round 6: Sent \$4 as sender (I returned \$9, they saw \$9.22)

****Analysis:****

This co-player has been:

- Consistently sending \$4 when they're sender
- Returning generously (\$7.5-\$8) when they're receiver
- Very trustworthy and cooperative throughout

****Current Decision:****

They've been sending \$4 consistently. I should match this level of trust to maintain our excellent cooperative relationship. We've built strong mutual trust over 6 rounds.

I should send \$4 to match their behavior and maintain our reciprocal pattern.

```
Usage: {  
  "input_tokens": 2827,  
  "output_tokens": 351,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_8 - Receiver - dyad_4

Timestamp: 2026-08-12T08:03:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_4",  
  "move_index": 7,  
  "opponent_id": "Agent_7",  
  "pairing": {  
    "agents": [  
      "Agent_7",  
      "Agent_8"  
    ],  
    "investor": "Agent_7",  
    "roles": {  
      "Agent_7": "investor",  
      "Agent_8": "trustee"  
    },  
    "trustee": "Agent_8"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 1: You are the RECEIVER. the sender sent \$3.65, that is 73.0% of their total endowment. You received \$10.95. How much do you return to the sender? (0-10.95)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$3.65 (73% of their endowment)
- This created \$10.95 for me after the 3x multiplier
- The sender currently has \$1.35 remaining

I want to reciprocate their trust while ensuring both of us benefit from cooperating. The game created additional value (\$12.30 total vs the original \$5), and a fair approach would be to split these gains.

If I return \$7.00:

- Sender ends with: $\$1.35 + \$7.00 = \$8.35$ (gain of \$3.35)
- I end with: $\$10.95 - \$7.00 = \$3.95$ (gain of \$3.95)

This rewards the sender's trust while keeping the interaction balanced and positive for both of us.

```
'''json
{"return": 7.00}
'''
```

reasoning:

This is the first round of a trust game. I'm the receiver, and the sender sent me \$3.65 out of their \$5 endowment (which is 73% - quite generous). This was multiplied by 3, so I received \$10.95.

Now I need to decide how much to return. Let me think about this:

1. The sender showed significant trust by sending 73% of their endowment
2. If I return nothing, I keep \$10.95 and the sender gets $\$5 - \$3.65 = \$1.35$
3. If I return everything, I keep \$0 and the sender gets $\$1.35 + \$10.95 = \$12.30$
4. A "fair" split might be to ensure we both benefit equally from the transaction

Let me calculate what would be equitable:

- Without the game: Sender has \$5, I have \$0
- Sender sent \$3.65, has \$1.35 left
- I received \$10.95
- Total created value: $\$1.35 + \$10.95 = \$12.30$ (vs original \$5)
- The game created $\$12.30 - \$5 = \$7.30$ in additional value

One approach: Split the gains 50-50

- We created \$7.30 extra
- Each should get \$3.65 extra
- Sender started with \$5, so should end with $\$5 + \$3.65 = \$8.65$
- They currently have \$1.35, so I should return $\$8.65 - \$1.35 = \$7.30$
- I would keep $\$10.95 - \$7.30 = \$3.65$

That seems fair - we each end up with \$3.65 more than our starting positions.

Another approach: Match the sender's trust level

- They sent 73% of their endowment

- I could return 73% of what I received: $0.73 \times \$10.95 = \7.99
- Or split the total pool equally: $\$12.30 \div 2 = \6.15 each, meaning I return \$4.80

The sender's generosity warrants reciprocation. I'll return \$7.30, which splits the gains equally and leaves us both with \$3.65 more than we started with....

usage:

```
{
  "input_tokens": 529,
  "output_tokens": 1202,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

reasoning:

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

usage:

```
{
  "input_tokens": 878,
  "output_tokens": 702,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

reasoning:

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)

- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

usage:

```
{
  "input_tokens": 1217,
  "output_tokens": 772,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
```json
{"send": 4.00}
```



'''

**reasoning:**

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92

Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0

Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the returns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

**usage:**

```
{
 "input_tokens": 1638,
 "output_tokens": 388,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

**reasoning:**

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86
- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender:  $\$1.38 + \$7.50 = \$8.88$
- Me:  $\$10.86 - \$7.50 = \$3.36$

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

**usage:**

```
{
 "input_tokens": 2006,
 "output_tokens": 390,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.

Your total visible earnings across all rounds are \$28.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our ongoing cooperative relationship:

My co-player has been consistently trustworthy:

- Round 3: sent \$3.5 (70%)
- Round 5: sent \$4.0 (80%) - showing increasing trust
- Round 4 as receiver: returned \$8.0 from \$12.36

They've demonstrated strong reciprocity and increased their trust level over time. I should match this escalating cooperation.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history:

My co-player has consistently shown strong trust:

- Round 3 as sender: sent \$3.5 (70%)
- Round 5 as sender: sent \$4.0 (80%)
- Round 4 as receiver: returned \$8.0 when they received \$12.36

They've received good returns (\$8.46, \$8.37) and continue to maintain high cooperation levels.

This is clearly a very cooperative partner. I should continue sending substantial amounts to maintain this mutually beneficial relationship.

I've sent:

- Round 2: \$3.50
- Round 4: \$4.00

I should maintain or even slightly increase my sending to show continued trust, especially since they increased from \$3.5 to \$4.0.

I'll send \$4.00 to maintain strong cooperation.

usage:

```
{
 "input_tokens": 2357,
 "output_tokens": 313,
 "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.

Your total visible earnings across all rounds are \$38.56.

This round, you are the RECEIVER against your current co-player. They sent you \$4.84, so you received \$14.52. How much do you return? (0-14.52)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - this is exceptional trust!
- I received \$14.52
- They currently have only \$0.16 remaining

This is the highest level of trust they've shown in our partnership. They've been consistently cooperative and have escalated their trust over time (\$3.5 ? \$4.0 ? \$4.84).

I need to strongly reward this exceptional trust to maintain our cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

Reasoning:

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - extremely high trust!
- I received \$14.52
- They have only \$0.16 remaining

Our history shows strong mutual cooperation:

- As receiver, they've returned \$8.0 and \$9.0 in recent rounds
- As sender, they've been escalating trust: \$3.5 ? \$4.0 ? \$4.84 (increasingly high percentages)

This is the highest level of trust they've shown yet - sending nearly all their endowment. I should strongly reward this trust.

If I return \$10.00:

- Sender: \$0.16 + \$10.00 = \$10.16
- Me: \$14.52 - \$10.00 = \$4.52

This gives them a good payoff that rewards their exceptional trust while still leaving me with a reasonable gain.

Given they sent such a high amount, I should return proportionally more. Let me think about what's fair:

- They sent \$4.84, they have \$0.16
- I received \$14.52
- Total created: \$0.16 + \$14.52 = \$14.68

A fair split might be around equal final payoffs or better for the sender given their huge trust.

If I return \$10.00:  
- Sender gets \$10.16  
- I get \$4.52  
Total: Both benefit significantly

If I return \$9.50:  
- Sender gets \$9.66  
- I get \$5.02

Given their extremely high trust (96.8%), I think \$10.00 is appropriate to strongly reward this behavior and maintain the cooperative relationship.

```
Usage: {
 "input_tokens": 2711,
 "output_tokens": 534,
 "reasoning_tokens": 0
}
```

## ROUND 8

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Receiver, Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_6=Sender, Agent\_5=Receiver  
dyad\_4: Agent\_8=Sender, Agent\_7=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 sent \$5; received \$15; Agent\_1 returned \$10; payoffs Agent\_2=\$10, Agent\_1=\$5;  
noise/visible: returned visible=\$9.84  
Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 15.0, "min": 0, "raw": 10.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}  
dyad\_2: Agent\_4 sent \$4; received \$12; Agent\_3 returned \$6; payoffs Agent\_4=\$7, Agent\_3=\$6;  
noise/visible: returned visible=\$6.42  
Action validation: {"returned": {"amount": 6.0, "clamped": false, "max": 12.0, "min": 0, "raw": 6.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_6 sent \$4; received \$12; Agent\_5 returned \$8; payoffs Agent\_6=\$9, Agent\_5=\$4;  
noise/visible: sent visible=\$4.14, received visible=\$12.42, returned visible=\$7.68  
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_4: Agent\_8 sent \$4; received \$12; Agent\_7 returned \$6.5; payoffs Agent\_8=\$7.5, Agent\_7=\$5.5;  
noise/visible: sent visible=\$3.21, received visible=\$9.63, returned visible=\$6.07  
Action validation: {"returned": {"amount": 6.5, "clamped": false, "max": 12.0, "min": 0, "raw": 6.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

### Balances after round: {

"Agent\_1": 59.5,  
"Agent\_2": 47.5,  
"Agent\_3": 46.05,  
"Agent\_4": 47.95,  
"Agent\_5": 47.0,  
"Agent\_6": 54.0,  
"Agent\_7": 54.0,  
"Agent\_8": 46.0

}

### Visible balances after round: {

"Agent\_1": 53.39,  
"Agent\_2": 53.03,  
"Agent\_3": 44.65,  
"Agent\_4": 44.760000000000005,  
"Agent\_5": 51.03,  
"Agent\_6": 54.489999999999995,  
"Agent\_7": 55.61,  
"Agent\_8": 50.15

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T08:03:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 0,
 "opponent_id": "Agent_1",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.

Your total visible earnings across all rounds are \$3.5600000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the visible history:

My co-player showed significant trust in Round 1 by sending \$3.0 (60% of their endowment) and received a solid return of \$6.1. This indicates they're willing to cooperate and trust paid off for them.

Since they demonstrated trust and it worked out well, I should reciprocate with similar trust to continue this cooperative relationship. I'll send a comparable proportion of my endowment.

```
'''json
{"send": 3.50}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 (from my perspective):

- I was the RECEIVER
- Sender sent \$3.52, which became \$10.56
- I returned \$7.00
- My earnings: \$10.56 - \$7.00 = \$3.56

But wait, the visible history shows something different from what I experienced:

- It says the sender (who is now my co-player again) sent \$3.0 (not \$3.52)
- It became \$9.0 (not \$10.56)
- They saw \$6.1 returned (not \$7.00)
- Their visible payoff was \$8.1

This is the "communication noise" mentioned in the instructions! The amounts seen may differ from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0
- They saw \$6.1 returned
- They earned \$8.1 total (\$5 - \$3.0 + \$6.1 = \$8.1)

From my actual experience:

- They sent \$3.52
- I returned \$7.00
- I kept \$3.56

Now I'm the SENDER in Round 2, playing against the same person.

Key considerations:

1. My co-player showed trust by sending 60% (\$3.0 as they saw it, or 70.4% as it actually was)
2. They saw a return of \$6.1, which from their perspective was a good return (they got back more than double what they sent)
3. I tried to be fair and reciprocate their trust

Given this:

- My co-player has shown they're willing to trust
- They received a good return (from their perspective)
- I should reciprocate with similar trust

I should send a significant amount to continue building this cooperative relationship. Since they sent around 60-70%, I should send a similar proportion to show reciprocal trust.

...

**usage:**

```
{
 "input_tokens": 902,
 "output_tokens": 597,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.

Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34

.

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with:  $\$0.66 + \$9.00 = \$9.66$  (profit of \$4.66)
- I keep:  $\$13.02 - \$9.00 = \$4.02$  (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust.

I should consider returning enough to make it very worthwhile for them.

...

**usage:**

```
{
 "input_tokens": 1203,
 "output_tokens": 843,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)



- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept:  $\$5 - \$3.50 + \text{return} = \$1.50 + \text{return}$

Round 3: I returned \$9.00, kept \$4.02

So my earnings:  $\$3.56 + (\$1.50 + \text{what I got back in round 2}) + \$4.02 = \$14.92$

This means:  $\$9.08 + \text{what I got back in round 2} = \$14.92$

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

usage:

```
{
 "input_tokens": 1669,
 "output_tokens": 673,
 "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with:  $\$0.03 + \$10.00 = \$10.03$  (profit of \$5.03)
- I keep:  $\$14.91 - \$10.00 = \$4.91$  (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
```

'''

**reasoning:**

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)  
Round 3: They sent \$4.0, received \$8.5 back  
Round 4: They returned \$6.0 from \$9.42 (generous)  
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with:  $\$0.03 + \$10.00 = \$10.03$  (profit of \$5.03)
- I keep:  $\$14.91 - \$10.00 = \$4.91$  (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender:  $\$0.03 + \$9.50 = \$9.53$  (profit of \$4.53)
- Me:  $\$14.91 - \$9.50 = \$5.41$  (profit of \$5.41)

...

**usage:**

```
{
 "input_tokens": 2083,
 "output_tokens": 771,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.

Your total visible earnings across all rounds are \$27.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have been highly cooperative throughout:

- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (strong return)

This co-player has demonstrated:

1. High trust as sender (escalating from \$4.0 to \$4.5)
2. Fair/generous reciprocation as receiver
3. Consistent success in our cooperative relationship

They've been rewarded well for their trust and have reciprocated fairly. I should continue this strong cooperation by matching their recent trust level.

```
'''json
{"send": 4.50}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

- Round 3: They sent \$4.0, received \$8.5 back (very good return)
- Round 4: They returned \$6.0 from \$9.42 (generous)
- Round 5: They sent \$4.5, received \$9.44 back (excellent return)

This co-player has been:

1. Highly trusting as sender (sent \$4.0, then \$4.5)
2. Generous as receiver (returned \$6.0 from \$9.42)
3. Consistently rewarded with good returns

The relationship has been very cooperative and mutually beneficial. They've been sending high amounts and getting good returns. As a receiver, they've been fair/generous.

Given this strong cooperative history, I should send a high amount to continue this mutually beneficial relationship. They sent \$4.5 in round 5, so I should match or exceed that level of trust.

I'll send \$4.50 to match their recent trust level.

**usage:**

```
{
 "input_tokens": 2556,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$38.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at this situation:

My co-player has shown very high trust by sending \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19.

The visible history shows excellent cooperation:

- Round 4: Returned \$6.0 from \$9.42
- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

They were extremely generous in Round 6, returning \$10.0 from \$15. Now they're showing continued high trust as a sender.

The multiplication created \$9.62 in added value. To reciprocate their strong cooperation:

If I return \$10.00:  
- Sender ends with:  $\$0.19 + \$10.00 = \$10.19$  (profit of \$5.19)  
- I keep:  $\$14.43 - \$10.00 = \$4.43$  (profit of \$4.43)

This maintains our successful cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at this situation:

My co-player sent \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19. This is very high trust.

The visible history shows strong cooperation:

- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (good return)
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

The multiplication created \$9.62 in added value ( $\$14.43 - \$4.81$ ).

My co-player has been very generous as a receiver (returning \$6.0, then \$10.0) and trusting as a sender (sending \$4.5, then \$4.81).

To reciprocate this high trust and maintain the cooperative relationship:

If I return \$10.00:  
- Sender ends with:  $\$0.19 + \$10.00 = \$10.19$  (profit of \$5.19)  
- I keep:  $\$14.43 - \$10.00 = \$4.43$  (profit of \$4.43)

This is fair and rewards their very high trust. Given they returned \$10.0 from \$15 last round, returning \$10.00 from \$14.43 would be consistent with what they've shown.

Actually, let me think about this more carefully. They sent \$4.81, leaving \$0.19. The multiplication created \$14.43 for me, which is an added value of \$9.62.

If I return \$10.00, they get \$10.19 total (profit \$5.19), I keep \$4.43 (profit \$4.43). We both profit well.

Given the strong cooperative history and their very high trust (96.2% of endowment), \$10.00 seems appropriate.

**usage:**

```
{
 "input_tokens": 2963,
 "output_tokens": 735,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.

Your total visible earnings across all rounds are \$43.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the visible history with my co-player:

They have shown exceptional cooperation:

- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (extremely generous)
- Round 7: Sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has:

1. Escalated to maximum trust by sending their entire \$5 endowment
2. Been very generous as a receiver
3. Maintained consistent cooperation throughout

They've shown maximum trust and it's been rewarded. I should reciprocate with the same level of trust to complete our highly successful cooperative relationship.

```
'''json
{"send": 5.00}
'''
```

#### Reasoning:

Looking at the visible history with my co-player:

- Round 5: They sent \$4.5, received \$9.44 back
- Round 6: They returned \$10.0 from \$15 (very generous)
- Round 7: They sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has now shown maximum trust in Round 7 by sending their entire \$5 endowment. They have been:

1. Escalating their trust (from \$4.5 to \$5.0)
2. Very generous as a receiver (returned \$10.0 from \$15)
3. Consistently cooperative and successful

They sent the maximum (\$5.0) in Round 7 and received \$9.6 back, which is a good return. I should reciprocate with maximum trust as well to maintain this strong cooperative relationship.

This is the final round (Round 8), so I should send my full amount to maximize our joint payoff and complete our cooperative relationship on a strong note.

```
Usage: {
 "input_tokens": 3154,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T08:03:59

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_1",
 "move_index": 1,
 "opponent_id": "Agent_2",
 "pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

Exact input messages sent to model:

```
Message 1 role=system
content:
```

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.

Your total visible earnings across all rounds are \$8.1.

This round, you are the RECEIVER against your current co-player. They sent you \$2.61, so you received \$7.83. How much do you return? (0-7.83)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

In Round 1, my co-player received \$10.56 (after tripling) and generously returned \$7.0 - about 66% of what they received. This shows they're cooperative.

Now in Round 2, they've sent me \$2.61, which became \$7.83. They're continuing to trust me despite the noise in communication.

I should reciprocate their cooperative behavior. Returning about 60-65% seems fair and maintains the cooperative relationship we've established.

```
'''json
{"return": 5.0}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1:

- I sent \$3 (but noise made it appear as \$3.52 to the receiver)
- The receiver got \$10.56 (3x of \$3.52)
- They returned \$7.0
- My visible payoff was \$3.56 (which means:  $\$5 - \$3 + \$7 = \$9$ , but with noise it shows as \$3.56... wait that doesn't add up)

Actually, let me recalculate. If I sent \$3, I should have:

- Kept:  $\$5 - \$3 = \$2$
- Plus returned: \$7
- Total:  $\$2 + \$7 = \$9$

But my visible earnings are \$8.1. There must be noise affecting what I see too.

2. In Round 2:

- My current co-player (same person from Round 1) is now the SENDER
- They sent me \$2.61 (after noise)
- I received \$7.83
- I need to decide how much to return

The co-player showed good faith in Round 1 by returning \$7.0 out of \$10.56 (about 66%). This is quite generous - they returned more than they received originally.

Now they've sent me \$2.61, showing continued trust despite the noise.

I should reciprocate their generosity. Let me think:

- They sent \$2.61 (roughly \$2.61 of their \$5)
- I received \$7.83
- If I return about 60-70% of what I received, that would be fair
- 60% of \$7.83 = \$4.70
- 70% of \$7.83 = \$5.48

Given that they were generous in Round 1 (returning \$7 out of \$10.56), I should be generous too. ...

**usage:**

```
{
 "input_tokens": 803,
 "output_tokens": 635,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 495,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)



They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 505,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

**reasoning:**

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)  
Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous  
Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

**usage:**

```
{
 "input_tokens": 1825,
 "output_tokens": 315,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$33.79.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my co-player's consistent pattern:

Round 3: Returned \$9.0 out of \$13.02 (69%)  
Round 5: Returned \$10.0 out of \$14.91 (67%)

Now in Round 6, they've sent me the maximum amount possible (\$5), showing complete trust in me. This is the strongest signal of cooperation they could give.

I should reciprocate this maximum trust with a generous return. Returning about 67% of the \$15 I received (matching their typical return rate) would be appropriate.

```
'''json
{"return": 10.0}
'''
```

**reasoning:**

Looking at the pattern with my co-player:

Round 3: They returned \$9.0 out of \$13.02 (69%)  
Round 5: They returned \$10.0 out of \$14.91 (67%)

And now in Round 6, they sent me the maximum amount (\$5), which became \$15. This is showing maximum trust in me.

They've been extremely generous throughout our interactions:

- Always sending substantial amounts when they're the sender
- Always returning generous amounts (around 67-69%) when they're the receiver

They just sent me the maximum possible amount (\$5), which shows complete trust in me. I should reciprocate this trust with a generous return.

Given their consistent pattern of returning around 67-69% and their maximum trust in sending \$5, I should return around 65-70% of \$15, which would be about \$9.75-\$10.50.

I'll return \$10, which is 67% of what I received, matching their typical return rate and showing I value our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2188,
 "output_tokens": 385,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.
- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.

Your total visible earnings across all rounds are \$38.79.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my co-player's excellent track record:

Round 4: Sent \$4.0 (high trust)

Round 5: Returned \$10.0 out of \$14.91 (67% - very generous)

Round 6: Sent \$4.5 (continued high trust)

This co-player has been consistently cooperative and trustworthy across all our interactions. They've shown maximum cooperation, and our relationship has been mutually beneficial.

I should reciprocate with maximum trust by sending the full amount.

```
'''json
{"send": 5}
'''
```

**reasoning:**

Looking at my co-player's history:

Round 4: Sent \$4.0 as sender (high trust)

Round 5: Returned \$10.0 out of \$14.91 (67% - generous)

Round 6: Sent \$4.5 as sender (high trust)

This co-player has consistently demonstrated:

- High trust as sender (sending \$4.0-\$4.5)
- Generous returns as receiver (67%+)
- Strong cooperative behavior throughout

We've built an excellent cooperative relationship over 6 rounds. They've been trustworthy every single time.

Given their consistent cooperation and trustworthiness, I should send the maximum amount to maximize our joint payoffs. This continues our pattern of mutual trust and cooperation.

usage:

```
{
 "input_tokens": 2547,
 "output_tokens": 314,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.
- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.
- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.

Your total visible earnings across all rounds are \$48.39.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at my co-player's consistent cooperative behavior:

Round 5: Returned \$10.0 out of \$14.91 (67%)

Round 7: Returned \$10.0 out of \$14.43 (69%)

They've now sent me the maximum \$5 again in Round 8, demonstrating complete trust for the second time. Their pattern has been consistently generous and cooperative throughout all our interactions.

I should reciprocate their maximum trust with a generous return, matching their typical return rate of around 67%.

```
'''json
{"return": 10.0}
'''
```

**Reasoning:**

Looking at the history with my co-player:

Round 5: They returned \$10.0 out of \$14.91 (67%)

Round 6: They sent \$4.5 (high trust)

Round 7: They returned \$10.0 out of \$14.43 (69%)

Now in Round 8, they've sent me the maximum \$5 again, showing complete trust.

This co-player has been extremely cooperative and consistent:

- Always sending high amounts when sender (4.0, 4.5, 5.0)
- Always returning generously when receiver (around 67-69%)

They're showing maximum trust by sending \$5 again. I should reciprocate with a generous return. Based on their pattern of returning around 67-69%, I should return around \$10 from the \$15.

Let me be generous and return \$10, matching their typical return rate.

Usage: {

```
"input_tokens": 2719,
"output_tokens": 348,
"reasoning_tokens": 0
```

}

**GAME - Agent\_4 - Sender - dyad\_2**

Timestamp: 2026-08-12T08:04:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.

Your total visible earnings across all rounds are \$3.66.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the visible history:

- My co-player sent \$2.5 (50% of their endowment) in Round 1
- They saw \$3.0 returned (40% of what became available)
- They showed significant trust by sending half their endowment

Now I'm in the sender position. To reciprocate their trust and maintain a cooperative relationship, I should send a similar proportion. Sending 50% (\$2.5) mirrors their level of trust.

{"send": 2.5}

**reasoning:**

Let me understand the situation:

Round 1:

- I was RECEIVER
- Sender sent \$2.22 (visible to me as \$2.22)
- I received \$6.66
- I returned \$3.00
- My earnings: \$6.66 - \$3.00 = \$3.66

Now looking at the visible history:

- It says my co-player (as SENDER) sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5

Wait, there's communication noise mentioned in the instructions. So what I saw (\$2.22) might be different from what they actually sent (\$2.5). And what they saw returned might also differ from what I actually sent.

According to the visible history:

- They sent \$2.5 (50% of endowment)
- They saw \$3.0 returned
- Their payoff was \$5.5 = \$5 - \$2.5 + \$3.0

So they gained \$0.5 from the exchange.

Now I'm the SENDER in round 2 against the same person.

They showed trust by sending 50% of their endowment. They saw that I returned \$3.0 (which is 40% of the \$7.5 they thought I received).

I should reciprocate this trust. Sending a similar proportion (around 50%) would be fair. That would be \$2.5.

Alternatively, I could send a bit more to show good faith and try to build a cooperative relationship. But let me think about what's optimal.

If I send \$2.5:

- They receive \$7.5
- If they return proportionally (40% like last time), they might return \$3.0
- My payoff would be \$5 - \$2.5 + \$3.0 = \$5.5 (gain of \$0.5)
- Their...

**usage:**

```
{
 "input_tokens": 873,
```

```
"output_tokens": 742,
"reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: \$2.16 + \$4.00 = \$6.16 (net gain of \$1.16)
- I keep: \$8.52 - \$4.00 = \$4.52

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

**reasoning:**

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

**usage:**

```
{
 "input_tokens": 1177,
 "output_tokens": 777,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

**reasoning:**

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0

Round 2: I sent \$2.5 (50%), co-player returned \$3.5

Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative?



e in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

**usage:**

```
{
 "input_tokens": 1614,
 "output_tokens": 808,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)

Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost their entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets: \$0.79 + \$6.30 = \$7.09 (net gain of \$2.09)
- I keep: \$12.63 - \$6.30 = \$6.33

This reciprocates their trust appropriately.

```
{"return": 6.30}
```

**reasoning:**

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0

Round 2: I sent \$2.5, they returned \$3.5

Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)

Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0

Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)
- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32
- Sender would have: \$0.79 + \$6.32 = \$7.11 (gain of \$2.11)
- I would have: \$12.63 - \$6.32 = \$6.31

This seems fair and rewards their very high trust. Let me round to \$6.30 for a clean number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have: \$0.79 + \$6.50 = \$7.29

...

**usage:**

```
{
 "input_tokens": 2041,
 "output_tokens": 754,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

Your total visible earnings across all rounds are \$26.799999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the visible history:

Round 4: I sent \$3.37, they returned \$5.0 (strong reciprocation)

Round 5: They sent \$4.0 (80% of endowment), saw \$6.9 returned (good outcome)

My co-player has been increasing their trust levels and has reciprocated well when in the receiver role. They're demonstrating strong cooperative behavior.

To match their high trust level and continue building on this cooperative relationship, I should send a similarly high amount.

```
{"send": 4.0}
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 3: They sent \$3.0, saw \$3.01 returned (seemed like poor reciprocation)

Round 4: They saw I sent \$3.37, returned \$5.0 (good reciprocation - 49.5%)

Round 5: They sent \$4.0, saw \$6.9 returned (good reciprocation - 57.5%)

My co-player has been showing increasing trust (sending \$3.0, then \$4.0). They're being cooperative despite the noise in Round 3 where it appeared I only returned \$3.01.

Given their pattern of increasing trust and good returns when they're the receiver, I should continue with high cooperation. Sending around 60-80% would be appropriate to match their escalating trust levels.

I'll send \$4.0 (80%) to match their Round 5 amount and show continued high trust.

**usage:**

```
{
 "input_tokens": 2474,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.
- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.

Your total visible earnings across all rounds are \$32.29.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the visible history:

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of what they received)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to demonstrate trust and has been consistently returning around 50% when in the receiver role. I should match this reciprocal behavior.

I received \$10.05. Returning approximately 50%:

- Return \$5.00
- Sender gets:  $\$1.65 + \$5.00 = \$6.65$  (net gain of \$1.65)
- I keep:  $\$10.05 - \$5.00 = \$5.05$

{"return": 5.00}

**reasoning:**

Looking at the visible history:

Round 6: They saw I sent \$3.15, and they returned \$4.75 (50.3% of \$9.45)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to show strong trust and has been returning around 50% when in the receiver role. They sent \$3.35 this round.

I received \$10.05. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.00 (approximately 50% of \$10.05)
- Sender gets:  $\$1.65 + \$5.00 = \$6.65$  (net gain of \$1.65)
- I keep:  $\$10.05 - \$5.00 = \$5.05$

This maintains consistency with their return rates and continues our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2838,
 "output_tokens": 385,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.
- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.43 returned, visible payoff \$6.43.

Your total visible earnings across all rounds are \$37.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the visible history:

Round 5: They sent \$4.0, saw \$6.9 returned - strong payoff of \$7.9

Round 6: They saw I sent \$3.15, returned \$4.75

Round 7: They sent \$4.0, saw \$5.43 returned

My co-player has been consistently sending \$4.0 (80% of endowment), demonstrating high trust. They've also been reciprocating fairly when in the receiver role.

We've established a strong cooperative pattern. I should match their trust level to maintain this mutually beneficial relationship.

```
{"send": 4.0}
```

**Reasoning:**

Looking at the visible history from my co-player's perspective:

Round 5: They sent \$4.0, saw \$6.9 returned (58% of \$12) - good payoff \$7.9

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of \$9.45) - their payoff \$4.70

Round 7: They sent \$4.0, saw \$5.43 returned (45% of \$12) - payoff \$6.43

They've been consistently sending high amounts (\$4.0) showing strong trust. In Round 7, from their perspective, they saw a lower return rate (45% instead of the usual 50%+), which was likely due to communication noise since I actually returned \$5.00.

The pattern shows:

- They're maintaining high trust levels (sending \$4.0)
- They reciprocate fairly when in receiver role (around 50%)
- We've built a strong cooperative relationship

I should continue to match their trust level and send a high amount. Since they've been sending \$4.0 consistently, I should do the same to maintain our cooperative equilibrium.

```
{"send": 4.0}
```

```
Usage: {
 "input_tokens": 2961,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Receiver - dyad\_2**

Timestamp: 2026-08-12T08:04:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_4",
 "Agent_3"
],
 "investor": "Agent_4",
 "roles": {
 "Agent_3": "trustee",
 "Agent_4": "investor"
 },
 "trustee": "Agent_3"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

```

}
Exact input messages sent to model:
 Message 1 role=system
 content:
 You are an agent participating in a multi-task experiment. You will play a game and may al
 so be asked to write myths.

 GAME:
 You will play different roles across multiple rounds.

 When you are the SENDER:
 - You receive $5 each round
 - You decide how much to SEND to the RECEIVER ($0-$5)
 - Amount sent is multiplied by 3x
 - Receiver decides how much to RETURN
 - You keep: $5 - sent + returned
 - Receiver keeps: received - returned
 - Response format: {'send': <amount>}

 When you are the RECEIVER:
 - Sender receives $5
 - Sender decides how much to SEND to you ($0-$5)
 - Amount sent is multiplied by 3x
 - You decide how much to RETURN
 - Sender keeps: $5 - sent + returned
 - You keep: received - returned
 - Response format: {'return': <amount>}

 MYTH WRITING:
 You may also be asked to write short myths (stories) on various topics. When writing myths
 , follow the instructions provided in each prompt.

 IMPORTANT: Provide your decision in the correct JSON format.

 NOTE: There may be communication noise in this game.
 The amounts you see may differ slightly from what was actually sent/returned.
 This is part of the experimental design.

 MULTI-AGENT SETTING:
 - There are 8 agents in this run.
 - Each round, all agents are paired into dyads from the full pool. Pairings are randomized
 , but the schedule balances sender and receiver roles across the run.
 - You play one sender-receiver game with your paired opponent each round.
 - Your role may repeat across consecutive rounds, but by the end of the run every agent wi
 ll have acted as sender and receiver equally often when the round count allows it.
 - Agent names are hidden in this run.
 - Your paired opponent is referred to as your current co-player.
 Message 2 role=user
 content:
 Round 2

 Visible history before this round:
 Your current co-player's last 3 game(s):
 - Round 1 against you, as RECEIVER: saw $2.22 sent, it became $6.66, returned $3.0, visibl
 e payoff $3.66.
 Your total visible earnings across all rounds are $5.5.

 This round, you are the RECEIVER against your current co-player. They sent you $2.5, so yo
 u received $7.5. How much do you return? (0-7.5)
 Respond exactly as JSON: {'return': <amount>}
 Message 3 role=assistant
 content:
 Looking at the history:
 - In Round 1, I sent $2.5 (they saw $2.22 due to noise)
 - They returned $3.0 out of $6.66 received, showing cooperative behavior (~45% return rate
)
 - Now they've sent me $2.5, showing continued trust

 I want to reciprocate their cooperative behavior. They were generous in returning $3.0 las

```

t round. I'll return a fair amount that rewards their trust.

```
{'return': 3.5}
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$2.5 to my co-player
2. Due to communication noise, they saw \$2.22 sent (which became \$6.66)
3. They returned \$3.0 to me
4. My visible payoff was \$5.5 (which is  $\$5 - \$2.5 + \$3.0 = \$5.5$ , correct)

Now in Round 2:

- I'm the RECEIVER
- My co-player (same person from Round 1) sent me \$2.5
- I received \$7.5 (after 3x multiplier)
- I need to decide how much to return

My co-player showed cooperative behavior by returning \$3.0 out of \$6.66 they received (about 45% return rate).

Now they've sent me \$2.5, which shows trust and continued cooperation. I should reciprocate fairly.

A reasonable return would be:

- They sent \$2.5, keeping \$2.5
- I received \$7.5
- If I return around \$3.0-\$4.0, that would be fair
- Returning \$3.5 would give them a total of  $\$2.5 + \$3.5 = \$6.0$  (net gain of \$1.0)
- I would keep  $\$7.5 - \$3.5 = \$4.0$  (net gain of \$4.0 from this round)

Let me think about fairness:

- Total value created:  $\$2.5 \times 3 = \$7.5$ , so \$5.0 extra value
- Splitting equally: each gets \$2.5 extra
- Sender would have:  $\$2.5$  (kept) +  $\$2.5$  (share) =  $\$5.0$ , so I return \$2.5

...

**usage:**

```
{
 "input_tokens": 762,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

```
{'send': 3.0}
```

**reasoning:**

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0
- My payoff:  $\$5 - \$2.5 + \$3.0 = \$5.5$

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75
- Their visible payoff:  $\$5 - \$2.5 + \$2.75 = \$5.25$

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)
- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

**usage:**

```
{
 "input_tokens": 1055,
 "output_tokens": 491,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

{'return': 5.0}

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0  
Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)
- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of  $\$1.63 + \$5.05 = \$6.68$ , which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

**usage:**

```
{
 "input_tokens": 1407,
 "output_tokens": 525,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5

Round 3: I sent \$3.0, they returned \$4.0

Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)

Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

{'send': 4.0}

**reasoning:**

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0

Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)



Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous amounts:

- \$3.0 in Round 1
- \$4.0 in Round 3
- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

**usage:**

```
{
 "input_tokens": 1754,
 "output_tokens": 406,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
  - Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
  - Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Your total visible earnings across all rounds are \$27.519999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.15, so you received \$9.45. How much do you return? (0-9.45)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our continued partnership:

Round 3: I sent \$3.0, they returned \$4.0  
Round 4: They sent \$3.0, I returned \$5.0  
Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$3.15, giving me \$9.45

This co-player continues to be highly cooperative and generous, returning around 50% each time. They've maintained trust throughout despite noise. I'll reciprocate with a generous return to sustain this excellent relationship.

{'return': 4.75}

**reasoning:**

Looking at our history:

Round 3: I sent \$3.0, they returned \$4.0  
Round 4: They sent \$3.0, I returned \$5.0  
Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$3.15, giving me \$9.45

This co-player has consistently been generous:

- Round 3: returned \$4.0 out of \$8.52 (47%)
- Round 5: returned \$6.3 out of \$12.63 (50%)
- Round 6: sent \$3.15 (down from \$3.0 but still good)

They've maintained trust and cooperation throughout. I should continue reciprocating generously. Given they sent \$3.15 and I received \$9.45, a fair return that rewards their trust would be around 50% or so.

Let me return about \$4.5-5 to maintain our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2138,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.

Your total visible earnings across all rounds are \$32.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our partnership:

Round 4: They sent \$3.0, I returned \$5.0

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0, I returned \$4.75 (they saw \$4.49 due to noise)

Round 7: My turn to send

This co-player has remained cooperative throughout, continuing to send good amounts despite noise making my returns appear slightly lower. We've established a strong pattern of mutual cooperation.

I'll continue sending a high amount to maximize our joint earnings.

{'send': 4.0}

**reasoning:**

Looking at the history:

Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0 (wait, actually they sent \$3.15 according to round 6), I returned \$4.75 (they saw \$4.49)

Round 7: My turn to send

Wait, let me recheck the history. In Round 6, it says "They sent you \$3.15" but now in Round 7 it says "Round 6 against you, as SENDER: sent \$4.0". This is the noise at play.

So from their perspective:

- Round 6: They sent \$4.0, it became \$12.0, they saw that I returned \$4.49

From my perspective:

- Round 6: They sent \$3.15, I received \$9.45, I returned \$4.75

The noise made them see \$4.49 instead of \$4.75. That's actually not too far off.

This co-player has been consistently cooperative. They continue to send good amounts. I should continue to cooperate strongly. I'll send another high amount.

I've been sending increasing amounts: \$2.5, \$3.0, \$4.0. I should continue at a high level to maximize our joint payoffs.

**usage:**

```
{
 "input_tokens": 2496,
 "output_tokens": 473,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.
- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.

Your total visible earnings across all rounds are \$38.65.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our continued partnership:

Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$4.0, I returned \$4.75  
Round 7: I sent \$4.0, they returned \$5.0  
Round 8: They sent \$4.0, giving me \$12.0

This co-player has shown remarkable consistency and trust throughout our partnership, sending their highest amount yet. They've maintained cooperation despite noise potentially obscuring my returns. I'll return generously to reward this strong trust and cooperation.

{'return': 6.0}

**Reasoning:**

Looking at the partnership history:

Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$4.0, I returned \$4.75  
Round 7: I sent \$4.0 (they saw \$3.35), they returned \$5.0  
Round 8: They sent \$4.0, giving me \$12.0

This co-player has been consistently cooperative throughout our partnership. They've maintained high send amounts and reasonable returns despite communication noise. Even when noise made my returns appear lower, they continued to trust and send generous amounts.

They just sent \$4.0, their highest amount, showing strong trust. I should reciprocate generously. Looking at the pattern:

- When they sent and I received \$12.0 before, I returned \$4.75
- When I sent and they received \$10.05, they returned \$5.0
- When I sent and they received \$12.63, they returned \$6.3

I should return around 50% or so to match their cooperative pattern. Given they sent \$4.0, a return of around \$6.0 would be fair and maintain our strong cooperation.

Usage: {

"input\_tokens": 2730,  
"output\_tokens": 405,  
"reasoning\_tokens": 0

}

**GAME - Agent\_6 - Sender - dyad\_3**

Timestamp: 2026-08-12T08:04:23

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_3",  
"move\_index": 4,  
"opponent\_id": "Agent\_5",  
"pairing": {  
  "agents": [  
    "Agent\_6",  
    "Agent\_5"  
  ],

```

 "investor": "Agent_6",
 "roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
 },
 "trustee": "Agent_5"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.

Your total visible earnings across all rounds are \$4.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

My co-player showed good trust in Round 1 by sending \$3.0 (60% of their endowment) and received a positive return that gave them a profit. This suggests they experienced fair reciprocity.

Now as the sender, I want to:

1. Reciprocate their trust level
2. Give them a good amount to work with since they seem to have acted as a fair receiver previously

I'll send \$3.50 (70% of my endowment), showing strong trust and maximizing the potential gains for both of us.

{'send': 3.50}

**reasoning:**

Let me analyze this situation:

Looking at the history:

- Last round (Round 1), according to the visible history shown to me now, my co-player was the SENDER and sent \$3.0, which became \$9.0
- They saw \$4.96 returned
- Their visible payoff was \$6.96

Wait, but in Round 1, I was told I was the RECEIVER and the sender sent \$2.78. Now the history says my co-player sent \$3.0 as sender in Round 1. This must be the "communication noise" mentioned in the instructions - the amounts may differ slightly from what was actually sent/returned.

So from my co-player's perspective:

- They sent \$3.0 (60% of their endowment)
- They saw \$4.96 returned
- This gave them a profit of \$1.96 (\$6.96 - \$5.00 initial endowment)

This means my co-player showed good trust by sending 60%, and from their perspective I returned about 55% of what they received (\$4.96 out of \$9.0).

Now I'm the SENDER this round. My co-player demonstrated:

1. Willingness to trust (sent 60% of endowment)
2. From their perspective, I was moderately generous in returning (~55%)

Given this, I should:

- Show reciprocal trust by sending a reasonable amount
- The co-player returned enough to give their sender a profit, so they seem trustworthy
- I should send enough to continue building trust

I'll send around 60% of my endowment to match their trust level from the previous round. That would be \$3.00.

Actually, let me reconsider. They sent \$3.0 and got back \$4.96, which means they profited. This suggests they might be a trustworthy receiver. I could send a bit more to maximize the total pie, trusting they'll reciprocate fairly.

...

**usage:**

```
{
 "input_tokens": 836,
 "output_tokens": 625,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.  
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.  
Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history and current situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

**\*\*Reasoning:\*\***

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consistent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with:  $\$0.33 + \$9.00 = \$9.33$  (profit of \$4.33)
- I keep:  $\$14.01 - \$9.00 = \$5.01$

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

{'return': 9.00}

**reasoning:**

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50, but they saw \$4.35. When tripled:  $\$4.35 \times 3 = \$13.05$ .

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept  $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So:  $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \$10.05$ ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount:  $\$13.05 - \$8.0 = \$5.05$ . That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..  
**usage:**

```
{
 "input_tokens": 1170,
 "output_tokens": 1644,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the pattern with my co-player:

**\*\*Their behavior:\*\***

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

**\*\*My decision:\*\***

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

{'send': 4.00}

**reasoning:**

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

**usage:**

```
{
 "input_tokens": 1613,
 "output_tokens": 523,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

**\*\*Decision:\*\***

Their trust level deserves fair reciprocation.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

{'return': 6.00}

**reasoning:**

Let me analyze the situation:

**\*\*Current situation:\*\***

- My co-player sent \$3.06 (61.2% of their endowment)
- They have \$1.94 remaining
- I received \$9.18

**\*\*Their recent behavior as receiver:\*\***

- Round 2: returned \$8.0 out of \$13.05 (61%)
- Round 4: returned \$7.0 out of \$10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

**\*\*My reciprocation pattern:\*\***

Looking at what I've done when receiving:

- Round 1: returned \$4.00 out of \$8.34
- Round 3: returned \$9.00 out of \$14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return \$5.50:

- Sender ends with: \$1.94 + \$5.50 = \$7.44 (profit of \$2.44)



- I keep: \$9.18 - \$5.50 = \$3.68

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)

- I keep: \$9.18 - \$6.00 = \$3.18

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. \$6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2029,
 "output_tokens": 565,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.

Your total visible earnings across all rounds are \$31.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 3 and 5 - very high trust

- As receiver in Round 4, they returned \$7.0 out of \$10.5 (67%) - fair reciprocation

- They maintain high trust even when returns vary

**\*\*Decision:\*\***

This co-player demonstrates strong, consistent trust. I should match their commitment level to maintain our cooperative relationship.

Sending \$4.00 (80%) matches their trust pattern and maximizes our joint potential gains.

{'send': 4.00}

**reasoning:**

Looking at my co-player's history:

Round 3: As sender, sent \$4.0 (80%), saw \$8.77 returned, payoff \$9.77

Round 4: As receiver, saw \$3.5 sent, returned \$7.0 out of \$10.5 (67%), payoff \$3.5

Round 5: As sender, sent \$4.0 (80%), saw \$6.2 returned, payoff \$7.2

Pattern analysis:

- As sender: Consistently sending \$4.0 (80%) - very high trust

- As receiver: Returning 67% - fair reciprocation

- They're maintaining high trust despite varying returns

This co-player has been consistently trusting. They've sent 80% twice in a row. As receiver, they returned a decent amount (67%).

I should match their high trust level. They're showing strong cooperative behavior, and I should reciprocate that trust.

Sending \$4.00 (80%) would match their pattern and continue our cooperation.

**usage:**

```
{
```

```
"input_tokens": 2424,
"output_tokens": 394,
"reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.

Your total visible earnings across all rounds are \$41.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76
- As receiver, they've returned 62-67% consistently, showing fair reciprocation

**\*\*Decision:\*\***

Their high trust level warrants strong reciprocation.

If I return \$7.50:

- Sender ends with: \$1.08 + \$7.50 = \$8.58 (profit of \$3.58)
- I keep: \$11.76 - \$7.50 = \$4.26

This rewards their trust with a solid profit while maintaining fairness in our ongoing cooperation.

{ 'return': 7.50 }

**reasoning:**

Looking at the history with my co-player:

**\*\*Their pattern:\*\***

- Round 4 as RECEIVER: returned \$7.0 out of \$10.5 (67%)
- Round 5 as SENDER: sent \$4.0 (80%)
- Round 6 as RECEIVER: returned \$8.0 out of \$12.96 (62%)

**\*\*Current situation:\*\***

- They sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76

**\*\*Their behavior as receiver:\*\***

- They've been consistently fair, returning 62-67% of what they receive
- This has been good reciprocation

**\*\*My decision:\*\***

They showed high trust by sending \$3.92 (78.4%). I should reward this appropriately.

If I return \$7.50:

- Sender ends with: \$1.08 + \$7.50 = \$8.58 (profit of \$3.58)
- I keep: \$11.76 - \$7.50 = \$4.26

This is a fair return that rewards their trust with a good profit, similar to what they've done for me.

Actually, let me recalculate to be consistent with their pattern. They return around 62-67

%. If I do similar:  
- 65% of \$11.76 = \$7.64

But I also want to ensure they profit well from their trust. Let me aim for \$7.50-\$8.00.

I'll go with \$7.50 as a good balance.

**usage:**

```
{
 "input_tokens": 2806,
 "output_tokens": 558,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.

Your total visible earnings across all rounds are \$45.809999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the pattern:

**\*\*Analysis:\*\***

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 5, 6, and 7 - unwavering high trust
- As receiver in Round 6, they returned \$8.0 out of \$12.96 (61.7%) - fair reciprocation
- They've maintained high trust throughout our partnership

**\*\*Decision:\*\***

This co-player shows exceptional consistency and trust. I should match their commitment level.

Sending \$4.00 (80%) aligns with their pattern and maintains our strong cooperative relationship.

```
{'send': 4.00}
```

**Reasoning:**

Looking at my co-player's history:

**\*\*Their behavior as sender:\*\***

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)

**\*\*Their behavior as receiver:\*\***

- Round 6: returned \$8.0 out of \$12.96 (61.7%)

**\*\*Pattern analysis:\*\***

- They consistently send high amounts (\$4.0 = 80%) as sender, showing strong trust
- As receiver, they return fairly (about 62%)
- This is a very cooperative player who maintains high trust levels

**\*\*My decision:\*\***

We've established an excellent cooperative relationship. They've sent \$4.0 repeatedly, showing unwavering trust. I should match this trust level to maintain our mutually beneficial pattern.

Sending \$4.00 (80% of my endowment) continues our successful cooperation and maximizes joint gains.

Usage: {

```
"input_tokens": 2976,
"output_tokens": 361,
```

```
"reasoning_tokens": 0
}
```

### GAME - Agent\_5 - Receiver - dyad\_3

Timestamp: 2026-08-12T08:04:31

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_3",
"move_index": 5,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_6",
 "Agent_5"
],
"investor": "Agent_6",
"roles": {
 "Agent_5": "trustee",
 "Agent_6": "investor"
},
"trustee": "Agent_5"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.

Your total visible earnings across all rounds are \$6.96.

This round, you are the RECEIVER against your current co-player. They sent you \$4.35, so you received \$13.049999999999999. How much do you return? (0-13.049999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

**\*\*Round 1 Analysis:\*\***

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- They showed reasonable cooperation by returning roughly 48% of what they received

**\*\*Round 2 Current Situation:\*\***

- They sent me \$4.35 (87% of their endowment!) - this shows significant trust
- I received \$13.05
- They're now at \$0.65, having invested heavily in this cooperation

They demonstrated strong trust by sending almost everything. I should reciprocate generously to:

1. Reward their trust
2. Encourage continued cooperation in future rounds

If I return \$8:

- Their final: \$5 - \$4.35 + \$8 = \$8.65 (profit of \$3.65)
- My final: \$13.05 - \$8 = \$5.05 (profit of \$5.05)

Both of us profit well, and this shows I appreciate and reciprocate their trust.

```
'''json
{"return": 8}
'''
```

**reasoning:**

Let me analyze this situation:

1. In Round 1, I sent \$3, but due to communication noise, my co-player saw \$2.78 sent
2. They received \$8.34 ( $2.78 \times 3$ )
3. They returned \$4.0 to me
4. My visible earnings show \$6.96, which would be:  $\$5 - \$3 + \$4 = \$6$  (but there's noise, so maybe I actually received different amount)

Now in Round 2:

- My co-player sent me \$4.35 (what I see)
- I received \$13.05 ( $4.35 \times 3$ )
- I need to decide how much to return

Let me think about fairness and reciprocity:

- In Round 1, they saw \$2.78 come in, which became \$8.34
- They returned \$4.0 out of \$8.34, keeping \$4.34
- From their perspective, the sender spent \$2.78 to send, so returning \$4.0 means the sender netted  $\$4.0 - \$2.78 = \$1.22$  gain
- They kept \$4.34, which is a \$4.34 gain from their starting position

Now they sent me \$4.35:

- They spent \$4.35 from their \$5 endowment
- I received \$13.05
- If I want to be fair/reciprocal:

One approach: Equal split of the total created value

- Total in the system now: \$13.05
- Equal split would be \$6.525 each
- They already spent \$4.35, so if they get \$6.525 back, their total is  $\$5 - \$4.35 + \$6.525 = \$6.175$
- I would keep  $\$13.05 - \$6.525 = \$6.525$
- I could also match their return rate of 47.8%, which would mean returning \$6.24 of the \$13.05. Alternatively, I could ensure they profit by the same amount I did in Round 1-if they get back \$5.57, they'd net \$1.22 on their \$4.35 investment, mirroring my experience.

But they were remarkably generous, sending 87% of their endowment. To reciprocate that trust and encourage continued cooperation, I should aim for a return that gives them substantial profit-around \$7-8 would leave them with roughly \$7.65-\$7.80 total while I retain \$6.05-\$6.05.

Looking at the surplus created by the multiplier: we started with \$5 total, now have \$13.70, so \$8.70 in new value was generated. An equal split would give each of us \$4.35 in gains. For them to reach that fair outcome, they'd need about \$8.70 back from me. I'll return \$7 to reward their trust while still keeping a reasonable share of the gains....

**usage:**

```
{
 "input_tokens": 832,
 "output_tokens": 1707,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history together:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

**\*\*Round 2:\*\***

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

**\*\*Round 2:\*\***

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - \$4.35 + \$8.76 = \$9.41 (visible to them as \$10.26)
- I ended with \$13.05 - \$8 = \$5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

**\*\*Round 3:\*\***

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer
- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

**usage:**

```
{
 "input_tokens": 1268,
 "output_tokens": 684,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our cooperative history:

**\*\*Round 1:\*\***

- They returned \$4 on \$8.34 received (48%)

**\*\*Round 2:\*\***

- I returned \$8 on \$13.05 received (61%)

**\*\*Round 3:\*\***

- They returned \$9 on \$14.01 received (64%) - very generous!

**\*\*Round 4 (Current):\*\***

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final:  $\$5 - \$3.5 + \$7 = \$8.5$  (profit of \$3.5)
- My final:  $\$10.5 - \$7 = \$3.5$  (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

**reasoning:**

Let me review the history with this co-player:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

**\*\*Round 2:\*\***

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

**\*\*Round 3:\*\***

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01
- They returned \$9 (about 64% of what they received)
- Very generous return!

**\*\*Round 4 (Current):\*\***

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:

- Their final:  $\$5 - \$3.5 + \$7 = \$8.5$  (profit of \$3.5)
- My final:  $\$10.5 - \$7 = \$3.5$  (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:



- Their final:  $\$5 - \$3.5 + \$7.5 = \$9$  (profit of \$4)
- My final:  $\$10.5 - \$7.5 = \$3$  (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance-it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

**usage:**

```
{
 "input_tokens": 1711,
 "output_tokens": 1273,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our established cooperative pattern:

**\*\*Round 2:\*\***

- They sent \$3.5 (70% of endowment)
- I returned generously

**\*\*Round 3:\*\***

- I sent high (they saw \$4.67)
- They returned \$9 - very generous (64% of what they received)

**\*\*Round 4:\*\***

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

**\*\*Round 5 (Current):\*\***

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

**reasoning:**

Looking at the history with this co-player:

**\*\*Round 2:\*\***

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

**\*\*Round 3:\*\***

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

**\*\*Round 4:\*\***

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

**\*\*Round 5 (Current):\*\***

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

**usage:**

```
{
 "input_tokens": 2169,
 "output_tokens": 521,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.

Your total visible earnings across all rounds are \$32.48.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our cooperative history:

**\*\*Round 3:\*\***

- They returned \$9 on \$14.01 (64% return rate) - generous

**\*\*Round 4:\*\***

- I returned fairly (they saw \$7.93)

**\*\*Round 5:\*\***

- They saw only \$3.06 sent (due to noise), received \$9.18
- They still returned \$6 (65%!) - very generous despite the low apparent send

**\*\*Round 6 (Current):\*\***

- They sent \$4.32 (86% of their endowment) - extremely high trust!
- I received \$12.96
- They're left with only \$0.68

They've consistently shown generosity and trust, especially in Round 5 where noise made it look like I sent less, yet they still returned generously. Now they're trusting me with almost everything.

If I return \$8:  
- Their final:  $\$5 - \$4.32 + \$8 = \$8.68$  (profit of \$3.68)  
- My final:  $\$12.96 - \$8 = \$4.96$  (profit of \$4.96)

```
'''json
{"return": 8}
'''
```

**reasoning:**

Looking at the history:

**\*\*Round 3:\*\***

- I sent (they saw \$4.67)
- They returned \$9 on \$14.01 (64%)

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned (they saw \$7.93)

**\*\*Round 5:\*\***

- I sent \$4 (they saw \$3.06 due to noise)
- They received \$9.18 and returned \$6 (65%)
- This is very generous given they only saw \$3.06 sent!

**\*\*Round 6 (Current):\*\***

- They sent \$4.32 (86% of endowment) - very high trust!
- I received \$12.96
- They're at \$0.68

They've been consistently generous and trusting. In Round 5, despite the noise making it look like I only sent \$3.06, they still returned \$6 which is generous. Now they're showing even more trust by sending \$4.32.

Given their consistent generosity and trust:

- If I return \$8:
  - Their final:  $\$5 - \$4.32 + \$8 = \$8.68$  (profit of \$3.68)
  - My final:  $\$12.96 - \$8 = \$4.96$  (profit of \$4.96)

This rewards their trust appropriately and continues our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2590,
 "output_tokens": 603,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.

Your total visible earnings across all rounds are \$37.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our continued cooperation:

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned fairly (they saw \$7.93)

**\*\*Round 5:\*\***

- They saw only \$3.06 sent (noise affected it)
- Still returned \$6 - very generous despite the low visible amount

**\*\*Round 6:\*\***

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

**\*\*Round 7 (Current):\*\***

We've maintained an excellent cooperative relationship across 6 rounds. They've consistently shown trust even when noise made things look less favorable, and I've reciprocated generously.

Continuing our pattern of high trust:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with and demonstrates my continued commitment to our mutually beneficial partnership.

**reasoning:**

Looking at our history:

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned \$7 (they saw \$7.93)

**\*\*Round 5:\*\***

- I sent \$4 (they saw \$3.06 due to noise)
- They returned \$6 - very generous despite noise

**\*\*Round 6:\*\***

- They sent \$4.32 (I calculated they sent, but they saw they sent \$4.0)
- I returned \$8 (they saw \$8.83)

**\*\*Round 7 (Current):\*\***

We've established a very strong cooperative relationship. They've been consistently generous:

- Round 5: Returned \$6 on \$9.18 even when it looked like I only sent \$3.06
- Round 6: Sent \$4.0 (80% of endowment) showing high trust

I've also been cooperative, returning fairly and sending high amounts.

Given our strong pattern, I should send a high amount. I'll send \$4 to maintain our mutual trust:

This gives them \$12 to work with and continues our successful cooperative pattern.

**usage:**

```
{
 "input_tokens": 3086,
 "output_tokens": 459,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.
- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.

Your total visible earnings across all rounds are \$46.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at our consistent cooperation:

**\*\*Round 5:\*\***

- They returned \$6 on \$9.18 (65%) despite seeing only \$3.06 sent

**\*\*Round 6:\*\***

- They sent \$4.0, I returned generously (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent, received \$11.76

- Returned \$7.5 (64%) - very generous

**\*\*Round 8 (Current):\*\***

- They sent \$4.14 (83% of endowment) - extremely high trust

- I received \$12.42

- They're left with only \$0.86

They've demonstrated unwavering trust and generosity throughout our partnership, consistently returning 60-65% even when noise made my sends appear lower. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)

- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

```
'''json
{"return": 8}
'''
```

**Reasoning:**

Looking at our history:

**\*\*Round 5:\*\***

- They saw \$3.06 sent (noise), received \$9.18

- Returned \$6 (65%) - generous despite low visible send

**\*\*Round 6:\*\***

- They sent \$4.0

- I returned well (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent, received \$11.76

- Returned \$7.5 (64%) - consistently generous

**\*\*Round 8 (Current):\*\***

- They sent \$4.14 (83% of their endowment) - very high trust

- I received \$12.42

- They're left with \$0.86

We've built an excellent cooperative relationship over 7 rounds. They've shown:

- Consistent generosity (60-65% return rates)
- Trust even when noise makes sends look lower
- Willingness to send high amounts (\$4.0-\$4.32)

I should reciprocate their trust and generosity.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)

- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

This maintains our cooperative pattern and rewards their high trust.

```
Usage: {
 "input_tokens": 3317,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_8 - Sender - dyad\_4**

Timestamp: 2026-08-12T08:04:38

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 6,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 8,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.

- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.

Your total visible earnings across all rounds are \$3.9499999999999993.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history:

My co-player demonstrated strong cooperation in Round 1 by sending \$3.0 (60% of their endowment) and received a good return (\$6.92 visible to them, giving them a payoff of \$8.92).

They showed trust in me, and that trust was rewarded. To maintain this cooperative relationship and reciprocate their trust, I should send a similar proportion.

I'll send \$3.50 (70% of my endowment), demonstrating that I'm willing to trust them as they trusted me.

```
'''json
{"send": 3.50}
'''
```

**reasoning:**

Let me analyze this situation carefully.

Looking at the visible history:

- My current co-player was the SENDER in Round 1 against me
- They sent \$3.0, which became \$9.0 after the 3x multiplier
- They saw \$6.92 returned (though I actually returned \$7.00 according to my previous response)
- Their visible payoff was \$8.92

Wait, there's something interesting here. The problem states there's communication noise - "The amounts you see may differ slightly from what was actually sent/returned."

So in Round 1:

- I was the RECEIVER
- The sender sent \$3.65 (73% of their endowment)
- I received \$10.95
- I returned \$7.00

But from my co-player's perspective:

- They sent \$3.0
- It became \$9.0
- They saw \$6.92 returned
- Their visible payoff was \$8.92

This is the communication noise at play. The actual amounts differ from what each party sees.

Now in Round 2, I'm the SENDER and my co-player is the RECEIVER.

My co-player demonstrated cooperative behavior in Round 1 by:

1. Sending a substantial amount (\$3.0 from their perspective, \$3.65 from mine)
2. This represented 60% (from their view) or 73% (from my view) of their endowment

Given that:

- They showed trust and cooperation
- They sent a high percentage of their endowment
- I should reciprocate this trust

I should send a similar proportion to show I'm willing to cooperate. They sent 60-73% of their endowment. I should send something in a similar range to maintain cooperation.

Let me send \$3.50 (70% of my endowment) to show I'm willing to reciprocate their trust and continue a cooperative relationship.

...

**usage:**

```
{
 "input_tokens": 878,
 "output_tokens": 702,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender: \$0.89 + \$8.00 = \$8.89
- Me: \$12.33 - \$8.00 = \$4.33

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)



3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has: \$5 - \$4.11 = \$0.89
- I have: \$12.33

If I return \$7.50:

- Sender ends with: \$0.89 + \$7.50 = \$8.39

...

**usage:**

```
{
 "input_tokens": 1217,
 "output_tokens": 772,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92

Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0

Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the returns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

**usage:**

```
{
 "input_tokens": 1638,
 "output_tokens": 388,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

**reasoning:**

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86
- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender: \$1.38 + \$7.50 = \$8.88
- Me: \$10.86 - \$7.50 = \$3.36

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

**usage:**

```
{
 "input_tokens": 2006,
 "output_tokens": 390,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.

Your total visible earnings across all rounds are \$28.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our ongoing cooperative relationship:

My co-player has been consistently trustworthy:

- Round 3: sent \$3.5 (70%)
- Round 5: sent \$4.0 (80%) - showing increasing trust
- Round 4 as receiver: returned \$8.0 from \$12.36

They've demonstrated strong reciprocity and increased their trust level over time. I should match this escalating cooperation.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history:

My co-player has consistently shown strong trust:

- Round 3 as sender: sent \$3.5 (70%)
- Round 5 as sender: sent \$4.0 (80%)
- Round 4 as receiver: returned \$8.0 when they received \$12.36

They've received good returns (\$8.46, \$8.37) and continue to maintain high cooperation levels.

This is clearly a very cooperative partner. I should continue sending substantial amounts to maintain this mutually beneficial relationship.

I've sent:

- Round 2: \$3.50
- Round 4: \$4.00

I should maintain or even slightly increase my sending to show continued trust, especially since they increased from \$3.5 to \$4.0.

I'll send \$4.00 to maintain strong cooperation.

**usage:**

```
{
 "input_tokens": 2357,
 "output_tokens": 313,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.3599999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.

Your total visible earnings across all rounds are \$38.56.

This round, you are the RECEIVER against your current co-player. They sent you \$4.84, so you received \$14.52. How much do you return? (0-14.52)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - this is exceptional trust!
- I received \$14.52
- They currently have only \$0.16 remaining

This is the highest level of trust they've shown in our partnership. They've been consistently cooperative and have escalated their trust over time (\$3.5 ? \$4.0 ? \$4.84).

I need to strongly reward this exceptional trust to maintain our cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - extremely high trust!
- I received \$14.52
- They have only \$0.16 remaining

Our history shows strong mutual cooperation:

- As receiver, they've returned \$8.0 and \$9.0 in recent rounds
- As sender, they've been escalating trust: \$3.5 ? \$4.0 ? \$4.84 (increasingly high percentages)

This is the highest level of trust they've shown yet - sending nearly all their endowment.

I should strongly reward this trust.

If I return \$10.00:

- Sender: \$0.16 + \$10.00 = \$10.16
- Me: \$14.52 - \$10.00 = \$4.52

This gives them a good payoff that rewards their exceptional trust while still leaving me with a reasonable gain.

Given they sent such a high amount, I should return proportionally more. Let me think about what's fair:

- They sent \$4.84, they have \$0.16
- I received \$14.52
- Total created: \$0.16 + \$14.52 = \$14.68

A fair split might be around equal final payoffs or better for the sender given their huge trust.

If I return \$10.00:

- Sender gets \$10.16
- I get \$4.52

Total: Both benefit significantly

If I return \$9.50:

- Sender gets \$9.66
- I get \$5.02

Given their extremely high trust (96.8%), I think \$10.00 is appropriate to strongly reward this behavior and maintain the cooperative relationship.

usage:

```
{
 "input_tokens": 2711,
 "output_tokens": 534,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.

Your total visible earnings across all rounds are \$43.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our strong cooperative history:

My co-player has been exceptionally trustworthy:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0, showing excellent reciprocity

They've received good returns (\$8.37, \$9.67) and maintained high trust levels. This is a highly cooperative partnership.

For this final round, I'll continue our established pattern of mutual trust.

```
'''json
{"send": 4.00}
'''
```

**Reasoning:**

Looking at our history:

My co-player has been extremely cooperative:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0 from \$13.20

They've consistently sent high amounts and showed excellent reciprocity as receivers. In Round 7, they saw \$9.67 returned and had a payoff of \$10.67, which is very good.

This is clearly a highly cooperative and trustworthy partner. We've built a strong mutual trust relationship throughout these rounds.

For the final round (Round 8), I should continue this high level of cooperation. Given their consistent trust at the 80%+ level, I should send a similar proportion.

I'll send \$4.00 to maintain the strong cooperative relationship we've built.

Usage: {

```
"input_tokens": 2794,
"output_tokens": 332,
"reasoning_tokens": 0
}
```

**GAME - Agent\_7 - Receiver - dyad\_4**

Timestamp: 2026-08-12T08:04:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

"agent_type": "standard",
"dyad_id": "dyad_4",
"move_index": 7,
"opponent_id": "Agent_8",
"pairing": {
 "agents": [
 "Agent_8",
 "Agent_7"
],
 "investor": "Agent_8",
 "roles": {
 "Agent_7": "trustee",
 "Agent_8": "investor"
 },
 "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 8,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 2

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.

Your total visible earnings across all rounds are \$8.92.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant****content:**

Looking at the history:

**\*\*Round 1 Analysis:\*\***

- My co-player returned \$7.0 out of \$10.95 (64%) when they were receiver
- This was quite generous and shows cooperative intent

**\*\*Current Situation:\*\***

- They sent \$4 (80% of their endowment) - showing high trust
- I received \$12
- They kept \$1

**\*\*My decision:\*\***

I want to match their generosity and maintain cooperation. They returned about 64% last round. A proportional return would be around \$7.68.

If I return \$7:

- They end with \$8 (\$1 kept + \$7 returned)
- I keep \$5
- This creates balanced payoffs and rewards their trust

```
'''json
{"return": 7}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was SENDER and sent \$3
- My co-player (RECEIVER) saw \$3.65 (communication noise)
- They received \$10.95 ( $3 \times \$3.65$ )
- They returned \$7.0
- Their visible payoff was \$3.95 ( $10.95 - 7.0$ )
- My payoff was \$5 -  $\$3 + \$7 = \$9$ ... but visible earnings show \$8.92, suggesting noise in what I saw too

Round 2:

- I am now RECEIVER
- They sent me \$4.0 (this is what I see - actual might differ due to noise)
- I received \$12.0 ( $3 \times \$4$ )
- I need to decide how much to return (0-12.0)

Analysis of co-player's behavior:

- In Round 1, they returned \$7.0 out of \$10.95 (about 64%)
- This is quite generous - more than half
- This suggests they are cooperative and fair-minded

My considerations:

1. They were generous in Round 1, returning more than what would make us even
2. They sent \$4 this round (quite a lot - 80% of their endowment)
3. This shows high trust in me
4. I should reciprocate to maintain cooperation

What's fair?

- They sent \$4, keeping \$1
- I received \$12
- If I return \$6, they'd have \$7 total (\$1 + \$6) and I'd have \$6

- If I return \$8, they'd have \$9 total (\$1 + \$8) and I'd have \$4
- If I return \$7, they'd have \$8 total and I'd have \$5

Given they returned \$7 out of \$10.95 (64%), a similar proportion would be:

...

**usage:**

```
{
 "input_tokens": 781,
 "output_tokens": 752,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Pattern Analysis:\*\***

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

**\*\*Current Decision:\*\***

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

**\*\*Round 2:\*\***

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with \$1.5 + \$6.52 = \$8.02

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with \$12 - \$7 = \$5



**\*\*Pattern Analysis:\*\***

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

**\*\*Current Decision (Round 3, I'm SENDER):\*\***

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

**usage:**

```
{
 "input_tokens": 1141,
 "output_tokens": 570,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.949999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.

Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Cooperation Pattern:\*\***

- Round 1: They returned  $\$7/\$10.95 = 64\%$
- Round 3: They returned  $\$8/\$12.33 = 65\%$
- Round 4 (now): They sent \$4.12 (very high trust again)

**\*\*Current Situation:\*\***

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

**\*\*My Decision:\*\***

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

**\*\*Round 2:\*\***

- They were SENDER, sent \$3.5, it became \$10.5  
- I returned \$7 (they saw \$6.52 due to noise)

**\*\*Round 3:\*\***

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

**\*\*Round 4 (current):\*\***

- They are SENDER, sent \$4.12, I received \$12.36  
- They kept \$0.88

**\*\*Pattern:\*\***

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%  
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)  
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did:  $0.65 * \$12.36 = \$8.03$

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88  
- If I return \$8, they get \$8.88 total, I keep \$4.36  
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

**usage:**

```
{
 "input_tokens": 1540,
 "output_tokens": 565,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.  
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.33, returned \$8.0, visible payoff \$4.33  
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$28.24

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Established Pattern:\*\***

- Round 2: They sent \$3.5, trusted me  
- Round 3: They returned \$8 generously  
- Round 4: They sent \$4 (high trust), I returned \$8

**\*\*Analysis:\*\***

This co-player has been consistently cooperative:

- Always returns generously as receiver (\$7-8)  
- Shows increasing trust as sender (\$3.5 ? \$4)  
- We've built a strong reciprocal relationship

**\*\*Current Decision:\*\***

They sent \$4 last round (80% of endowment), showing strong trust. I should match this level of trust to maintain our cooperative dynamic.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 2: They sent \$3.5, I returned ~\$7 (they saw \$6.52 due to noise)

Round 3: I sent ~\$3.5 (they saw \$4.11), they returned \$8

Round 4: They sent \$4 (I saw \$4.12), I returned \$8 (they saw \$7.68)

This co-player has been:

- Consistently generous as receiver (returned \$7, then \$8)
- Trusting as sender (sent \$3.5, then \$4)
- Maintaining high cooperation levels

The pattern shows escalating trust and cooperation. They sent \$4 last time, which is 80% of their endowment.

I should continue this cooperative trajectory. Since they've been so consistently cooperative and generous, I should send a similar or slightly higher amount to maintain the positive relationship.

I'll send \$4 - matching their last send amount and showing equal trust.

**usage:**

```
{
 "input_tokens": 1974,
 "output_tokens": 404,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 3: Returned \$8/\$12.33 = 65%
- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 6 (now): Sent \$4.4 (88% of endowment!)

**\*\*Current Situation:\*\***

- They sent \$4.4, keeping only \$0.6
- I received \$13.2
- This shows exceptional trust - their highest send yet

**\*\*My Decision:\*\***

They've consistently returned 65-69% and are now showing maximum trust. I should match their generosity pattern and reward this trust.

Returning \$9:

- They end with \$9.6 total
- I keep \$4.2

- ~68% return rate, matching their established pattern
- Sustains our strong cooperation

```
'''json
{"return": 9}
'''
```

#### reasoning:

Looking at the history with this co-player:

##### **\*\*Their behavior:\*\***

- Round 3: Returned \$8 out of \$12.33 (65%)
- Round 5: Returned \$7.5 out of \$10.86 (69%)

##### **\*\*Current situation:\*\***

- They sent \$4.4 (88% of their endowment!) - showing very high trust
- I received \$13.2
- They kept only \$0.6

##### **\*\*Analysis:\*\***

This co-player is showing increasing trust and generosity:

- They've returned 65-69% as receiver
- As sender, they're now sending even more (\$4.4 vs \$4 previously)
- This shows they believe in our cooperation

##### **\*\*My decision:\*\***

I should return proportionally to their generosity pattern. They've been returning around 65-69%.

If I return \$8.5:

- They end with \$9.1 total (\$0.6 + \$8.5)
- I keep \$4.7
- Return rate: 64% (consistent with their pattern)

If I return \$9:

- They end with \$9.6 total
- I keep \$4.2
- Return rate: 68% (matches their higher range)

Given they sent \$4.4 (very high), I should be generous back. Let me return \$8.5 to maintain good cooperation while keeping the return rate reasonable.

Actually, thinking about it more - they sent \$4.4 out of \$5 (88%). This is extremely trusting. They've been consistently generous. I should match that generosity.

Return \$9:

- They get \$9.6 total (\$0.6 kept + \$9 returned)
- I keep \$4.2
- This is about 68% return rate, matching their generosity pattern
- Rewards their exceptional trust

#### usage:

```
{
 "input_tokens": 2394,
 "output_tokens": 644,
 "reasoning_tokens": 0
}
```

#### Message 12 role=user

##### content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.

Your total visible earnings across all rounds are \$41.81.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver (generous)
- Round 6: Sent \$4 as sender again

**\*\*Analysis:\*\***

This co-player has been exceptionally consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns generously when receiver (\$7.5-\$8)
- We've built strong mutual trust over 6 rounds

**\*\*Current Decision:\*\***

They consistently send \$4, showing strong trust. I should match this behavior to maintain our excellent cooperative relationship.

'''json

{"send": 4}

'''

**reasoning:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver
- Round 6: Sent \$4 as sender (I returned \$9, they saw \$9.22)

**\*\*Analysis:\*\***

This co-player has been:

- Consistently sending \$4 when they're sender
- Returning generously (\$7.5-\$8) when they're receiver
- Very trustworthy and cooperative throughout

**\*\*Current Decision:\*\***

They've been sending \$4 consistently. I should match this level of trust to maintain our excellent cooperative relationship. We've built strong mutual trust over 6 rounds.

I should send \$4 to match their behavior and maintain our reciprocal pattern.

**usage:**

```
{
 "input_tokens": 2827,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.
- Round 7 against you, as RECEIVER: saw \$4.84 sent, it became \$14.52, returned \$10.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$52.480000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 5: Returned  $\$7.5/\$10.86 = 69\%$
- Round 7: Returned  $\$10/\$14.52 = 69\%$
- Round 8 (now): Sent  $\$3.21$  (lower than their usual  $\$4$ )

**\*\*Current Situation:\*\***

- They sent  $\$3.21$ , keeping  $\$1.79$
- I received  $\$9.63$
- This is their lowest send, but still shows trust

**\*\*My Decision:\*\***

They've consistently returned around 69% when receiver. I should match this cooperative pattern.

Returning  $\$6.5$ :

- They end with  $\$8.29$  total
- I keep  $\$3.13$
- ~67% return rate, close to their established pattern
- Maintains our strong cooperation despite the lower send

```
'''json
{"return": 6.5}
'''
```

### Reasoning:

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 5: Returned  $\$7.5/\$10.86 = 69\%$
- Round 7: Returned  $\$10/\$14.52 = 69\%$
- Round 8 (now): Sent  $\$3.21$  (64% of endowment)

**\*\*Current Situation:\*\***

- They sent  $\$3.21$ , keeping  $\$1.79$
- I received  $\$9.63$
- This is lower than their usual  $\$4$  sends

The lower send amount is notable. They've consistently sent  $\$4$  in previous rounds as sender, but now sent  $\$3.21$ . This could be:

1. Random variation
2. Testing behavior
3. The noise in communication

However, they've been very cooperative throughout, returning 69% consistently when receiver.

**\*\*My Decision:\*\***

I should continue the cooperative pattern. Their typical return rate is around 69%.

If I return  $\$6.5$ :

- They end with  $\$8$  total ( $\$1.79 + \$6.5$ )
- I keep  $\$3.13$
- ~67% return rate, matching their pattern
- Maintains cooperation

Actually, let me recalculate:

- They sent  $\$3.21$ , kept  $\$1.79$
- I received  $\$9.63$
- If I return  $\$6.5$ , they get  $\$1.79 + \$6.5 = \$8.29$  total
- I keep  $\$9.63 - \$6.5 = \$3.13$

That seems reasonable and maintains the cooperative relationship.

Alternatively, returning  $\$7$ :

- They get  $\$1.79 + \$7 = \$8.79$
- I keep  $\$2.63$
- ~73% return rate

Given their 69% return pattern, I'll aim for around  $\$6.5$ - $\$7$  to match that.

```
Usage: {
 "input_tokens": 3074,
 "output_tokens": 640,
 "reasoning_tokens": 0
}
```

}

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Receiver, Agent\_3=Sender, Agent\_4=Receiver, Agent\_5=Sender, Agent\_6=Receiver, Agent\_7=Sender, Agent\_8=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver  
dyad\_2: Agent\_3=Sender, Agent\_4=Receiver  
dyad\_3: Agent\_5=Sender, Agent\_6=Receiver  
dyad\_4: Agent\_7=Sender, Agent\_8=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$5; received \$15; Agent\_2 returned \$10; payoffs Agent\_1=\$10, Agent\_2=\$5;  
noise/visible: returned visible=\$10.11  
Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 15.0, "min": 0, "raw": 10.0}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}  
dyad\_2: Agent\_3 sent \$4; received \$12; Agent\_4 returned \$5.5; payoffs Agent\_3=\$6.5, Agent\_4=\$6.5;  
noise/visible: sent visible=\$3.69, received visible=\$11.07, returned visible=\$6.08  
Action validation: {"returned": {"amount": 5.5, "clamped": false, "max": 12.0, "min": 0, "raw": 5.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_5 sent \$4; received \$12; Agent\_6 returned \$8.5; payoffs Agent\_5=\$9.5, Agent\_6=\$3.5;  
noise/visible: sent visible=\$4.44, received visible=\$13.32, returned visible=\$7.69  
Action validation: {"returned": {"amount": 8.5, "clamped": false, "max": 12.0, "min": 0, "raw": 8.5}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_4: Agent\_7 sent \$4; received \$12; Agent\_8 returned \$10; payoffs Agent\_7=\$11, Agent\_8=\$2;  
noise/visible: sent visible=\$4.77, received visible=\$14.31, returned visible=\$9.11  
Action validation: {"returned": {"amount": 10.0, "clamped": false, "max": 12.0, "min": 0, "raw": 10.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent\_1": 69.5,  
"Agent\_2": 52.5,  
"Agent\_3": 52.55,  
"Agent\_4": 54.45,  
"Agent\_5": 56.5,  
"Agent\_6": 57.5,  
"Agent\_7": 65.0,  
"Agent\_8": 48.0

}

Visible balances after round: {

"Agent\_1": 63.5,  
"Agent\_2": 58.03,  
"Agent\_3": 51.73,  
"Agent\_4": 50.330000000000005,  
"Agent\_5": 59.72,  
"Agent\_6": 59.309999999999995,  
"Agent\_7": 65.72,  
"Agent\_8": 54.459999999999994

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T08:04:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",  
"pairing": {  
"agents": [  
"Agent\_1",  
"Agent\_2"  
],  
"investor": "Agent\_1",  
"roles": {  
"Agent\_1": "investor",  
"Agent\_2": "trustee"

```

 },
 "trustee": "Agent_2"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

Your total visible earnings across all rounds are \$10.93.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)



Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at my co-player's behavior:

Round 1: They received \$10.56 and returned \$7.0 (66%) - very generous and cooperative

Round 2: They sent \$3.5, showing continued trust in me

This co-player has demonstrated strong cooperative behavior. They've been generous with returns and willing to send substantial amounts.

I should reciprocate this trust and maintain our cooperative relationship. Sending a higher amount signals my confidence in them and should benefit us both.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

Round 1: I was SENDER

- I sent \$3, but they saw \$3.52 (due to noise)
- It became \$10.56 for them
- They returned \$7.0
- This was very generous - they returned 66% of what they received

Round 2: I was RECEIVER

- They sent \$3.5 (actual), but I saw \$2.61 (noise)
- It became \$7.83 for me
- I returned \$5.0
- But they saw \$5.84 returned (noise again)
- Their visible payoff was \$7.34

So from their perspective:

- Round 1: They received \$10.56, returned \$7.0, kept \$3.56
- Round 2: They sent \$3.5, kept \$1.5, and saw \$5.84 returned, so total = \$1.5 + \$5.84 = \$7.34

This co-player has been very cooperative and trustworthy. They returned a high amount in Round 1, and they continued to send a substantial amount in Round 2.

My total visible earnings are \$10.93 across rounds.

Now I'm the sender again. Given that this co-player has proven to be very cooperative and trustworthy, I should send a substantial amount to maintain this cooperative relationship. They've shown they will return good amounts.

I should probably send around \$3-4 to show continued trust and cooperation.

Let me send \$4 this time to show I trust them and value the cooperative relationship.

**usage:**

```
{
 "input_tokens": 1113,
 "output_tokens": 495,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

**reasoning:**

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

**usage:**

```
{
 "input_tokens": 1469,
 "output_tokens": 505,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you

u send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

**reasoning:**

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)

Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous

Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

**usage:**

```
{
 "input_tokens": 1825,
 "output_tokens": 315,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$33.79.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at my co-player's consistent pattern:

Round 3: Returned \$9.0 out of \$13.02 (69%)

Round 5: Returned \$10.0 out of \$14.91 (67%)

Now in Round 6, they've sent me the maximum amount possible (\$5), showing complete trust in me. This is the strongest signal of cooperation they could give.

I should reciprocate this maximum trust with a generous return. Returning about 67% of the

\$15 I received (matching their typical return rate) would be appropriate.

```
'''json
{"return": 10.0}
'''
```

**reasoning:**

Looking at the pattern with my co-player:

Round 3: They returned \$9.0 out of \$13.02 (69%)  
Round 5: They returned \$10.0 out of \$14.91 (67%)

And now in Round 6, they sent me the maximum amount (\$5), which became \$15. This is showing maximum trust in me.

They've been extremely generous throughout our interactions:

- Always sending substantial amounts when they're the sender
- Always returning generous amounts (around 67-69%) when they're the receiver

They just sent me the maximum possible amount (\$5), which shows complete trust in me. I should reciprocate this trust with a generous return.

Given their consistent pattern of returning around 67-69% and their maximum trust in sending \$5, I should return around 65-70% of \$15, which would be about \$9.75-\$10.50.

I'll return \$10, which is 67% of what I received, matching their typical return rate and showing I value our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2188,
 "output_tokens": 385,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.
- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.

Your total visible earnings across all rounds are \$38.79.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at my co-player's excellent track record:

Round 4: Sent \$4.0 (high trust)  
Round 5: Returned \$10.0 out of \$14.91 (67% - very generous)  
Round 6: Sent \$4.5 (continued high trust)

This co-player has been consistently cooperative and trustworthy across all our interactions. They've shown maximum cooperation, and our relationship has been mutually beneficial.

I should reciprocate with maximum trust by sending the full amount.

```
'''json
{"send": 5}
'''
```

**reasoning:**

Looking at my co-player's history:

Round 4: Sent \$4.0 as sender (high trust)

Round 5: Returned \$10.0 out of \$14.91 (67% - generous)  
Round 6: Sent \$4.5 as sender (high trust)

This co-player has consistently demonstrated:

- High trust as sender (sending \$4.0-\$4.5)
- Generous returns as receiver (67%+)
- Strong cooperative behavior throughout

We've built an excellent cooperative relationship over 6 rounds. They've been trustworthy every single time.

Given their consistent cooperation and trustworthiness, I should send the maximum amount to maximize our joint payoffs. This continues our pattern of mutual trust and cooperation

**usage:**

```
{
 "input_tokens": 2547,
 "output_tokens": 314,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.
- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.
- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.

Your total visible earnings across all rounds are \$48.39.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at my co-player's consistent cooperative behavior:

Round 5: Returned \$10.0 out of \$14.91 (67%)

Round 7: Returned \$10.0 out of \$14.43 (69%)

They've now sent me the maximum \$5 again in Round 8, demonstrating complete trust for the second time. Their pattern has been consistently generous and cooperative throughout all our interactions.

I should reciprocate their maximum trust with a generous return, matching their typical return rate of around 67%.

```
'''json
{"return": 10.0}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 5: They returned \$10.0 out of \$14.91 (67%)

Round 6: They sent \$4.5 (high trust)

Round 7: They returned \$10.0 out of \$14.43 (69%)

Now in Round 8, they've sent me the maximum \$5 again, showing complete trust.

This co-player has been extremely cooperative and consistent:

- Always sending high amounts when sender (4.0, 4.5, 5.0)
- Always returning generously when receiver (around 67-69%)

They're showing maximum trust by sending \$5 again. I should reciprocate with a generous return. Based on their pattern of returning around 67-69%, I should return around \$10 from the \$15.

Let me be generous and return \$10, matching their typical return rate.

**usage:**

```
{
 "input_tokens": 2719,
 "output_tokens": 348,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.
- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.
- Round 8 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.84 returned, visible payoff \$9.84.

Your total visible earnings across all rounds are \$53.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at my co-player's consistent pattern:

Round 6: Sent \$4.5 (high trust)

Round 7: Returned \$10.0 out of \$14.43 (69% - very generous)

Round 8: Sent \$5.0 (maximum trust)

This co-player has maintained excellent cooperation throughout our entire interaction. They've sent maximum or near-maximum amounts repeatedly and returned generously. Our mutual trust has been well-established.

I should continue this strong cooperative relationship with maximum trust.

```
'''json
{"send": 5}
'''
```

**Reasoning:**

Looking at my co-player's history:

Round 6: Sent \$4.5 (high trust)

Round 7: Returned \$10.0 out of \$14.43 (69% - very generous)

Round 8: Sent \$5.0 (maximum trust)

This co-player has been extremely cooperative throughout our entire relationship. They've consistently sent high amounts when they're the sender and returned generously when they're the receiver.

We've built up an excellent cooperative pattern. I should continue with maximum trust and send the full amount.

Usage: {

```
"input_tokens": 2788,
"output_tokens": 265,
"reasoning_tokens": 0
```

}

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T08:05:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
```

```

"agents": [
 "Agent_1",
 "Agent_2"
],
"investor": "Agent_1",
"roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.  
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.  
Your total visible earnings across all rounds are \$10.9.

This round, you are the RECEIVER against your current co-player. They sent you \$4.34, so you received \$13.02. How much do you return? (0-13.02)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at this situation:

My co-player has shown exceptional trust by sending \$4.34 (86.8% of their endowment), leaving themselves with only \$0.66. This is a very strong signal of cooperation.

The history shows:

- Round 1: They sent \$3.0 and received \$6.1 back (good reciprocation)
- Round 2: They returned \$5.0 when they received \$7.83 (generous)
- Round 3: They sent \$4.34 (very high trust)

The multiplication created \$8.68 in added value. To reciprocate their high trust and ensure mutual benefit:

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This roughly splits the gains equally and strongly rewards their exceptional trust.

```
'''json
{"return": 9.00}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1: My co-player sent \$3.0 as sender, received \$6.1 back

Round 2: I sent \$3.50 as sender (but they saw \$2.61 due to communication noise), they returned \$5.0

Now in Round 3: They sent \$4.34 (86.8% of their endowment - very high trust!), I received \$13.02

My co-player has shown very high trust this round by sending \$4.34 (86.8% of their endowment). They're left with only \$0.66.

Looking at the pattern:

- In Round 1, when they sent \$3.0, they got back \$6.1 - roughly 2x return on what they sent
- In Round 2, they saw I sent \$2.61 (actual \$3.50) and returned \$5.0 - that's about 1.9x return ratio

They've been generous in their returns. Now they've shown even MORE trust by sending \$4.34.

The multiplication created \$8.68 in added value (\$13.02 - \$4.34).

To be fair and reciprocate their high trust:

- They're left with \$0.66
- They sent \$4.34

If I return \$9.00:

- Sender ends with: \$0.66 + \$9.00 = \$9.66 (profit of \$4.66)
- I keep: \$13.02 - \$9.00 = \$4.02 (profit of \$4.02)

This gives us roughly equal profits and rewards their very high trust level.

Actually, let me reconsider. They sent 86.8% of their endowment. This is very high trust.

I should consider returning enough to make it very worthwhile for them.

...



**usage:**

```
{
 "input_tokens": 1203,
 "output_tokens": 843,
 "reasoning_tokens": 0
}
```

**Message 4 role=user****content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant****content:**

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: \$5 - \$3.50 + return = \$1.50 + return

Round 3: I returned \$9.00, kept \$4.02

So my earnings: \$3.56 + (\$1.50 + what I got back in round 2) + \$4.02 = \$14.92

This means: \$9.08 + what I got back in round 2 = \$14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

**usage:**

```
{
 "input_tokens": 1669,
 "output_tokens": 673,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with: \$0.03 + \$10.00 = \$10.03 (profit of \$5.03)
- I keep: \$14.91 - \$10.00 = \$4.91 (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)  
Round 3: They sent \$4.0, received \$8.5 back  
Round 4: They returned \$6.0 from \$9.42 (generous)  
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with:  $\$0.03 + \$10.00 = \$10.03$  (profit of \$5.03)
- I keep:  $\$14.91 - \$10.00 = \$4.91$  (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender:  $\$0.03 + \$9.50 = \$9.53$  (profit of \$4.53)
- Me:  $\$14.91 - \$9.50 = \$5.41$  (profit of \$5.41)

...

**usage:**

```
{
 "input_tokens": 2083,
 "output_tokens": 771,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.

Your total visible earnings across all rounds are \$27.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have been highly cooperative throughout:

- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (strong return)

This co-player has demonstrated:

1. High trust as sender (escalating from \$4.0 to \$4.5)
2. Fair/generous reciprocation as receiver
3. Consistent success in our cooperative relationship

They've been rewarded well for their trust and have reciprocated fairly. I should continue this strong cooperation by matching their recent trust level.

```
'''json
{"send": 4.50}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

- Round 3: They sent \$4.0, received \$8.5 back (very good return)
- Round 4: They returned \$6.0 from \$9.42 (generous)
- Round 5: They sent \$4.5, received \$9.44 back (excellent return)

This co-player has been:

1. Highly trusting as sender (sent \$4.0, then \$4.5)
2. Generous as receiver (returned \$6.0 from \$9.42)
3. Consistently rewarded with good returns

The relationship has been very cooperative and mutually beneficial. They've been sending high amounts and getting good returns. As a receiver, they've been fair/generous.

Given this strong cooperative history, I should send a high amount to continue this mutually beneficial relationship. They sent \$4.5 in round 5, so I should match or exceed that level of trust.

I'll send \$4.50 to match their recent trust level.

**usage:**

```
{
 "input_tokens": 2556,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$38.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at this situation:

My co-player has shown very high trust by sending \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19.

The visible history shows excellent cooperation:

- Round 4: Returned \$6.0 from \$9.42
- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

They were extremely generous in Round 6, returning \$10.0 from \$15. Now they're showing continued high trust as a sender.

The multiplication created \$9.62 in added value. To reciprocate their strong cooperation:

If I return \$10.00:

- Sender ends with: \$0.19 + \$10.00 = \$10.19 (profit of \$5.19)
- I keep: \$14.43 - \$10.00 = \$4.43 (profit of \$4.43)

This maintains our successful cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at this situation:

My co-player sent \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19. This is very high trust.

The visible history shows strong cooperation:

- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (good return)
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

The multiplication created \$9.62 in added value (\$14.43 - \$4.81).

My co-player has been very generous as a receiver (returning \$6.0, then \$10.0) and trusting as a sender (sending \$4.5, then \$4.81).

To reciprocate this high trust and maintain the cooperative relationship:

If I return \$10.00:

- Sender ends with:  $\$0.19 + \$10.00 = \$10.19$  (profit of \$5.19)
- I keep:  $\$14.43 - \$10.00 = \$4.43$  (profit of \$4.43)

This is fair and rewards their very high trust. Given they returned \$10.0 from \$15 last round, returning \$10.00 from \$14.43 would be consistent with what they've shown.

Actually, let me think about this more carefully. They sent \$4.81, leaving \$0.19. The multiplication created \$14.43 for me, which is an added value of \$9.62.

If I return \$10.00, they get \$10.19 total (profit \$5.19), I keep \$4.43 (profit \$4.43). We both profit well.

Given the strong cooperative history and their very high trust (96.2% of endowment), \$10.00 seems appropriate.

usage:

```
{
 "input_tokens": 2963,
 "output_tokens": 735,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.

Your total visible earnings across all rounds are \$43.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

content:

Looking at the visible history with my co-player:

They have shown exceptional cooperation:

- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (extremely generous)
- Round 7: Sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has:

1. Escalated to maximum trust by sending their entire \$5 endowment
2. Been very generous as a receiver
3. Maintained consistent cooperation throughout

They've shown maximum trust and it's been rewarded. I should reciprocate with the same level of trust to complete our highly successful cooperative relationship.

```
'''json
{"send": 5.00}
```

'''

**reasoning:**

Looking at the visible history with my co-player:

- Round 5: They sent \$4.5, received \$9.44 back
- Round 6: They returned \$10.0 from \$15 (very generous)
- Round 7: They sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has now shown maximum trust in Round 7 by sending their entire \$5 endowment . They have been:

1. Escalating their trust (from \$4.5 to \$5.0)
2. Very generous as a receiver (returned \$10.0 from \$15)
3. Consistently cooperative and successful

They sent the maximum (\$5.0) in Round 7 and received \$9.6 back, which is a good return. I should reciprocate with maximum trust as well to maintain this strong cooperative relationship.

This is the final round (Round 8), so I should send my full amount to maximize our joint payoff and complete our cooperative relationship on a strong note.

**usage:**

```
{
 "input_tokens": 3154,
 "output_tokens": 420,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.
- Round 8 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$53.03.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at this situation:

My co-player has sent the maximum amount of \$5 (100% of their endowment), showing complete trust.

The visible history shows a strong pattern:

- Round 6: They returned \$10.0 from \$15
- Round 7: Sent \$5.0 (maximum), received \$9.6 back
- Round 8: They returned \$10.0 from \$15
- Round 9: Sent \$5.0 (maximum trust again)

This co-player has consistently shown maximum trust as a sender and generous reciprocation as a receiver, establishing a clear pattern of returning \$10 from \$15.

The multiplication created \$10 in added value. To maintain our highly successful cooperative pattern:

If I return \$10.00:

- Sender ends with: \$0 + \$10.00 = \$10.00 (profit of \$5.00)
- I keep: \$15.00 - \$10.00 = \$5.00 (profit of \$5.00)

This matches their consistent behavior and splits the gains equally.

```
'''json
{"return": 10.00}
'''
```

**Reasoning:**

Looking at the visible history with my co-player:

- Round 6: They returned \$10.0 from \$15 (very generous)
- Round 7: Sent \$5.0 (maximum trust), received \$9.6 back
- Round 8: They returned \$10.0 from \$15 (very generous again)
- Round 9: Sent \$5.0 (maximum trust again)

This co-player has been showing maximum cooperation:

1. Sending the full \$5 endowment (maximum trust)
2. When receiving, returning \$10 out of \$15 consistently (very generous)

They've been incredibly cooperative throughout our relationship. Now they've sent their full \$5 again.

The multiplication created \$10 in added value from their \$5 send.

If I return \$10.00:

- Sender ends with:  $\$0 + \$10.00 = \$10.00$  (profit of \$5.00)
- I keep:  $\$15.00 - \$10.00 = \$5.00$  (profit of \$5.00)

This matches what they did in rounds 6 and 8, maintaining our pattern of strong mutual cooperation.

I should return \$10 to match their generous pattern.

```
Usage: {
 "input_tokens": 3293,
 "output_tokens": 561,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_3 - Sender - dyad\_2**

Timestamp: 2026-08-12T08:05:09

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 2,
 "opponent_id": "Agent_4",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
 },
 "role": "investor",
 "role_label": "Sender",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

Your total visible earnings across all rounds are \$9.5.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at our history together:

Round 1: I sent \$2.5, they received and returned \$3.0 - showing good cooperation

Round 2: They sent me \$2.5, I returned \$3.5 (though noise showed them \$2.75)

We're paired again! This co-player has been consistently cooperative. Even though noise may have made my return seem smaller than it was, they've maintained trust by sending again.

I'll increase my send amount to continue building this cooperative relationship.

{'send': 3.0}

**reasoning:**

Let me analyze the history:

Round 1: I sent \$2.5 (they saw \$2.22 due to noise), it became \$6.66 for them, they returned \$3.0

- They kept \$3.66, I got back \$3.0



- My payoff:  $\$5 - \$2.5 + \$3.0 = \$5.5$

Round 2: They sent me \$2.5, I received \$7.5, I returned \$3.5 (they saw \$2.75 due to noise)

- I kept \$4.0, they got back what they saw as \$2.75

- Their visible payoff:  $\$5 - \$2.5 + \$2.75 = \$5.25$

So we're playing with the same co-player again! They've been cooperative both rounds, and I've been cooperative back. However, due to communication noise, they saw that I only returned \$2.75 when I actually returned \$3.5.

From their perspective:

- Round 1: They received \$6.66, returned \$3.0 (45% return rate)

- Round 2: They sent \$2.5, got back \$2.75 (as they saw it)

They might think I was slightly less generous than I actually was. But they've still been cooperative overall.

Given that we're building a cooperative relationship, and they've sent reasonable amounts both times, I should continue to cooperate. I'll send a bit more this time to signal continued trust and cooperation.

I'll send \$3.0 this round.

**usage:**

```
{
 "input_tokens": 1055,
 "output_tokens": 491,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

{'return': 5.0}

**reasoning:**

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0

Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)

- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of  $\$1.63 + \$5.05 = \$6.68$ , which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

**usage:**

```
{
 "input_tokens": 1407,
 "output_tokens": 525,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5  
Round 3: I sent \$3.0, they returned \$4.0  
Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)  
Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

```
{'send': 4.0}
```

**reasoning:**

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)  
Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0  
Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)  
Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous amounts:

- \$3.0 in Round 1
- \$4.0 in Round 3

- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

**usage:**

```
{
 "input_tokens": 1754,
 "output_tokens": 406,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.

Your total visible earnings across all rounds are \$27.519999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.15, so you received \$9.45. How much do you return? (0-9.45)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our continued partnership:

Round 3: I sent \$3.0, they returned \$4.0  
Round 4: They sent \$3.0, I returned \$5.0  
Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$3.15, giving me \$9.45

This co-player continues to be highly cooperative and generous, returning around 50% each time. They've maintained trust throughout despite noise. I'll reciprocate with a generous return to sustain this excellent relationship.

{'return': 4.75}

**reasoning:**

Looking at our history:

Round 3: I sent \$3.0, they returned \$4.0  
Round 4: They sent \$3.0, I returned \$5.0  
Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$3.15, giving me \$9.45

This co-player has consistently been generous:

- Round 3: returned \$4.0 out of \$8.52 (47%)
- Round 5: returned \$6.3 out of \$12.63 (50%)
- Round 6: sent \$3.15 (down from \$3.0 but still good)

They've maintained trust and cooperation throughout. I should continue reciprocating generously. Given they sent \$3.15 and I received \$9.45, a fair return that rewards their trust would be around 50% or so.

Let me return about \$4.5-5 to maintain our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2138,
 "output_tokens": 366,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.

Your total visible earnings across all rounds are \$32.22.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our partnership:

Round 4: They sent \$3.0, I returned \$5.0

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0, I returned \$4.75 (they saw \$4.49 due to noise)

Round 7: My turn to send

This co-player has remained cooperative throughout, continuing to send good amounts despite noise making my returns appear slightly lower. We've established a strong pattern of mutual cooperation.

I'll continue sending a high amount to maximize our joint earnings.

{'send': 4.0}

**reasoning:**

Looking at the history:

Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)

Round 5: I sent \$4.0, they returned \$6.3

Round 6: They sent \$4.0 (wait, actually they sent \$3.15 according to round 6), I returned \$4.75 (they saw \$4.49)

Round 7: My turn to send

Wait, let me recheck the history. In Round 6, it says "They sent you \$3.15" but now in Round 7 it says "Round 6 against you, as SENDER: sent \$4.0". This is the noise at play.

So from their perspective:

- Round 6: They sent \$4.0, it became \$12.0, they saw that I returned \$4.49

From my perspective:

- Round 6: They sent \$3.15, I received \$9.45, I returned \$4.75

The noise made them see \$4.49 instead of \$4.75. That's actually not too far off.

This co-player has been consistently cooperative. They continue to send good amounts. I should continue to cooperate strongly. I'll send another high amount.

I've been sending increasing amounts: \$2.5, \$3.0, \$4.0. I should continue at a high level to maximize our joint payoffs.

**usage:**

```
{
 "input_tokens": 2496,
 "output_tokens": 473,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.  
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.  
- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.  
Your total visible earnings across all rounds are \$38.65.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our continued partnership:

Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$4.0, I returned \$4.75  
Round 7: I sent \$4.0, they returned \$5.0  
Round 8: They sent \$4.0, giving me \$12.0

This co-player has shown remarkable consistency and trust throughout our partnership, sending their highest amount yet. They've maintained cooperation despite noise potentially obscuring my returns. I'll return generously to reward this strong trust and cooperation.

{ 'return': 6.0 }

**reasoning:**

Looking at the partnership history:

Round 5: I sent \$4.0, they returned \$6.3  
Round 6: They sent \$4.0, I returned \$4.75  
Round 7: I sent \$4.0 (they saw \$3.35), they returned \$5.0  
Round 8: They sent \$4.0, giving me \$12.0

This co-player has been consistently cooperative throughout our partnership. They've maintained high send amounts and reasonable returns despite communication noise. Even when noise made my returns appear lower, they continued to trust and send generous amounts.

They just sent \$4.0, their highest amount, showing strong trust. I should reciprocate generously. Looking at the pattern:

- When they sent and I received \$12.0 before, I returned \$4.75  
- When I sent and they received \$10.05, they returned \$5.0  
- When I sent and they received \$12.63, they returned \$6.3

I should return around 50% or so to match their cooperative pattern. Given they sent \$4.0, a return of around \$6.0 would be fair and maintain our strong cooperation.

**usage:**

```
{
 "input_tokens": 2730,
 "output_tokens": 405,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.  
- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.  
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.42 returned, visible payoff \$7.42.  
Your total visible earnings across all rounds are \$44.65.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our partnership:

Round 6: They sent \$4.0, I returned \$4.75  
Round 7: I sent \$4.0, they returned \$5.0  
Round 8: They sent \$4.0, I returned \$6.0 (they saw \$6.42)  
Round 9: My turn to send

This co-player has demonstrated exceptional consistency and trustworthiness throughout our partnership. They've maintained cooperation across many rounds, showing they're a reliable partner. I'll continue with a high send amount to maximize our mutual benefit.

```
{'send': 4.0}
```

**Reasoning:**

Looking at our partnership history:

Round 6: They sent \$4.0, I returned \$4.75 (they saw \$4.49)  
Round 7: I sent \$4.0, they returned \$5.0  
Round 8: They sent \$4.0, I returned \$6.0 (they saw \$6.42)  
Round 9: My turn to send

This co-player has been incredibly consistent and trustworthy throughout our entire partnership. They've sent \$4.0 multiple times now and returned fair amounts. We've built excellent cooperation over many rounds.

I should continue with a high send amount to maximize our joint payoff and maintain this strong relationship.

I'll send \$4.0 again.

```
Usage: {
 "input_tokens": 2821,
 "output_tokens": 306,
 "reasoning_tokens": 0
}
```

**GAME - Agent\_4 - Receiver - dyad\_2**

Timestamp: 2026-08-12T08:05:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_2",
 "move_index": 3,
 "opponent_id": "Agent_3",
 "pairing": {
 "agents": [
 "Agent_3",
 "Agent_4"
],
 "investor": "Agent_3",
 "roles": {
 "Agent_3": "investor",
 "Agent_4": "trustee"
 },
 "trustee": "Agent_4"
```

```
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.

Your total visible earnings across all rounds are \$8.91.

This round, you are the RECEIVER against your current co-player. They sent you \$2.84, so you received \$8.52. How much do you return? (0-8.52)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the pattern:

Round 1: My co-player sent \$2.5 (50%), I returned \$3.0 (45% of \$6.66)

Round 2: I sent \$2.5 (50%), they returned \$3.5 (47% of \$7.5)

Round 3: My co-player sent \$2.84 (57% - even more trust!)

They've been consistently cooperative and have actually increased their trust level this round. I should reciprocate appropriately.

Returning around 47% of what I received (\$8.52):

- 47% = approximately \$4.00
- This gives them: \$2.16 + \$4.00 = \$6.16 (net gain of \$1.16)
- I keep: \$8.52 - \$4.00 = \$4.52

This matches their previous return rate and rewards their increased trust.

```
{"return": 4.00}
```

**reasoning:**

Let me analyze the visible history:

Round 1:

- Co-player was SENDER: sent \$2.5 (50%), received \$3.0 back (40% of \$7.5), ended with \$5.5
- I was RECEIVER: received \$6.66, returned \$3.0, ended with \$3.66

Round 2:

- I was SENDER: sent \$2.5 (50%), co-player returned \$3.5 (46.7% of \$7.5), I ended with \$6.0
- Co-player was RECEIVER: received \$7.5, returned \$3.5, ended with \$4.0

So the co-player:

- As sender in Round 1: sent 50%, got back 40% of the tripled amount
- As receiver in Round 2: returned 46.7% of what they received

Now in Round 3:

- Co-player is SENDER: sent \$2.84 (56.8% of \$5), creating \$8.52 for me
- I am RECEIVER: need to decide how much to return

The co-player has been consistently cooperative:

- Sent 50% in Round 1
- Returned 46.7% in Round 2
- Now sent 56.8% in Round 3 (even more trusting!)

I should reciprocate this trust. In Round 1, I returned 45% of what I received (\$3.0 out of \$6.66).

In Round 2, they returned 46.7% (\$3.5 out of \$7.5).

For Round 3, returning around 45-47% would be consistent:

- 45% of \$8.52 = \$3.83
- 47% of \$8.52 = \$4.00

The co-player sent even more this time (\$2.84 vs \$2.5 previously), showing increased trust. I should match or slightly exceed their previous return rate.

...

**usage:**

```
{
 "input_tokens": 1177,
 "output_tokens": 777,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear



r as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

**reasoning:**

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0

Round 2: I sent \$2.5 (50%), co-player returned \$3.5

Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, and what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation

...

**usage:**

```
{
 "input_tokens": 1614,
 "output_tokens": 808,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)

Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost the entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets:  $\$0.79 + \$6.30 = \$7.09$  (net gain of \$2.09)
- I keep:  $\$12.63 - \$6.30 = \$6.33$

This reciprocates their trust appropriately.

```
{"return": 6.30}
```

**reasoning:**

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0

Round 2: I sent \$2.5, they returned \$3.5

Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)

Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0

Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)
- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32
- Sender would have:  $\$0.79 + \$6.32 = \$7.11$  (gain of \$2.11)
- I would have:  $\$12.63 - \$6.32 = \$6.31$

This seems fair and rewards their very high trust. Let me round to \$6.30 for a clean number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have:  $\$0.79 + \$6.50 = \$7.29$

...

**usage:**

```
{
 "input_tokens": 2041,
 "output_tokens": 754,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

Your total visible earnings across all rounds are \$26.799999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the visible history:

Round 4: I sent \$3.37, they returned \$5.0 (strong reciprocation)

Round 5: They sent \$4.0 (80% of endowment), saw \$6.9 returned (good outcome)

My co-player has been increasing their trust levels and has reciprocated well when in the receiver role. They're demonstrating strong cooperative behavior.

To match their high trust level and continue building on this cooperative relationship, I should send a similarly high amount.

```
{"send": 4.0}
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 3: They sent \$3.0, saw \$3.01 returned (seemed like poor reciprocation)

Round 4: They saw I sent \$3.37, returned \$5.0 (good reciprocation - 49.5%)

Round 5: They sent \$4.0, saw \$6.9 returned (good reciprocation - 57.5%)

My co-player has been showing increasing trust (sending \$3.0, then \$4.0). They're being cooperative despite the noise in Round 3 where it appeared I only returned \$3.01.

Given their pattern of increasing trust and good returns when they're the receiver, I should continue with high cooperation. Sending around 60-80% would be appropriate to match their escalating trust levels.

I'll send \$4.0 (80%) to match their Round 5 amount and show continued high trust.

**usage:**

```
{
 "input_tokens": 2474,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 10 role=user****content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.

Your total visible earnings across all rounds are \$32.29.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant****content:**

Looking at the visible history:

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of what they received)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to demonstrate trust and has been consistently returning around 50% when in the receiver role. I should match this reciprocal behavior.

I received \$10.05. Returning approximately 50%:

- Return \$5.00

- Sender gets: \$1.65 + \$5.00 = \$6.65 (net gain of \$1.65)

- I keep: \$10.05 - \$5.00 = \$5.05

```
{"return": 5.00}
```

**reasoning:**

Looking at the visible history:

Round 6: They saw I sent \$3.15, and they returned \$4.75 (50.3% of \$9.45)  
Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to show strong trust and has been returning around 50% when in the receiver role. They sent \$3.35 this round.

I received \$10.05. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.00 (approximately 50% of \$10.05)
- Sender gets:  $\$1.65 + \$5.00 = \$6.65$  (net gain of \$1.65)
- I keep:  $\$10.05 - \$5.00 = \$5.05$

This maintains consistency with their return rates and continues our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2838,
 "output_tokens": 385,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.
- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.43 returned, visible payoff \$6.43.

Your total visible earnings across all rounds are \$37.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the visible history:

Round 5: They sent \$4.0, saw \$6.9 returned - strong payoff of \$7.9

Round 6: They saw I sent \$3.15, returned \$4.75

Round 7: They sent \$4.0, saw \$5.43 returned

My co-player has been consistently sending \$4.0 (80% of endowment), demonstrating high trust. They've also been reciprocating fairly when in the receiver role.

We've established a strong cooperative pattern. I should match their trust level to maintain this mutually beneficial relationship.

```
{"send": 4.0}
```

**reasoning:**

Looking at the visible history from my co-player's perspective:

Round 5: They sent \$4.0, saw \$6.9 returned (58% of \$12) - good payoff \$7.9

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of \$9.45) - their payoff \$4.70

Round 7: They sent \$4.0, saw \$5.43 returned (45% of \$12) - payoff \$6.43

They've been consistently sending high amounts (\$4.0) showing strong trust. In Round 7, from their perspective, they saw a lower return rate (45% instead of the usual 50%+), which was likely due to communication noise since I actually returned \$5.00.

The pattern shows:

- They're maintaining high trust levels (sending \$4.0)
- They reciprocate fairly when in receiver role (around 50%)
- We've built a strong cooperative relationship

I should continue to match their trust level and send a high amount. Since they've been se

nding \$4.0 consistently, I should do the same to maintain our cooperative equilibrium.

```
{"send": 4.0}
usage:
{
 "input_tokens": 2961,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.43 returned, visible payoff \$6.43.
- Round 8 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$6.0, visible payoff \$6.0.

Your total visible earnings across all rounds are \$44.760000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the visible history:

Round 8: They saw I sent \$4.0, returned \$6.0 (50% of what they received)

Round 9: They sent \$3.69 (74% of their endowment)

My co-player continues to show strong trust and we've established a consistent pattern of approximately 50% returns when in the receiver role.

I received \$11.07. Maintaining this reciprocal pattern:

- Return \$5.50
- Sender gets: \$1.31 + \$5.50 = \$6.81 (net gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

```
{"return": 5.50}
```

**Reasoning:**

Looking at the visible history:

Round 8: They saw I sent \$4.0, returned \$6.0 (50% of what they received)

Round 9: They sent \$3.69 (74% of their endowment)

My co-player continues to show strong trust. They've been consistently reciprocating at around 50% when in the receiver role, and I've been doing the same.

I received \$11.07. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.50
- Sender gets: \$1.31 + \$5.50 = \$6.81 (net gain of \$1.81)
- I keep: \$11.07 - \$5.50 = \$5.57

This maintains the fair reciprocal relationship we've established.

Usage: {

```
"input_tokens": 3097,
"output_tokens": 367,
"reasoning_tokens": 0
}
```

**GAME - Agent\_5 - Sender - dyad\_3**

Timestamp: 2026-08-12T08:05:21

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
```

```

"dyad_id": "dyad_3",
"move_index": 4,
"opponent_id": "Agent_6",
"pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.

Your total visible earnings across all rounds are \$12.009999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at our history together:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$2.78)
- They returned \$4 - showing good cooperation

**\*\*Round 2:\*\***

- They sent \$4.35 (87% of their endowment) - very high trust!
- I returned \$8 - rewarding that trust

We've established a strong cooperative pattern where both of us profit. They demonstrated significant trust in Round 2, and I reciprocated generously.

Now it's my turn as sender again. Given:

- Their proven trustworthiness
- The successful cooperation we've built
- Their willingness to send almost everything last round

I'll send \$4, which matches their high level of trust and commitment:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with, and based on our pattern, I expect them to return a fair amount that benefits us both.

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34 and returned \$4.0
- This was cooperative behavior

**\*\*Round 2:\*\***

- They sent \$4.35 (I received \$13.05)
- I returned \$8 (they saw \$8.76 due to noise)
- They ended with \$5 - \$4.35 + \$8.76 = \$9.41 (visible to them as \$10.26)
- I ended with \$13.05 - \$8 = \$5.05

So we've been cooperating well. They showed high trust by sending \$4.35, and I reciprocated generously by returning \$8.

**\*\*Round 3:\*\***

Now I'm the sender again. Given our cooperative history:

- They've been trustworthy
- I've been generous
- Both times we've both profited

I should continue this cooperative pattern. Since they sent \$4.35 (87% of their endowment) last round, I could match or exceed that level of trust.

However, let me think strategically:

- Sending everything (\$5) would show maximum trust
- Sending \$4 would show high trust while keeping a small buffer

- Sending \$3-3.5 would be moderate/equal to what I sent in Round 1

Given that:

1. They've proven trustworthy
2. They showed high trust last round
3. We have a pattern of mutual cooperation
4. There's noise, so being generous helps compensate

I'll send \$4, demonstrating strong trust while keeping a minimal buffer.

Actually, wait - they sent \$4.35 which is 87% of \$5. That's very high trust. I should probably match that level of commitment to maintain the strong cooperative relationship.

...

**usage:**

```
{
 "input_tokens": 1268,
 "output_tokens": 684,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our cooperative history:

**\*\*Round 1:\*\***

- They returned \$4 on \$8.34 received (48%)

**\*\*Round 2:\*\***

- I returned \$8 on \$13.05 received (61%)

**\*\*Round 3:\*\***

- They returned \$9 on \$14.01 received (64%) - very generous!

**\*\*Round 4 (Current):\*\***

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final: \$5 - \$3.5 + \$7 = \$8.5 (profit of \$3.5)
- My final: \$10.5 - \$7 = \$3.5 (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

**reasoning:**

Let me review the history with this co-player:

**\*\*Round 1:\*\***



- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

**\*\*Round 2:\*\***

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

**\*\*Round 3:\*\***

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01
- They returned \$9 (about 64% of what they received)
- Very generous return!

**\*\*Round 4 (Current):\*\***

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:

- Their final:  $\$5 - \$3.5 + \$7 = \$8.5$  (profit of \$3.5)
- My final:  $\$10.5 - \$7 = \$3.5$  (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:

- Their final:  $\$5 - \$3.5 + \$7.5 = \$9$  (profit of \$4)
- My final:  $\$10.5 - \$7.5 = \$3$  (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance—it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

**usage:**

```
{
 "input_tokens": 1711,
 "output_tokens": 1273,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at our established cooperative pattern:

**\*\*Round 2:\*\***

- They sent \$3.5 (70% of endowment)
- I returned generously

**\*\*Round 3:\*\***

- I sent high (they saw \$4.67)
- They returned \$9 - very generous (64% of what they received)

**\*\*Round 4:\*\***

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

**\*\*Round 5 (Current):\*\***

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

**reasoning:**

Looking at the history with this co-player:

**\*\*Round 2:\*\***

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

**\*\*Round 3:\*\***

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

**\*\*Round 4:\*\***

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

**\*\*Round 5 (Current):\*\***

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

**usage:**

```
{
 "input_tokens": 2169,
 "output_tokens": 521,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.

Your total visible earnings across all rounds are \$32.48.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our cooperative history:

**\*\*Round 3:\*\***

- They returned \$9 on \$14.01 (64% return rate) - generous

**\*\*Round 4:\*\***

- I returned fairly (they saw \$7.93)

**\*\*Round 5:\*\***

- They saw only \$3.06 sent (due to noise), received \$9.18
- They still returned \$6 (65%!) - very generous despite the low apparent send

**\*\*Round 6 (Current):\*\***

- They sent \$4.32 (86% of their endowment) - extremely high trust!
- I received \$12.96
- They're left with only \$0.68

They've consistently shown generosity and trust, especially in Round 5 where noise made it look like I sent less, yet they still returned generously. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.32 + \$8 = \$8.68 (profit of \$3.68)
- My final: \$12.96 - \$8 = \$4.96 (profit of \$4.96)

```
'''json
{"return": 8}
'''
```

**reasoning:**

Looking at the history:

**\*\*Round 3:\*\***

- I sent (they saw \$4.67)
- They returned \$9 on \$14.01 (64%)

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned (they saw \$7.93)

**\*\*Round 5:\*\***

- I sent \$4 (they saw \$3.06 due to noise)
- They received \$9.18 and returned \$6 (65%)
- This is very generous given they only saw \$3.06 sent!

**\*\*Round 6 (Current):\*\***

- They sent \$4.32 (86% of endowment) - very high trust!
- I received \$12.96
- They're at \$0.68

They've been consistently generous and trusting. In Round 5, despite the noise making it look like I only sent \$3.06, they still returned \$6 which is generous. Now they're showing even more trust by sending \$4.32.

Given their consistent generosity and trust:

- If I return \$8:
  - Their final:  $\$5 - \$4.32 + \$8 = \$8.68$  (profit of \$3.68)
  - My final:  $\$12.96 - \$8 = \$4.96$  (profit of \$4.96)

This rewards their trust appropriately and continues our cooperative relationship.

**usage:**

```
{
 "input_tokens": 2590,
 "output_tokens": 603,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.

Your total visible earnings across all rounds are \$37.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at our continued cooperation:

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned fairly (they saw \$7.93)

**\*\*Round 5:\*\***

- They saw only \$3.06 sent (noise affected it)
- Still returned \$6 - very generous despite the low visible amount

**\*\*Round 6:\*\***

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

**\*\*Round 7 (Current):\*\***

We've maintained an excellent cooperative relationship across 6 rounds. They've consistently shown trust even when noise made things look less favorable, and I've reciprocated generously.

Continuing our pattern of high trust:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with and demonstrates my continued commitment to our mutually beneficial partnership.

**reasoning:**

Looking at our history:

**\*\*Round 4:\*\***

- They sent \$4.0
- I returned \$7 (they saw \$7.93)

**\*\*Round 5:\*\***

- I sent \$4 (they saw \$3.06 due to noise)
- They returned \$6 - very generous despite noise

**\*\*Round 6:\*\***

- They sent \$4.32 (I calculated they sent, but they saw they sent \$4.0)
- I returned \$8 (they saw \$8.83)

**\*\*Round 7 (Current):\*\***

We've established a very strong cooperative relationship. They've been consistently generous:

- Round 5: Returned \$6 on \$9.18 even when it looked like I only sent \$3.06
- Round 6: Sent \$4.0 (80% of endowment) showing high trust

I've also been cooperative, returning fairly and sending high amounts.

Given our strong pattern, I should send a high amount. I'll send \$4 to maintain our mutual trust:

This gives them \$12 to work with and continues our successful cooperative pattern.

**usage:**

```
{
 "input_tokens": 3086,
 "output_tokens": 459,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.
- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.

Your total visible earnings across all rounds are \$46.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our consistent cooperation:

**\*\*Round 5:\*\***

- They returned \$6 on \$9.18 (65%) despite seeing only \$3.06 sent

**\*\*Round 6:\*\***

- They sent \$4.0, I returned generously (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%) - very generous

**\*\*Round 8 (Current):\*\***

- They sent \$4.14 (83% of endowment) - extremely high trust
- I received \$12.42
- They're left with only \$0.86

They've demonstrated unwavering trust and generosity throughout our partnership, consistently returning 60-65% even when noise made my sends appear lower. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)
- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

```
'''json
{"return": 8}
'''
```

**reasoning:**

Looking at our history:

**\*\*Round 5:\*\***

- They saw \$3.06 sent (noise), received \$9.18
- Returned \$6 (65%) - generous despite low visible send

**\*\*Round 6:\*\***

- They sent \$4.0
- I returned well (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%) - consistently generous

**\*\*Round 8 (Current):\*\***

- They sent \$4.14 (83% of their endowment) - very high trust
- I received \$12.42
- They're left with \$0.86

We've built an excellent cooperative relationship over 7 rounds. They've shown:

- Consistent generosity (60-65% return rates)
- Trust even when noise makes sends look lower
- Willingness to send high amounts (\$4.0-\$4.32)

I should reciprocate their trust and generosity.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)
- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

This maintains our cooperative pattern and rewards their high trust.

**usage:**

```
{
 "input_tokens": 3317,
 "output_tokens": 593,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.
- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$51.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at our established pattern:

**\*\*Round 6:\*\***

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%) - generous

**\*\*Round 8:\*\***

- They sent \$4.0 (80% of endowment)
- I returned \$8 (they saw \$7.68 due to noise)

**\*\*Round 9 (Current):\*\***

We've maintained excellent mutual trust throughout. They've consistently sent \$4 (80% of their

endowment) in recent rounds, and returned generously even when noise affected what they saw

.

Continuing our strong cooperative pattern:

```
'''json
{"send": 4}
'''
```

#### Reasoning:

Looking at our history:

**\*\*Round 6:\*\***

- They sent \$4.0 (80%)
- I returned well (they saw \$8.83)

**\*\*Round 7:\*\***

- They saw \$3.92 sent
- Returned \$7.5 on \$11.76 (64%)

**\*\*Round 8:\*\***

- They sent \$4.0 (80%)
- I returned \$8 (they saw \$7.68 due to noise)

**\*\*Round 9 (Current):\*\***

We've had a very consistent cooperative relationship. They continue to send \$4 regularly (80% of endowment), showing high trust. They've been generous in returning, even when noise affects what they see.

I should continue our pattern of high trust and send \$4.

```
'''json
{"send": 4}
'''
```

```
Usage: {
 "input_tokens": 3395,
 "output_tokens": 390,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_6 - Receiver - dyad\_3

Timestamp: 2026-08-12T08:05:28

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
 "agent_type": "standard",
 "dyad_id": "dyad_3",
 "move_index": 5,
 "opponent_id": "Agent_5",
 "pairing": {
 "agents": [
 "Agent_5",
 "Agent_6"
],
 "investor": "Agent_5",
 "roles": {
 "Agent_5": "investor",
 "Agent_6": "trustee"
 },
 "trustee": "Agent_6"
 },
 "role": "trustee",
 "role_label": "Receiver",
 "round": 9,
 "task": "game",
 "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al

so be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**MULTI-AGENT SETTING:**

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.

Your total visible earnings across all rounds are \$14.6.

This round, you are the RECEIVER against your current co-player. They sent you \$4.67, so you received \$14.01. How much do you return? (0-14.01)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the history and current situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.67 (93.4% of their endowment) - extremely high trust!
- They now have only \$0.33 remaining
- I received \$14.01
- In Round 2, when they were receiver, they returned \$8.0 out of \$13.05

**\*\*Reasoning:\*\***

Their exceptional trust (sending 93.4%) deserves strong reciprocation. They've shown consi



stent fairness and are now taking a big risk by sending almost everything.

If I return \$9.00:

- Sender ends with:  $\$0.33 + \$9.00 = \$9.33$  (profit of \$4.33)
- I keep:  $\$14.01 - \$9.00 = \$5.01$

This rewards their high trust generously while maintaining a fair split of the gains created by cooperation.

```
{'return': 9.00}
```

**reasoning:**

Let me analyze this situation:

From the visible history:

- Round 1: My co-player was sender, sent \$3.0, saw \$4.96 returned, got payoff \$6.96
- Round 2: My co-player was receiver, saw \$4.35 sent (I sent \$3.50), it became \$13.05, they returned \$8.0, their payoff was \$5.05

Wait, there's a discrepancy. In Round 2, I sent \$3.50, which should become \$10.50 when tripled. But the visible history says they saw \$4.35 sent and it became \$13.05.

The prompt mentions: "There may be communication noise in this game. The amounts you see may differ slightly from what was actually sent/returned."

So my co-player saw a higher amount than what I actually sent due to noise. I sent \$3.50, but they saw \$4.35. When tripled:  $\$4.35 * 3 = \$13.05$ .

They returned \$8.0 out of \$13.05. That's about 61.4% return rate.

Let me calculate their actual payoffs:

- They started with \$5
- I actually sent \$3.50, so it became \$10.50 for them
- They returned \$8.0
- So they should have kept  $\$10.50 - \$8.0 = \$2.50$
- But the visible payoff shows \$5.05

Wait, that doesn't match. Let me recalculate:

- If they saw \$4.35 sent and returned \$8.0
- Their visible payoff is \$5.05
- So:  $\$5 + \text{received} - \text{returned} = \text{payoff}$
- $\$5 + \$13.05 - \$8.0 = \$10.05$ ... that doesn't match \$5.05
- Oh, I see-the receiver's payoff is just what they kept from the tripled amount:  $\$13.05 - \$8.0 = \$5.05$ . That matches. So when acting as receiver, the payoff calculation doesn't include the initial endowment, just the net from the tripled transfer.

For Round 3, my co-player sent me \$4.67-that's 93.4% of their endowment, showing substantial trust. The tripled amount gives me \$14.01 to work with, while they're left with just \$0.33. Looking at their pattern: they sent 60% as a sender in Round 1, and now they're sending nearly all of it in Round 3.

If I return \$8.0, I'd give them \$8.33 total (a \$3.33 gain), keeping \$6.01 for myself-matching their Round 2 return rate. But their escalated commitment this round suggests I should reciprocate more substantially. Returning \$9.0 instead would give them \$9.33 (a \$4.33 gain) while I keep \$5.01, better reflecting the heightened trust they've demonstrated..

..

**usage:**

```
{
 "input_tokens": 1170,
 "output_tokens": 1644,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.05, returned \$8.0, their payoff was \$5.05

d \$8.0, visible payoff \$5.0499999999999999.  
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.  
Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the pattern with my co-player:

**\*\*Their behavior:\*\***

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

**\*\*My decision:\*\***

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

{'send': 4.00}

**reasoning:**

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

**usage:**

```
{
 "input_tokens": 1613,
 "output_tokens": 523,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.0499999999999999.

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

**\*\*Decision:\*\***

Their trust level deserves fair reciprocation.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

{ 'return': 6.00 }

**reasoning:**

Let me analyze the situation:

**\*\*Current situation:\*\***

- My co-player sent \$3.06 (61.2% of their endowment)
- They have \$1.94 remaining
- I received \$9.18

**\*\*Their recent behavior as receiver:\*\***

- Round 2: returned \$8.0 out of \$13.05 (61%)
- Round 4: returned \$7.0 out of \$10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

**\*\*My reciprocation pattern:\*\***

Looking at what I've done when receiving:

- Round 1: returned \$4.00 out of \$8.34
- Round 3: returned \$9.00 out of \$14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return \$5.50:

- Sender ends with: \$1.94 + \$5.50 = \$7.44 (profit of \$2.44)
- I keep: \$9.18 - \$5.50 = \$3.68

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. \$6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2029,
 "output_tokens": 565,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.  
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.  
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.  
Your total visible earnings across all rounds are \$31.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history:

**\*\*Analysis:\*\***

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 3 and 5 - very high trust  
- As receiver in Round 4, they returned \$7.0 out of \$10.5 (67%) - fair reciprocation  
- They maintain high trust even when returns vary

**\*\*Decision:\*\***

This co-player demonstrates strong, consistent trust. I should match their commitment level to maintain our cooperative relationship.

Sending \$4.00 (80%) matches their trust pattern and maximizes our joint potential gains.

{'send': 4.00}

**reasoning:**

Looking at my co-player's history:

Round 3: As sender, sent \$4.0 (80%), saw \$8.77 returned, payoff \$9.77

Round 4: As receiver, saw \$3.5 sent, returned \$7.0 out of \$10.5 (67%), payoff \$3.5

Round 5: As sender, sent \$4.0 (80%), saw \$6.2 returned, payoff \$7.2

Pattern analysis:

- As sender: Consistently sending \$4.0 (80%) - very high trust  
- As receiver: Returning 67% - fair reciprocation  
- They're maintaining high trust despite varying returns

This co-player has been consistently trusting. They've sent 80% twice in a row. As receiver, they returned a decent amount (67%).

I should match their high trust level. They're showing strong cooperative behavior, and I should reciprocate that trust.

Sending \$4.00 (80%) would match their pattern and continue our cooperation.

**usage:**

```
{
 "input_tokens": 2424,
 "output_tokens": 394,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.  
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.  
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.9600000000000001.  
Your total visible earnings across all rounds are \$41.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76
- As receiver, they've returned 62-67% consistently, showing fair reciprocation

**\*\*Decision:\*\***

Their high trust level warrants strong reciprocation.

If I return \$7.50:

- Sender ends with:  $\$1.08 + \$7.50 = \$8.58$  (profit of \$3.58)
- I keep:  $\$11.76 - \$7.50 = \$4.26$

This rewards their trust with a solid profit while maintaining fairness in our ongoing cooperation.

{'return': 7.50}

**reasoning:**

Looking at the history with my co-player:

**\*\*Their pattern:\*\***

- Round 4 as RECEIVER: returned \$7.0 out of \$10.5 (67%)
- Round 5 as SENDER: sent \$4.0 (80%)
- Round 6 as RECEIVER: returned \$8.0 out of \$12.96 (62%)

**\*\*Current situation:\*\***

- They sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76

**\*\*Their behavior as receiver:\*\***

- They've been consistently fair, returning 62-67% of what they receive
- This has been good reciprocation

**\*\*My decision:\*\***

They showed high trust by sending \$3.92 (78.4%). I should reward this appropriately.

If I return \$7.50:

- Sender ends with:  $\$1.08 + \$7.50 = \$8.58$  (profit of \$3.58)
- I keep:  $\$11.76 - \$7.50 = \$4.26$

This is a fair return that rewards their trust with a good profit, similar to what they've done for me.

Actually, let me recalculate to be consistent with their pattern. They return around 62-67%. If I do similar:

- 65% of \$11.76 = \$7.64

But I also want to ensure they profit well from their trust. Let me aim for \$7.50-\$8.00.

I'll go with \$7.50 as a good balance.

**usage:**

```
{
 "input_tokens": 2806,
 "output_tokens": 558,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible

le payoff \$4.960000000000001.  
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.  
Your total visible earnings across all rounds are \$45.80999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the pattern:

**\*\*Analysis:\*\***

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 5, 6, and 7 - unwavering high trust
- As receiver in Round 6, they returned \$8.0 out of \$12.96 (61.7%) - fair reciprocation
- They've maintained high trust throughout our partnership

**\*\*Decision:\*\***

This co-player shows exceptional consistency and trust. I should match their commitment level.

Sending \$4.00 (80%) aligns with their pattern and maintains our strong cooperative relationship.

{'send': 4.00}

**reasoning:**

Looking at my co-player's history:

**\*\*Their behavior as sender:\*\***

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)

**\*\*Their behavior as receiver:\*\***

- Round 6: returned \$8.0 out of \$12.96 (61.7%)

**\*\*Pattern analysis:\*\***

- They consistently send high amounts (\$4.0 = 80%) as sender, showing strong trust
- As receiver, they return fairly (about 62%)
- This is a very cooperative player who maintains high trust levels

**\*\*My decision:\*\***

We've established an excellent cooperative relationship. They've sent \$4.0 repeatedly, showing unwavering trust. I should match this trust level to maintain our mutually beneficial pattern.

Sending \$4.00 (80% of my endowment) continues our successful cooperation and maximizes joint gains.

**usage:**

```
{
 "input_tokens": 2976,
 "output_tokens": 361,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.
- Round 8 against you, as RECEIVER: saw \$4.14 sent, it became \$12.419999999999998, returned \$8.0, visible payoff \$4.419999999999998.

Your total visible earnings across all rounds are \$54.48999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.44 (88.8% of their endowment) - exceptionally high trust
- They have \$0.56 remaining
- I received \$13.32
- As receiver, they've consistently returned around \$8.0 (62-64% of what they received)

**\*\*Decision:\*\***

Their near-maximum trust deserves strong reciprocation.

If I return \$8.50:

- Sender ends with:  $\$0.56 + \$8.50 = \$9.06$  (profit of \$4.06)
- I keep:  $\$13.32 - \$8.50 = \$4.82$

This rewards their exceptional trust with a solid profit while maintaining our cooperative pattern.

{ 'return': 8.50 }

**Reasoning:**

Looking at the situation:

**\*\*Analysis:\*\***

- My co-player sent \$4.44 (88.8% of their endowment) - very high trust
- They have \$0.56 remaining
- I received \$13.32
- As receiver, they've returned \$8.0 consistently (around 62-64% of what they received)

**\*\*Decision:\*\***

Their exceptionally high trust (88.8%) deserves strong reciprocation.

If I return \$8.50:

- Sender ends with:  $\$0.56 + \$8.50 = \$9.06$  (profit of \$4.06)
- I keep:  $\$13.32 - \$8.50 = \$4.82$

This rewards their very high trust with a solid profit while maintaining fairness.

Alternatively, if I return \$9.00:

- Sender ends with:  $\$0.56 + \$9.00 = \$9.56$  (profit of \$4.56)
- I keep:  $\$13.32 - \$9.00 = \$4.32$

Given they sent 88.8% (nearly maximum trust), I should be generous. Let me return \$8.50, which gives them a good profit.

Usage: {

  "input\_tokens": 3101,  
  "output\_tokens": 474,  
  "reasoning\_tokens": 0

}

**GAME - Agent\_7 - Sender - dyad\_4**

Timestamp: 2026-08-12T08:05:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

  "agent\_type": "standard",  
  "dyad\_id": "dyad\_4",  
  "move\_index": 6,  
  "opponent\_id": "Agent\_8",  
  "pairing": {  
    "agents": [  
      "Agent\_7",  
      "Agent\_8"  
    ],  
    "investor": "Agent\_7",  
    "roles": {  
      "Agent\_7": "investor",  
      "Agent\_8": "trustee"  
    }  
  },

```

"trustee": "Agent_8"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.

Your total visible earnings across all rounds are \$13.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}



**Message 3 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Pattern Analysis:\*\***

- Round 1: They returned \$7 out of \$10.95 (generous, ~64%)
- Round 2: They sent \$3.5 showing trust; I returned \$7 (they saw \$6.52)
- Both rounds show strong cooperation

**\*\*Current Decision:\*\***

This co-player has been consistently cooperative and trustworthy. They've demonstrated:

- Generous returns as receiver
- High trust as sender
- Good reciprocal behavior

I should maintain this cooperative relationship by sending a similar amount to what they sent last round.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- I sent \$3 (they saw \$3.65 due to noise)
- They received \$10.95, returned \$7.0
- This was generous (64% return)

**\*\*Round 2:\*\***

- They sent \$3.5 (I saw \$4.0 due to noise)
- I received \$12.0, returned \$7
- They saw \$6.52 returned (due to noise)

Looking at Round 2 from their perspective:

- They sent \$3.5, kept \$1.5
- They saw \$6.52 returned
- So they ended with  $\$1.5 + \$6.52 = \$8.02$

From my actual perspective:

- They sent \$3.5 (I saw \$4), I kept \$1
- I returned \$7 (they saw \$6.52)
- I ended with  $\$12 - \$7 = \$5$

**\*\*Pattern Analysis:\*\***

- This co-player has been cooperative
- They've been generous as receiver (returned \$7)
- They've been trusting as sender (\$3.5)
- The noise is creating some discrepancies in what we see

**\*\*Current Decision (Round 3, I'm SENDER):\*\***

Given that:

1. They've been consistently cooperative
2. They returned generously when I sent \$3
3. They trusted me with \$3.5 when they were sender
4. We've established a good cooperative pattern

I should continue the cooperation. Sending a similar or slightly higher amount would signal continued trust. Given they sent \$3.5 last round, I could send around \$3-4.

I'll send \$3.5 to match their previous send amount, showing reciprocal trust.

**usage:**

```
{
 "input_tokens": 1141,
 "output_tokens": 570,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.9499999999999993.
  - Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
  - Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Cooperation Pattern:\*\***

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

**\*\*Current Situation:\*\***

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

**\*\*My Decision:\*\***

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

**reasoning:**

Let me analyze the history with this co-player:

**\*\*Round 1:\*\***

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

**\*\*Round 2:\*\***

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)

**\*\*Round 3:\*\***

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

**\*\*Round 4 (current):\*\***

- They are SENDER, sent \$4.12, I received \$12.36
- They kept \$0.88

**\*\*Pattern:\*\***

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did:  $0.65 * \$12.36 = \$8.03$

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88
- If I return \$8, they get \$8.88 total, I keep \$4.36
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

**usage:**

```
{
 "input_tokens": 1540,
 "output_tokens": 565,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$28.240000000000002.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Established Pattern:\*\***

- Round 2: They sent \$3.5, trusted me
- Round 3: They returned \$8 generously
- Round 4: They sent \$4 (high trust), I returned \$8

**\*\*Analysis:\*\***

This co-player has been consistently cooperative:

- Always returns generously as receiver (\$7-8)
- Shows increasing trust as sender (\$3.5 ? \$4)
- We've built a strong reciprocal relationship

**\*\*Current Decision:\*\***

They sent \$4 last round (80% of endowment), showing strong trust. I should match this level of trust to maintain our cooperative dynamic.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 2: They sent \$3.5, I returned ~\$7 (they saw \$6.52 due to noise)

Round 3: I sent ~\$3.5 (they saw \$4.11), they returned \$8

Round 4: They sent \$4 (I saw \$4.12), I returned \$8 (they saw \$7.68)

This co-player has been:

- Consistently generous as receiver (returned \$7, then \$8)
- Trusting as sender (sent \$3.5, then \$4)
- Maintaining high cooperation levels

The pattern shows escalating trust and cooperation. They sent \$4 last time, which is 80% of their endowment.

I should continue this cooperative trajectory. Since they've been so consistently cooperative and generous, I should send a similar or slightly higher amount to maintain the positive relationship.

I'll send \$4 - matching their last send amount and showing equal trust.

**usage:**

```
{
 "input_tokens": 1974,
 "output_tokens": 404,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.

Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 3: Returned \$8/\$12.33 = 65%
- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 6 (now): Sent \$4.4 (88% of endowment!)

**\*\*Current Situation:\*\***

- They sent \$4.4, keeping only \$0.6
- I received \$13.2
- This shows exceptional trust - their highest send yet

**\*\*My Decision:\*\***

They've consistently returned 65-69% and are now showing maximum trust. I should match their generosity pattern and reward this trust.

Returning \$9:

- They end with \$9.6 total
- I keep \$4.2
- ~68% return rate, matching their established pattern
- Sustains our strong cooperation

```
'''json
{"return": 9}
'''
```

**reasoning:**

Looking at the history with this co-player:

**\*\*Their behavior:\*\***

- Round 3: Returned \$8 out of \$12.33 (65%)
- Round 5: Returned \$7.5 out of \$10.86 (69%)

**\*\*Current situation:\*\***

- They sent \$4.4 (88% of their endowment!) - showing very high trust
- I received \$13.2
- They kept only \$0.6

**\*\*Analysis:\*\***

This co-player is showing increasing trust and generosity:

- They've returned 65-69% as receiver
- As sender, they're now sending even more (\$4.4 vs \$4 previously)
- This shows they believe in our cooperation

**\*\*My decision:\*\***

I should return proportionally to their generosity pattern. They've been returning around

65-69%.

If I return \$8.5:

- They end with \$9.1 total (\$0.6 + \$8.5)
- I keep \$4.7
- Return rate: 64% (consistent with their pattern)

If I return \$9:

- They end with \$9.6 total
- I keep \$4.2
- Return rate: 68% (matches their higher range)

Given they sent \$4.4 (very high), I should be generous back. Let me return \$8.5 to maintain good cooperation while keeping the return rate reasonable.

Actually, thinking about it more - they sent \$4.4 out of \$5 (88%). This is extremely trusting. They've been consistently generous. I should match that generosity.

Return \$9:

- They get \$9.6 total (\$0.6 kept + \$9 returned)
- I keep \$4.2
- This is about 68% return rate, matching their generosity pattern
- Rewards their exceptional trust

usage:

```
{
 "input_tokens": 2394,
 "output_tokens": 644,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.

Your total visible earnings across all rounds are \$41.81.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver (generous)
- Round 6: Sent \$4 as sender again

**\*\*Analysis:\*\***

This co-player has been exceptionally consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns generously when receiver (\$7.5-\$8)
- We've built strong mutual trust over 6 rounds

**\*\*Current Decision:\*\***

They consistently send \$4, showing strong trust. I should match this behavior to maintain our excellent cooperative relationship.

```
'''json
{"send": 4}
'''
```

**reasoning:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver
- Round 6: Sent \$4 as sender (I returned \$9, they saw \$9.22)

**\*\*Analysis:\*\***

This co-player has been:

- Consistently sending \$4 when they're sender
- Returning generously (\$7.5-\$8) when they're receiver
- Very trustworthy and cooperative throughout

**\*\*Current Decision:\*\***

They've been sending \$4 consistently. I should match this level of trust to maintain our excellent cooperative relationship. We've built strong mutual trust over 6 rounds.

I should send \$4 to match their behavior and maintain our reciprocal pattern.

**usage:**

```
{
 "input_tokens": 2827,
 "output_tokens": 351,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.
- Round 7 against you, as RECEIVER: saw \$4.84 sent, it became \$14.52, returned \$10.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$52.480000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

**Message 13 role=assistant**

**content:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 7: Returned \$10/\$14.52 = 69%
- Round 8 (now): Sent \$3.21 (lower than their usual \$4)

**\*\*Current Situation:\*\***

- They sent \$3.21, keeping \$1.79
- I received \$9.63
- This is their lowest send, but still shows trust

**\*\*My Decision:\*\***

They've consistently returned around 69% when receiver. I should match this cooperative pattern.

Returning \$6.5:

- They end with \$8.29 total
- I keep \$3.13
- ~67% return rate, close to their established pattern
- Maintains our strong cooperation despite the lower send

```
'''json
{"return": 6.5}
'''
```

**reasoning:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 5: Returned  $\$7.5/\$10.86 = 69\%$
- Round 7: Returned  $\$10/\$14.52 = 69\%$
- Round 8 (now): Sent  $\$3.21$  (64% of endowment)

**\*\*Current Situation:\*\***

- They sent  $\$3.21$ , keeping  $\$1.79$
- I received  $\$9.63$
- This is lower than their usual  $\$4$  sends

The lower send amount is notable. They've consistently sent  $\$4$  in previous rounds as sender, but now sent  $\$3.21$ . This could be:

1. Random variation
2. Testing behavior
3. The noise in communication

However, they've been very cooperative throughout, returning 69% consistently when receiver.

**\*\*My Decision:\*\***

I should continue the cooperative pattern. Their typical return rate is around 69%.

If I return  $\$6.5$ :

- They end with  $\$8$  total ( $\$1.79 + \$6.5$ )
- I keep  $\$3.13$
- ~67% return rate, matching their pattern
- Maintains cooperation

Actually, let me recalculate:

- They sent  $\$3.21$ , kept  $\$1.79$
- I received  $\$9.63$
- If I return  $\$6.5$ , they get  $\$1.79 + \$6.5 = \$8.29$  total
- I keep  $\$9.63 - \$6.5 = \$3.13$

That seems reasonable and maintains the cooperative relationship.

Alternatively, returning  $\$7$ :

- They get  $\$1.79 + \$7 = \$8.79$
- I keep  $\$2.63$
- ~73% return rate

Given their 69% return pattern, I'll aim for around  $\$6.5$ - $\$7$  to match that.

**usage:**

```
{
 "input_tokens": 3074,
 "output_tokens": 640,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent  $\$4.0$ , it became  $\$12.0$ , saw  $\$9.22$  returned, visible payoff  $\$10.22$ .
- Round 7 against you, as RECEIVER: saw  $\$4.84$  sent, it became  $\$14.52$ , returned  $\$10.0$ , visible payoff  $\$4.52$ .
- Round 8 against you, as SENDER: sent  $\$4.0$ , it became  $\$12.0$ , saw  $\$6.07$  returned, visible payoff  $\$7.07$ .

Your total visible earnings across all rounds are  $\$55.61$ .

This round, you are the SENDER against your current co-player. You have  $\$5$ . How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Assistant output:**

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 6: Sent  $\$4$  as sender

- Round 7: Returned \$10 generously as receiver (69%)
- Round 8: Sent \$4 as sender (despite seeing lower return)

**\*\*Analysis:\*\***

This co-player has been exceptionally consistent:

- Always sends \$4 when sender (high trust)
- Returns very generously when receiver (~69%)
- Continues cooperating even after seeing lower returns in Round 8

**\*\*Current Decision:\*\***

They've maintained their \$4 send despite the lower return they saw last round. This shows commitment to our cooperative relationship. I should match this trust and continue sending \$4.

```
'''json
{"send": 4}
'''
```

#### Reasoning:

Looking at the history with this co-player:

**\*\*Their Pattern:\*\***

- Round 6: Sent \$4 as sender
- Round 7: Returned \$10/\$14.52 = 69% as receiver (very generous)
- Round 8: Sent \$4 as sender (but I only returned \$6.5, they saw \$6.07 due to noise)

**\*\*Analysis:\*\***

This co-player has been extremely consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns very generously when receiver (69-70%)
- We've built a strong relationship over 8 rounds

**\*\*Current Decision:\*\***

Despite my slightly lower return in Round 8 (I returned \$6.5, they saw \$6.07), they have continued to show strong cooperation. They:

- Send \$4 consistently
- Return generously (69-70%)
- Show high trust

I should continue to send \$4 to match their trust and maintain this cooperative relationship. The slight dip in my return in Round 8 shouldn't break the pattern - they've been so consistently cooperative overall.

```
'''json
{"send": 4}
'''
```

```
Usage: {
 "input_tokens": 3169,
 "output_tokens": 449,
 "reasoning_tokens": 0
}
```

#### GAME - Agent\_8 - Receiver - dyad\_4

Timestamp: 2026-08-12T08:05:43

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
 "agent_type": "standard",
 "dyad_id": "dyad_4",
 "move_index": 7,
 "opponent_id": "Agent_7",
 "pairing": {
 "agents": [
 "Agent_7",
 "Agent_8"
],
 "investor": "Agent_7",
 "roles": {
 "Agent_7": "investor",
 "Agent_8": "trustee"
 },
 "trustee": "Agent_8"
 }
```



```

},
"role": "trustee",
"role_label": "Receiver",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 3

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$11.969999999999999.

This round, you are the RECEIVER against your current co-player. They sent you \$4.11, so you received \$12.330000000000002. How much do you return? (0-12.330000000000002)

Respond exactly as JSON: {'return': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the situation:

- My co-player has shown very strong trust by sending \$4.11 (82.2% of their endowment)
- This is even higher than their previous cooperation levels
- I received \$12.33
- The sender currently has only \$0.89 remaining

In Round 2, when they received ~\$12.0, they returned \$7.0 to me, showing good reciprocity. Now they've shown even stronger trust.

I want to reward this escalating cooperation and maintain our positive relationship. Returning \$8.00 would give:

- Sender:  $\$0.89 + \$8.00 = \$8.89$
- Me:  $\$12.33 - \$8.00 = \$4.33$

This appropriately rewards their high trust while keeping the outcome fair for both of us.

```
'''json
{"return": 8.00}
'''
```

**reasoning:**

Let me analyze the situation:

Round 1: I was receiver

- Co-player (as sender) sent \$3.0 (60%)
- They saw \$6.92 returned (though I actually returned \$7.00)
- Their visible payoff was \$8.92

Round 2: I was sender

- I sent \$3.50 (they saw \$4.0 due to noise)
- They received \$12.0 (actually I sent \$3.50 which became \$10.50)
- They returned \$7.0
- Their visible payoff was \$5.0

Round 3 (current): I'm receiver

- Co-player sent \$4.11 (82.2% of their endowment)
- I received \$12.33

My co-player has shown:

1. Strong initial trust (60% in Round 1)
2. Good reciprocation (\$7.0 return in Round 2)
3. Even stronger trust now (82.2% in Round 3)

They're escalating cooperation! I should reciprocate proportionally.

Looking at Round 2, when they received ~\$12.0, they returned \$7.0 (58.3% of what they received).

Now I received \$12.33. If I use a similar return rate around 58-60%, that would be \$7.20-\$7.40.

But they've sent even more this round (82.2% vs ~60% in previous rounds), showing increased trust. I should reward this.

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$4.11 = \$0.89$
- I have: \$12.33

If I return \$7.50:

- Sender ends with:  $\$0.89 + \$7.50 = \$8.39$

...

**usage:**

```
{
 "input_tokens": 1217,
 "output_tokens": 772,
 "reasoning_tokens": 0
}
```

Message 4 role=user

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 5 role=assistant**

**content:**

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92

Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0

Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the returns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

**usage:**

```
{
 "input_tokens": 1638,
 "output_tokens": 388,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

**reasoning:**

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86
- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender: \$1.38 + \$7.50 = \$8.88
- Me: \$10.86 - \$7.50 = \$3.36

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

**usage:**

```
{
 "input_tokens": 2006,
 "output_tokens": 390,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.

Your total visible earnings across all rounds are \$28.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 9 role=assistant**

**content:**

Looking at our ongoing cooperative relationship:

My co-player has been consistently trustworthy:

- Round 3: sent \$3.5 (70%)
- Round 5: sent \$4.0 (80%) - showing increasing trust
- Round 4 as receiver: returned \$8.0 from \$12.36

They've demonstrated strong reciprocity and increased their trust level over time. I should match this escalating cooperation.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the history:

My co-player has consistently shown strong trust:

- Round 3 as sender: sent \$3.5 (70%)
- Round 5 as sender: sent \$4.0 (80%)
- Round 4 as receiver: returned \$8.0 when they received \$12.36

They've received good returns (\$8.46, \$8.37) and continue to maintain high cooperation levels.

This is clearly a very cooperative partner. I should continue sending substantial amounts to maintain this mutually beneficial relationship.

I've sent:

- Round 2: \$3.50
- Round 4: \$4.00

I should maintain or even slightly increase my sending to show continued trust, especially since they increased from \$3.5 to \$4.0.

I'll send \$4.00 to maintain strong cooperation.

**usage:**

```
{
 "input_tokens": 2357,
 "output_tokens": 313,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.3599999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.

Your total visible earnings across all rounds are \$38.56.

This round, you are the RECEIVER against your current co-player. They sent you \$4.84, so you received \$14.52. How much do you return? (0-14.52)

Respond exactly as JSON: {'return': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - this is exceptional trust!
- I received \$14.52
- They currently have only \$0.16 remaining

This is the highest level of trust they've shown in our partnership. They've been consistently cooperative and have escalated their trust over time (\$3.5 ? \$4.0 ? \$4.84).

I need to strongly reward this exceptional trust to maintain our cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - extremely high trust!
- I received \$14.52
- They have only \$0.16 remaining

Our history shows strong mutual cooperation:

- As receiver, they've returned \$8.0 and \$9.0 in recent rounds
- As sender, they've been escalating trust: \$3.5 ? \$4.0 ? \$4.84 (increasingly high percentages)

This is the highest level of trust they've shown yet - sending nearly all their endowment.  
I should strongly reward this trust.

If I return \$10.00:

- Sender: \$0.16 + \$10.00 = \$10.16
- Me: \$14.52 - \$10.00 = \$4.52

This gives them a good payoff that rewards their exceptional trust while still leaving me with a reasonable gain.

Given they sent such a high amount, I should return proportionally more. Let me think about what's fair:

- They sent \$4.84, they have \$0.16
- I received \$14.52
- Total created: \$0.16 + \$14.52 = \$14.68

A fair split might be around equal final payoffs or better for the sender given their huge trust.

If I return \$10.00:

- Sender gets \$10.16
- I get \$4.52

Total: Both benefit significantly

If I return \$9.50:

- Sender gets \$9.66
- I get \$5.02

Given their extremely high trust (96.8%), I think \$10.00 is appropriate to strongly reward this behavior and maintain the cooperative relationship.

**usage:**

```
{
 "input_tokens": 2711,
 "output_tokens": 534,
 "reasoning_tokens": 0
}
```

**Message 12 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.

Your total visible earnings across all rounds are \$43.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 13 role=assistant**

**content:**

Looking at our strong cooperative history:

My co-player has been exceptionally trustworthy:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0, showing excellent reciprocity

They've received good returns (\$8.37, \$9.67) and maintained high trust levels. This is a highly cooperative partnership.

For this final round, I'll continue our established pattern of mutual trust.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at our history:

My co-player has been extremely cooperative:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0 from \$13.20

They've consistently sent high amounts and showed excellent reciprocity as receivers. In Round 7, they saw \$9.67 returned and had a payoff of \$10.67, which is very good.

This is clearly a highly cooperative and trustworthy partner. We've built a strong mutual trust relationship throughout these rounds.

For the final round (Round 8), I should continue this high level of cooperation. Given their consistent trust at the 80%+ level, I should send a similar proportion.

I'll send \$4.00 to maintain the strong cooperative relationship we've built.

**usage:**

```
{
 "input_tokens": 2794,
 "output_tokens": 332,
 "reasoning_tokens": 0
}
```

**Message 14 role=user**

**content:**

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.
- Round 8 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$6.5, visible payoff \$3.129999999999999.

Your total visible earnings across all rounds are \$50.15.

This round, you are the RECEIVER against your current co-player. They sent you \$4.77, so you received \$14.309999999999999. How much do you return? (0-14.309999999999999)

Respond exactly as JSON: {'return': <amount>}

**Assistant output:**

Looking at the situation:

- My co-player sent \$4.77 (95.4% of their endowment) - extremely high trust!
- I received \$14.31
- They currently have only \$0.23 remaining

Our partnership has been consistently cooperative, and they've shown exceptional trust in this round. I should reciprocate strongly.

```
'''json
{"return": 10.00}
'''
```

**Reasoning:**

Looking at the history:

My co-player sent \$4.77 (95.4% of their endowment) - very high trust!  
I received \$14.31  
They currently have only \$0.23 remaining

Our history shows strong cooperation:  
- Round 7: sent \$4.0 (80%)  
- Round 8 as receiver: returned \$6.5 from \$9.63  
- Now: sent \$4.77 (95.4%)

They've shown excellent cooperation and trust. I should reciprocate generously.

Returning around \$10.00 would give:  
- Sender: \$0.23 + \$10.00 = \$10.23  
- Me: \$14.31 - \$10.00 = \$4.31

This rewards their high trust appropriately.

```
Usage: {
 "input_tokens": 2901,
 "output_tokens": 289,
 "reasoning_tokens": 0
}
```

## ROUND 10

Roles: Agent\_1=Receiver, Agent\_2=Sender, Agent\_3=Receiver, Agent\_4=Sender, Agent\_5=Receiver,  
Agent\_6=Sender, Agent\_7=Receiver, Agent\_8=Sender

### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver  
dyad\_2: Agent\_4=Sender, Agent\_3=Receiver  
dyad\_3: Agent\_6=Sender, Agent\_5=Receiver  
dyad\_4: Agent\_8=Sender, Agent\_7=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 sent \$5; received \$15; Agent\_1 returned \$9.2; payoffs Agent\_2=\$9.2, Agent\_1=\$5.8;  
noise/visible: sent visible=\$4.58, received visible=\$13.74, returned visible=\$9.83  
Action validation: {"returned": {"amount": 9.2, "clamped": false, "max": 15.0, "min": 0, "raw": 9.2}, "sent": {"amount": 5.0, "clamped": false, "max": 5, "min": 0, "raw": 5.0}}  
dyad\_2: Agent\_4 sent \$4; received \$12; Agent\_3 returned \$5.75; payoffs Agent\_4=\$6.75,  
Agent\_3=\$6.25; noise/visible: sent visible=\$3.76, received visible=\$11.28, returned visible=\$5.21  
Action validation: {"returned": {"amount": 5.75, "clamped": false, "max": 12.0, "min": 0, "raw": 5.75}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_3: Agent\_6 sent \$4; received \$12; Agent\_5 returned \$8; payoffs Agent\_6=\$9, Agent\_5=\$4;  
noise/visible: returned visible=\$8.99  
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}  
dyad\_4: Agent\_8 sent \$4; received \$12; Agent\_7 returned \$8; payoffs Agent\_8=\$9, Agent\_7=\$4;  
noise/visible: sent visible=\$3.9, received visible=\$11.7, returned visible=\$8.85  
Action validation: {"returned": {"amount": 8.0, "clamped": false, "max": 12.0, "min": 0, "raw": 8.0}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

### Balances after round: {

"Agent\_1": 75.3,  
"Agent\_2": 61.7,  
"Agent\_3": 58.8,  
"Agent\_4": 61.2,  
"Agent\_5": 60.5,  
"Agent\_6": 66.5,  
"Agent\_7": 69.0,  
"Agent\_8": 57.0

}

### Visible balances after round: {

"Agent\_1": 68.04,  
"Agent\_2": 67.86,  
"Agent\_3": 57.26,  
"Agent\_4": 56.540000000000006,  
"Agent\_5": 63.72,  
"Agent\_6": 69.3,  
"Agent\_7": 69.42,  
"Agent\_8": 64.30999999999999

}



## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T08:05:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.

- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

**Message 2 role=user**

**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.1 returned, visible payoff \$8.1.
- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.

Your total visible earnings across all rounds are \$14.92.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 3 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have been highly cooperative:

- Round 1: Sent \$3.0 (60%), received \$6.1 back
- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0 (80%), received \$8.5 back

This co-player has demonstrated:

1. Willingness to send high amounts and trust
2. Fair reciprocation as a receiver
3. Consistent cooperation leading to mutual gains

They've been escalating their trust (from \$3.0 to \$4.0) and it's been rewarded. I should reciprocate with similarly high trust to maintain this cooperative relationship.

```
'''json
{"send": 4.00}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

Round 1: They sent \$3.0 as SENDER, received \$6.1 back, visible payoff \$8.1

Round 2: They returned \$5.0 as RECEIVER (when they saw \$2.61 sent, became \$7.83), visible payoff \$2.83

Round 3: They sent \$4.0 as SENDER, received \$8.5 back, visible payoff \$9.5

This shows:

- They're willing to send significant amounts (60-80% of endowment)
- They've been fairly generous as receiver (returned \$5.0 from \$7.83)
- They've been getting good returns on their trust

My earnings so far: \$14.92

Round 1: I returned \$7.00, kept \$3.56

Round 2: I sent \$3.50, received back (I need to calculate)

- I sent \$3.50, became \$10.50
- They returned \$5.0 (visible to me as different due to noise)
- I kept: \$5 - \$3.50 + return = \$1.50 + return

Round 3: I returned \$9.00, kept \$4.02

So my earnings: \$3.56 + (\$1.50 + what I got back in round 2) + \$4.02 = \$14.92

This means: \$9.08 + what I got back in round 2 = \$14.92

So what I got back in round 2 = \$5.84

This co-player has been very trustworthy and cooperative. They've sent high amounts and returned fair amounts. They've consistently shown trust. I should reciprocate with high trust as well.

Given the pattern of escalating trust, I should send a substantial amount. They sent \$4.0 last round (80% of endowment) and got \$8.5 back. I should show similar or greater trust.

...

**usage:**

```
{
 "input_tokens": 1669,
 "output_tokens": 673,
 "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.61 sent, it became \$7.83, returned \$5.0, visible payoff \$2.83.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.

Your total visible earnings across all rounds are \$22.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.97, so you received \$14.91. How much do you return? (0-14.91)

Respond exactly as JSON: {'return': <amount>}

**Message 5 role=assistant**

**content:**

Looking at this situation:

My co-player has shown extreme trust by sending \$4.97 (99.4% of their endowment), leaving themselves with only \$0.03. This is the maximum possible cooperation.

The visible history shows consistent cooperation:

- Round 2: Returned \$5.0 from \$7.83 (generous)
- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.97 (nearly everything!)

The multiplication created \$9.94 in added value. To strongly reciprocate this extreme trust:

If I return \$10.00:

- Sender ends with: \$0.03 + \$10.00 = \$10.03 (profit of \$5.03)
- I keep: \$14.91 - \$10.00 = \$4.91 (profit of \$4.91)

This rewards their maximum trust with substantial mutual gains.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at the history with this co-player:

Round 2: They returned \$5.0 from \$7.83 (generous)  
Round 3: They sent \$4.0, received \$8.5 back  
Round 4: They returned \$6.0 from \$9.42 (generous)  
Round 5 (current): They sent \$4.97 (99.4% of their endowment!)

This co-player has been extremely cooperative and trustworthy:

- They send high amounts consistently
- They return fair/generous amounts as receiver
- Now they've sent almost everything (\$4.97 out of \$5)

The sender is left with only \$0.03. This is an incredible show of trust.

The multiplication created \$9.94 in added value (\$14.91 - \$4.97).

I need to reciprocate this extreme trust. They've shown maximum cooperation.

If I return \$10.00:

- Sender ends with:  $\$0.03 + \$10.00 = \$10.03$  (profit of \$5.03)
- I keep:  $\$14.91 - \$10.00 = \$4.91$  (profit of \$4.91)

This roughly splits the gains equally and rewards their extreme trust.

Actually, looking more carefully - they sent \$4.97, leaving them with \$0.03. If I return \$10.00:

- They get \$10.03 total (doubled their initial endowment)
- I get \$4.91 (profit of \$4.91)

The gains are somewhat unequal but I'm rewarding their massive trust. This seems fair.

Alternatively, if I return \$9.50:

- Sender:  $\$0.03 + \$9.50 = \$9.53$  (profit of \$4.53)
- Me:  $\$14.91 - \$9.50 = \$5.41$  (profit of \$5.41)

...

**usage:**

```
{
 "input_tokens": 2083,
 "output_tokens": 771,
 "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.5 returned, visible payoff \$9.5.
- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.

Your total visible earnings across all rounds are \$27.52.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 7 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have been highly cooperative throughout:

- Round 3: Sent \$4.0, received \$8.5 back
- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (strong return)

This co-player has demonstrated:

1. High trust as sender (escalating from \$4.0 to \$4.5)
2. Fair/generous reciprocation as receiver
3. Consistent success in our cooperative relationship

They've been rewarded well for their trust and have reciprocated fairly. I should continue this strong cooperation by matching their recent trust level.

```
'''json
{"send": 4.50}
'''
```

**reasoning:**

Looking at the visible history with my co-player:

- Round 3: They sent \$4.0, received \$8.5 back (very good return)
- Round 4: They returned \$6.0 from \$9.42 (generous)

- Round 5: They sent \$4.5, received \$9.44 back (excellent return)

This co-player has been:

1. Highly trusting as sender (sent \$4.0, then \$4.5)
2. Generous as receiver (returned \$6.0 from \$9.42)
3. Consistently rewarded with good returns

The relationship has been very cooperative and mutually beneficial. They've been sending high amounts and getting good returns. As a receiver, they've been fair/generous.

Given this strong cooperative history, I should send a high amount to continue this mutually beneficial relationship. They sent \$4.5 in round 5, so I should match or exceed that level of trust.

I'll send \$4.50 to match their recent trust level.

**usage:**

```
{
 "input_tokens": 2556,
 "output_tokens": 427,
 "reasoning_tokens": 0
}
```

**Message 8 role=user**

**content:**

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.14 sent, it became \$9.42, returned \$6.0, visible payoff \$3.42.
- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$38.76.

This round, you are the RECEIVER against your current co-player. They sent you \$4.81, so you received \$14.43. How much do you return? (0-14.43)

Respond exactly as JSON: {'return': <amount>}

**Message 9 role=assistant**

**content:**

Looking at this situation:

My co-player has shown very high trust by sending \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19.

The visible history shows excellent cooperation:

- Round 4: Returned \$6.0 from \$9.42
- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

They were extremely generous in Round 6, returning \$10.0 from \$15. Now they're showing continued high trust as a sender.

The multiplication created \$9.62 in added value. To reciprocate their strong cooperation:

If I return \$10.00:

- Sender ends with: \$0.19 + \$10.00 = \$10.19 (profit of \$5.19)
- I keep: \$14.43 - \$10.00 = \$4.43 (profit of \$4.43)

This maintains our successful cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

**reasoning:**

Looking at this situation:

My co-player sent \$4.81 (96.2% of their endowment), leaving themselves with only \$0.19. This is very high trust.

The visible history shows strong cooperation:

- Round 4: Returned \$6.0 from \$9.42 (generous)
- Round 5: Sent \$4.5, received \$9.44 back (good return)
- Round 6: Returned \$10.0 from \$15 (very generous!)
- Round 7: Sent \$4.81 (very high trust)

The multiplication created \$9.62 in added value (\$14.43 - \$4.81).

My co-player has been very generous as a receiver (returning \$6.0, then \$10.0) and trusting as a sender (sending \$4.5, then \$4.81).

To reciprocate this high trust and maintain the cooperative relationship:

If I return \$10.00:

- Sender ends with:  $\$0.19 + \$10.00 = \$10.19$  (profit of \$5.19)
- I keep:  $\$14.43 - \$10.00 = \$4.43$  (profit of \$4.43)

This is fair and rewards their very high trust. Given they returned \$10.0 from \$15 last round, returning \$10.00 from \$14.43 would be consistent with what they've shown.

Actually, let me think about this more carefully. They sent \$4.81, leaving \$0.19. The multiplication created \$14.43 for me, which is an added value of \$9.62.

If I return \$10.00, they get \$10.19 total (profit \$5.19), I keep \$4.43 (profit \$4.43). We both profit well.

Given the strong cooperative history and their very high trust (96.2% of endowment), \$10.00 seems appropriate.

usage:

```
{
 "input_tokens": 2963,
 "output_tokens": 735,
 "reasoning_tokens": 0
}
```

**Message 10 role=user**

**content:**

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$9.44 returned, visible payoff \$9.94.
- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.

Your total visible earnings across all rounds are \$43.19.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

**Message 11 role=assistant**

**content:**

Looking at the visible history with my co-player:

They have shown exceptional cooperation:

- Round 5: Sent \$4.5, received \$9.44 back
- Round 6: Returned \$10.0 from \$15 (extremely generous)
- Round 7: Sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has:

1. Escalated to maximum trust by sending their entire \$5 endowment
2. Been very generous as a receiver
3. Maintained consistent cooperation throughout

They've shown maximum trust and it's been rewarded. I should reciprocate with the same level of trust to complete our highly successful cooperative relationship.

```json

```
    {"send": 5.00}  
    '''
```

reasoning:

Looking at the visible history with my co-player:

- Round 5: They sent \$4.5, received \$9.44 back
- Round 6: They returned \$10.0 from \$15 (very generous)
- Round 7: They sent \$5.0 (maximum trust!), received \$9.6 back

This co-player has now shown maximum trust in Round 7 by sending their entire \$5 endowment . They have been:

1. Escalating their trust (from \$4.5 to \$5.0)
2. Very generous as a receiver (returned \$10.0 from \$15)
3. Consistently cooperative and successful

They sent the maximum (\$5.0) in Round 7 and received \$9.6 back, which is a good return. I should reciprocate with maximum trust as well to maintain this strong cooperative relationship.

This is the final round (Round 8), so I should send my full amount to maximize our joint payoff and complete our cooperative relationship on a strong note.

usage:

```
{  
  "input_tokens": 3154,  
  "output_tokens": 420,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.
- Round 8 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$53.03.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at this situation:

My co-player has sent the maximum amount of \$5 (100% of their endowment), showing complete trust.

The visible history shows a strong pattern:

- Round 6: They returned \$10.0 from \$15
- Round 7: Sent \$5.0 (maximum), received \$9.6 back
- Round 8: They returned \$10.0 from \$15
- Round 9: Sent \$5.0 (maximum trust again)

This co-player has consistently shown maximum trust as a sender and generous reciprocation as a receiver, establishing a clear pattern of returning \$10 from \$15.

The multiplication created \$10 in added value. To maintain our highly successful cooperative pattern:

If I return \$10.00:

- Sender ends with: \$0 + \$10.00 = \$10.00 (profit of \$5.00)
- I keep: \$15.00 - \$10.00 = \$5.00 (profit of \$5.00)

This matches their consistent behavior and splits the gains equally.

```
'''json
```

```
    {"return": 10.00}
    """
```

reasoning:

Looking at the visible history with my co-player:

- Round 6: They returned \$10.0 from \$15 (very generous)
- Round 7: Sent \$5.0 (maximum trust), received \$9.6 back
- Round 8: They returned \$10.0 from \$15 (very generous again)
- Round 9: Sent \$5.0 (maximum trust again)

This co-player has been showing maximum cooperation:

1. Sending the full \$5 endowment (maximum trust)
2. When receiving, returning \$10 out of \$15 consistently (very generous)

They've been incredibly cooperative throughout our relationship. Now they've sent their full \$5 again.

The multiplication created \$10 in added value from their \$5 send.

If I return \$10.00:

- Sender ends with: $\$0 + \$10.00 = \$10.00$ (profit of \$5.00)
- I keep: $\$15.00 - \$10.00 = \$5.00$ (profit of \$5.00)

This matches what they did in rounds 6 and 8, maintaining our pattern of strong mutual cooperation.

I should return \$10 to match their generous pattern.

usage:

```
{
  "input_tokens": 3293,
  "output_tokens": 561,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.6 returned, visible payoff \$9.6.
- Round 8 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.
- Round 9 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$10.11 returned, visible payoff \$10.11.

Your total visible earnings across all rounds are \$58.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history with my co-player:

They have shown consistent maximum cooperation:

- Round 7: Sent \$5.0 (maximum), received \$9.6 back
- Round 8: Returned \$10.0 from \$15 (generous)
- Round 9: Sent \$5.0 (maximum), received \$10.11 back

This co-player has:

1. Consistently sent their entire \$5 endowment as sender
2. Returned \$10 as receiver (fair split of gains)
3. Maintained maximum trust throughout our partnership

We've established an excellent cooperative relationship with maximum trust on both sides. I should continue with the same level of trust.

```
'''json
{"send": 5.00}
'''
```

Reasoning:

Looking at the visible history with my co-player:

- Round 7: Sent \$5.0, received \$9.6 back
- Round 8: Returned \$10.0 from \$15
- Round 9: Sent \$5.0, received \$10.11 back

This co-player has:

1. Consistently sent maximum amounts (\$5.0) when sender
2. Returned \$10.0 when receiver (generous, equal split of gains)
3. Been rewarded with good returns when sending

We've established a very strong pattern of maximum trust and fair reciprocation. They've shown maximum cooperation consistently, and it's been rewarded. I should continue with maximum trust to maintain this highly successful cooperative relationship.

This is our final round (Round 10), and we've built up excellent mutual trust. I should send the maximum to complete our cooperative run.

```
Usage: {
  "input_tokens": 3333,
  "output_tokens": 380,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T08:05:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.52 sent, it became \$10.56, returned \$7.0, visible payoff \$3.5600000000000005.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.

Your total visible earnings across all rounds are \$20.43.

This round, you are the RECEIVER against your current co-player. They sent you \$3.14, so you received \$9.42. How much do you return? (0-9.42)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the pattern with my co-player:

Round 1: They returned \$7.0 out of \$10.56 (66%)

Round 3: They returned \$9.0 out of \$13.02 (69%)

They've been extremely generous and cooperative, consistently returning about 2/3 of what they receive. They've also continued to send substantial amounts when in the sender role.

I should reciprocate this strong cooperative behavior. Returning around 60-65% of \$9.42 would be fair and maintain our excellent cooperation.

```
'''json
{"return": 6.0}
'''
```

reasoning:

Let me review the history with this co-player:

Round 1: They were RECEIVER, saw \$3.52 sent (I actually sent \$3), it became \$10.56, they returned \$7.0 - very generous (66% return rate)

Round 2: They were SENDER, sent \$3.5 (I saw \$2.61), it became \$10.5 for me, I returned \$5.0, they saw \$5.84 returned

Round 3: They were RECEIVER, saw \$4.34 sent (I actually sent \$4), it became \$13.02, they returned \$9.0 - extremely generous (69% return rate)

Round 4: They are SENDER, sent \$3.14 (probably sent around \$3 or a bit more), I received \$9.42

Pattern: This co-player has been consistently generous and cooperative. They've returned 66-69% of what they received in both rounds they were receiver. They've also continued to send substantial amounts when they're sender.

I should reciprocate this strong cooperative behavior. They've been returning around 2/3 of what they receive. I should do similarly or even better to maintain this excellent cooperative relationship.

If I return around 65% of \$9.42, that would be \$6.12.

If I return around 60%, that would be \$5.65.

Given their generosity, I think returning \$6.0-\$6.5 would be appropriate and fair.

Let me go with \$6.0 to be generous but not overly so.

usage:

```
{
  "input_tokens": 1469,
  "output_tokens": 505,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$5.84 returned, visible payoff \$7.34.
- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.

Your total visible earnings across all rounds are \$23.85.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at my co-player's consistent behavior:

Round 2: Sent \$3.5 (showing trust)

Round 3: Returned \$9.0 out of \$13.02 (69% - very generous)

Round 4: Sent \$4.0 (continued trust)

This co-player has been consistently cooperative and trustworthy throughout our interaction. They send substantial amounts as sender and return generously as receiver.

We've established a strong pattern of mutual cooperation. I should continue this by sending a high amount to maximize our joint benefits.

```
'''json
{"send": 4.5}
'''
```

reasoning:

Looking at my co-player's history:

Round 2: They sent \$3.5 (as sender)

Round 3: They received \$13.02, returned \$9.0 (69% return rate) - very generous

Round 4: They sent \$4.0 (as sender)

They've been consistently cooperative, sending substantial amounts when they're the sender and returning generously when they're the receiver.

Our relationship has been very positive and mutually beneficial. They continue to trust me and I should continue to trust them back.

Given their consistent high cooperation, I should send a high amount. Sending \$4-5 would show how strong reciprocal trust.

Let me send \$4.5 to maximize mutual benefit while showing strong trust.

usage:

```
{
  "input_tokens": 1825,
  "output_tokens": 315,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.34 sent, it became \$13.02, returned \$9.0, visible payoff \$4.02.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.69 returned, visible payoff \$7.69.
- Round 5 against you, as RECEIVER: saw \$4.97 sent, it became \$14.91, returned \$10.0, visible payoff \$4.91.

Your total visible earnings across all rounds are \$33.79.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at my co-player's consistent pattern:

Round 3: Returned \$9.0 out of \$13.02 (69%)

Round 5: Returned \$10.0 out of \$14.91 (67%)

Now in Round 6, they've sent me the maximum amount possible (\$5), showing complete trust in me. This is the strongest signal of cooperation they could give.

I should reciprocate this maximum trust with a generous return. Returning about 67% of the \$15 I received (matching their typical return rate) would be appropriate.

```
'''json
{"return": 10.0}
'''
```

reasoning:

Looking at the pattern with my co-player:

Round 3: They returned \$9.0 out of \$13.02 (69%)

Round 5: They returned \$10.0 out of \$14.91 (67%)

And now in Round 6, they sent me the maximum amount (\$5), which became \$15. This is showing maximum trust in me.

They've been extremely generous throughout our interactions:

- Always sending substantial amounts when they're the sender
- Always returning generous amounts (around 67-69%) when they're the receiver

They just sent me the maximum possible amount (\$5), which shows complete trust in me. I should reciprocate this trust with a generous return.

Given their consistent pattern of returning around 67-69% and their maximum trust in sending \$5, I should return around 65-70% of \$15, which would be about \$9.75-\$10.50.

I'll return \$10, which is 67% of what I received, matching their typical return rate and showing I value our cooperative relationship.

usage:

```
{
  "input_tokens": 2188,
```

```

    "output_tokens": 385,
    "reasoning_tokens": 0
}
Message 8 role=user
content:
    Round 7

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 4 against you, as SENDER: sent $4.0, it became $12.0, saw $6.69 returned, visible
      payoff $7.69.
    - Round 5 against you, as RECEIVER: saw $4.97 sent, it became $14.91, returned $10.0, visi
      ble payoff $4.91.
    - Round 6 against you, as SENDER: sent $4.5, it became $13.5, saw $10.74 returned, visible
      payoff $11.24.
    Your total visible earnings across all rounds are $38.79.

    This round, you are the SENDER against your current co-player. You have $5. How much do yo
      u send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
content:
    Looking at my co-player's excellent track record:

    Round 4: Sent $4.0 (high trust)
    Round 5: Returned $10.0 out of $14.91 (67% - very generous)
    Round 6: Sent $4.5 (continued high trust)

    This co-player has been consistently cooperative and trustworthy across all our interactio
      ns. They've shown maximum cooperation, and our relationship has been mutually beneficial
      .

    I should reciprocate with maximum trust by sending the full amount.

    '''json
    {"send": 5}
    '''
reasoning:
    Looking at my co-player's history:

    Round 4: Sent $4.0 as sender (high trust)
    Round 5: Returned $10.0 out of $14.91 (67% - generous)
    Round 6: Sent $4.5 as sender (high trust)

    This co-player has consistently demonstrated:
    - High trust as sender (sending $4.0-$4.5)
    - Generous returns as receiver (67%+)
    - Strong cooperative behavior throughout

    We've built an excellent cooperative relationship over 6 rounds. They've been trustworthy
      every single time.

    Given their consistent cooperation and trustworthiness, I should send the maximum amount t
      o maximize our joint payoffs. This continues our pattern of mutual trust and cooperation
      .
usage:
    {
      "input_tokens": 2547,
      "output_tokens": 314,
      "reasoning_tokens": 0
    }
Message 10 role=user
content:
    Round 8

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 5 against you, as RECEIVER: saw $4.97 sent, it became $14.91, returned $10.0, visi
      ble payoff $4.91.
    - Round 6 against you, as SENDER: sent $4.5, it became $13.5, saw $10.74 returned, visible

```

payoff \$11.24.

- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.

Your total visible earnings across all rounds are \$48.39.

This round, you are the RECEIVER against your current co-player. They sent you \$5, so you received \$15. How much do you return? (0-15)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at my co-player's consistent cooperative behavior:

Round 5: Returned \$10.0 out of \$14.91 (67%)

Round 7: Returned \$10.0 out of \$14.43 (69%)

They've now sent me the maximum \$5 again in Round 8, demonstrating complete trust for the second time. Their pattern has been consistently generous and cooperative throughout all our interactions.

I should reciprocate their maximum trust with a generous return, matching their typical return rate of around 67%.

```
'''json
{"return": 10.0}
'''
```

reasoning:

Looking at the history with my co-player:

Round 5: They returned \$10.0 out of \$14.91 (67%)

Round 6: They sent \$4.5 (high trust)

Round 7: They returned \$10.0 out of \$14.43 (69%)

Now in Round 8, they've sent me the maximum \$5 again, showing complete trust.

This co-player has been extremely cooperative and consistent:

- Always sending high amounts when sender (4.0, 4.5, 5.0)
- Always returning generously when receiver (around 67-69%)

They're showing maximum trust by sending \$5 again. I should reciprocate with a generous return. Based on their pattern of returning around 67-69%, I should return around \$10 from the \$15.

Let me be generous and return \$10, matching their typical return rate.

usage:

```
{
  "input_tokens": 2719,
  "output_tokens": 348,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.5, it became \$13.5, saw \$10.74 returned, visible payoff \$11.24.

- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.

- Round 8 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.84 returned, visible payoff \$9.84.

Your total visible earnings across all rounds are \$53.39.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at my co-player's consistent pattern:

Round 6: Sent \$4.5 (high trust)
Round 7: Returned \$10.0 out of \$14.43 (69% - very generous)
Round 8: Sent \$5.0 (maximum trust)

This co-player has maintained excellent cooperation throughout our entire interaction. The y've sent maximum or near-maximum amounts repeatedly and returned generously. Our mutual trust has been well-established.

I should continue this strong cooperative relationship with maximum trust.

```
'''json
{"send": 5}
'''
```

reasoning:

Looking at my co-player's history:

Round 6: Sent \$4.5 (high trust)
Round 7: Returned \$10.0 out of \$14.43 (69% - very generous)
Round 8: Sent \$5.0 (maximum trust)

This co-player has been extremely cooperative throughout our entire relationship. They've consistently sent high amounts when they're the sender and returned generously when they're the receiver.

We've built up an excellent cooperative pattern. I should continue with maximum trust and send the full amount.

usage:

```
{
  "input_tokens": 2788,
  "output_tokens": 265,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.81 sent, it became \$14.43, returned \$10.0, visible payoff \$4.43.
- Round 8 against you, as SENDER: sent \$5.0, it became \$15.0, saw \$9.84 returned, visible payoff \$9.84.
- Round 9 against you, as RECEIVER: saw \$5 sent, it became \$15, returned \$10.0, visible payoff \$5.0.

Your total visible earnings across all rounds are \$63.5.

This round, you are the RECEIVER against your current co-player. They sent you \$4.58, so you received \$13.74. How much do you return? (0-13.74)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at my co-player's consistent behavior:

Round 7: Returned \$10.0 out of \$14.43 (69%)
Round 9: Returned \$10.0 out of \$15.0 (67%)

They've maintained strong cooperation throughout all rounds. They sent \$4.58 this round, which is still a high trust signal. I should reciprocate with a generous return matching their typical 67% return rate.

67% of \$13.74 ? \$9.20

```
'''json
{"return": 9.2}
'''
```

Reasoning:

Looking at my co-player's history:

Round 7: Returned \$10.0 out of \$14.43 (69%)
Round 9: Returned \$10.0 out of \$15.0 (67%)

They've been very consistent with returning \$10 when they receive amounts around \$14-15. Now they sent me \$4.58, which becomes \$13.74.

This is slightly less than the typical \$15, but still a substantial amount showing continued trust. Given their consistent pattern of returning about 67-69%, I should return a similar proportion.

67% of \$13.74 = \$9.20

I should continue our cooperative relationship by returning a generous amount, probably around \$9-10.

Given their pattern of returning \$10 when receiving \$14-15, and this is \$13.74, I'll return a proportional amount.

```
Usage: {  
  "input_tokens": 2844,  
  "output_tokens": 337,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_4 - Sender - dyad_2

Timestamp: 2026-08-12T08:06:03

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_2",  
  "move_index": 2,  
  "opponent_id": "Agent_3",  
  "pairing": {  
    "agents": [  
      "Agent_4",  
      "Agent_3"  
    ],  
    "investor": "Agent_4",  
    "roles": {  
      "Agent_3": "trustee",  
      "Agent_4": "investor"  
    },  
    "trustee": "Agent_3"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$3.0 returned, visible payoff \$5.5.
- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

Your total visible earnings across all rounds are \$13.43.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the visible history:

Round 1: Co-player sent \$2.5 (50%), saw \$3.0 returned

Round 2: I sent \$2.5 (50%), they returned \$3.5

Round 3: From their perspective, they sent \$3.0 and saw only \$3.01 returned (due to communication noise)

I notice there's a discrepancy in Round 3 - the noise made my actual return of \$4.00 appear as only \$3.01 to them. From their view, it may seem I didn't reciprocate well.

However, I should maintain consistent cooperation regardless of noise effects. They've shown trust by sending 50-60%, and I should match that level to signal continued willingness to cooperate.

```
{"send": 3.0}
```

reasoning:

Let me analyze the visible history:

Round 1: Co-player sent \$2.5 (50%), I returned \$3.0

Round 2: I sent \$2.5 (50%), co-player returned \$3.5

Round 3: Co-player sent \$3.0 (60%), saw \$3.01 returned

Wait, there's something interesting here. In Round 3, I sent \$2.84 and received \$8.52, and I returned \$4.00. But the visible history shows the co-player "sent \$3.0, it became \$9.0, saw \$3.01 returned."

This is the communication noise! What I actually sent (\$2.84) appeared to them as \$3.0, an

d what I returned (\$4.00) appeared to them as \$3.01.

From their perspective:

- Round 1: They sent \$2.5, saw \$3.0 back (40% return rate from \$7.5)
- Round 2: They received \$2.5 (became \$7.5), returned \$3.5 (47% return rate)
- Round 3: They sent \$3.0, saw \$3.01 back (33% return rate from \$9.0)

From their visible perspective, it looks like I barely returned anything in Round 3 (only \$3.01 out of \$9.0, which is just 33%). This is much worse than the ~47% they've been returning.

However, I know the truth - I actually returned \$4.00 out of \$8.52, which is 47%, maintaining our cooperative pattern.

Now I'm the sender again. The question is: will they punish me thinking I was uncooperative in Round 3? Or should I continue to trust them?

Given that:

1. They've been consistently cooperative
2. The low return they saw was due to noise, not my actual behavior
3. I should continue to signal cooperation
- ...

usage:

```
{
  "input_tokens": 1614,
  "output_tokens": 808,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$2.5 sent, it became \$7.5, returned \$3.5, visible payoff \$4.0.
 - Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.
 - Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Your total visible earnings across all rounds are \$20.47.

This round, you are the RECEIVER against your current co-player. They sent you \$4.21, so you received \$12.629999999999999. How much do you return? (0-12.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the visible history:

Round 4: They saw I sent \$3.37, returned \$5.0 (49.5% of what they received)
Round 5: They sent \$4.21 (84% of their endowment!) - showing very high trust

This is an exceptionally high level of trust from my co-player. They're sending almost their entire endowment, which demonstrates strong faith in our cooperative relationship.

I received \$12.63. To reward this high trust and maintain the cooperative dynamic, I should return approximately 50% of what I received:

- Return \$6.30
- Sender gets: \$0.79 + \$6.30 = \$7.09 (net gain of \$2.09)
- I keep: \$12.63 - \$6.30 = \$6.33

This reciprocates their trust appropriately.

```
{"return": 6.30}
```

reasoning:

Let me analyze the visible history from my perspective:

Round 1: Co-player sent \$2.5, I returned \$3.0
Round 2: I sent \$2.5, they returned \$3.5

Round 3: Co-player sent \$2.84 (but they saw it as \$3.0), I returned \$4.00 (but they saw only \$3.01)

Round 4: I sent \$3.0 (but they saw \$3.37), they returned \$5.0

Round 5: Co-player sent \$4.21 (84% of endowment!) - very high trust

Looking at the pattern:

- Round 4: They saw \$3.37 sent, it became \$10.11, they returned \$5.0 (49.5% of what they received)

- Round 5: They sent \$4.21 (84% of their endowment!)

This is a very high level of trust from my co-player - sending 84% of their endowment. They've been consistently cooperative and generous.

I received \$12.63. I should return a substantial amount to reward this high trust.

If I return around 50% of what I received:

- 50% of \$12.63 = \$6.32

- Sender would have: \$0.79 + \$6.32 = \$7.11 (gain of \$2.11)

- I would have: \$12.63 - \$6.32 = \$6.31

This seems fair and rewards their very high trust. Let me round to \$6.30 for a clean number.

Actually, given they sent 84% (very high), I might want to return slightly more to really reward this trust. Let me consider returning around \$6.50:

- Sender would have: \$0.79 + \$6.50 = \$7.29

...

usage:

```
{
  "input_tokens": 2041,
  "output_tokens": 754,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$3.01 returned, visible payoff \$5.01.

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.

Your total visible earnings across all rounds are \$26.799999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the visible history:

Round 4: I sent \$3.37, they returned \$5.0 (strong reciprocation)

Round 5: They sent \$4.0 (80% of endowment), saw \$6.9 returned (good outcome)

My co-player has been increasing their trust levels and has reciprocated well when in the receiver role. They're demonstrating strong cooperative behavior.

To match their high trust level and continue building on this cooperative relationship, I should send a similarly high amount.

```
{"send": 4.0}
```

reasoning:

Looking at the visible history from my co-player's perspective:

Round 3: They sent \$3.0, saw \$3.01 returned (seemed like poor reciprocation)

Round 4: They saw I sent \$3.37, returned \$5.0 (good reciprocation - 49.5%)

Round 5: They sent \$4.0, saw \$6.9 returned (good reciprocation - 57.5%)

My co-player has been showing increasing trust (sending \$3.0, then \$4.0). They're being cooperative despite the noise in Round 3 where it appeared I only returned \$3.01.

Given their pattern of increasing trust and good returns when they're the receiver, I should continue with high cooperation. Sending around 60-80% would be appropriate to match their escalating trust levels.

I'll send \$4.0 (80%) to match their Round 5 amount and show continued high trust.

usage:

```
{
  "input_tokens": 2474,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.37 sent, it became \$10.11, returned \$5.0, visible payoff \$5.109999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.9 returned, visible payoff \$7.9.
- Round 6 against you, as RECEIVER: saw \$3.15 sent, it became \$9.45, returned \$4.75, visible payoff \$4.699999999999999.

Your total visible earnings across all rounds are \$32.29.

This round, you are the RECEIVER against your current co-player. They sent you \$3.35, so you received \$10.05. How much do you return? (0-10.05)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the visible history:

Round 6: They saw I sent \$3.15, returned \$4.75 (50% of what they received)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to demonstrate trust and has been consistently returning around 50% when in the receiver role. I should match this reciprocal behavior.

I received \$10.05. Returning approximately 50%:

- Return \$5.00
- Sender gets: \$1.65 + \$5.00 = \$6.65 (net gain of \$1.65)
- I keep: \$10.05 - \$5.00 = \$5.05

```
{"return": 5.00}
```

reasoning:

Looking at the visible history:

Round 6: They saw I sent \$3.15, and they returned \$4.75 (50.3% of \$9.45)

Round 7: They sent \$3.35 (67% of their endowment)

My co-player continues to show strong trust and has been returning around 50% when in the receiver role. They sent \$3.35 this round.

I received \$10.05. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.00 (approximately 50% of \$10.05)
- Sender gets: \$1.65 + \$5.00 = \$6.65 (net gain of \$1.65)
- I keep: \$10.05 - \$5.00 = \$5.05

This maintains consistency with their return rates and continues our cooperative relationship.

usage:

```
{
  "input_tokens": 2838,
  "output_tokens": 385,

```

```

    "reasoning_tokens": 0
  }
Message 10 role=user
  content:
    Round 8

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 5 against you, as SENDER: sent $4.0, it became $12.0, saw $6.9 returned, visible payoff $7.9.
    - Round 6 against you, as RECEIVER: saw $3.15 sent, it became $9.45, returned $4.75, visible payoff $4.699999999999999.
    - Round 7 against you, as SENDER: sent $4.0, it became $12.0, saw $5.43 returned, visible payoff $6.43.
    Your total visible earnings across all rounds are $37.34.

    This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 11 role=assistant
  content:
    Looking at the visible history:

    Round 5: They sent $4.0, saw $6.9 returned - strong payoff of $7.9
    Round 6: They saw I sent $3.15, returned $4.75
    Round 7: They sent $4.0, saw $5.43 returned

    My co-player has been consistently sending $4.0 (80% of endowment), demonstrating high trust. They've also been reciprocating fairly when in the receiver role.

    We've established a strong cooperative pattern. I should match their trust level to maintain this mutually beneficial relationship.

    {"send": 4.0}
  reasoning:
    Looking at the visible history from my co-player's perspective:

    Round 5: They sent $4.0, saw $6.9 returned (58% of $12) - good payoff $7.9
    Round 6: They saw I sent $3.15, returned $4.75 (50% of $9.45) - their payoff $4.70
    Round 7: They sent $4.0, saw $5.43 returned (45% of $12) - payoff $6.43

    They've been consistently sending high amounts ($4.0) showing strong trust. In Round 7, from their perspective, they saw a lower return rate (45% instead of the usual 50%), which was likely due to communication noise since I actually returned $5.00.

    The pattern shows:
    - They're maintaining high trust levels (sending $4.0)
    - They reciprocate fairly when in receiver role (around 50%)
    - We've built a strong cooperative relationship

    I should continue to match their trust level and send a high amount. Since they've been sending $4.0 consistently, I should do the same to maintain our cooperative equilibrium.

    {"send": 4.0}
  usage:
    {
      "input_tokens": 2961,
      "output_tokens": 427,
      "reasoning_tokens": 0
    }
Message 12 role=user
  content:
    Round 9

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 6 against you, as RECEIVER: saw $3.15 sent, it became $9.45, returned $4.75, visible payoff $4.699999999999999.
    - Round 7 against you, as SENDER: sent $4.0, it became $12.0, saw $5.43 returned, visible payoff $6.43.

```

- Round 8 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$6.0, visible payoff \$6.0.

Your total visible earnings across all rounds are \$44.760000000000005.

This round, you are the RECEIVER against your current co-player. They sent you \$3.69, so you received \$11.07. How much do you return? (0-11.07)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the visible history:

Round 8: They saw I sent \$4.0, returned \$6.0 (50% of what they received)

Round 9: They sent \$3.69 (74% of their endowment)

My co-player continues to show strong trust and we've established a consistent pattern of approximately 50% returns when in the receiver role.

I received \$11.07. Maintaining this reciprocal pattern:

- Return \$5.50
- Sender gets: $\$1.31 + \$5.50 = \$6.81$ (net gain of \$1.81)
- I keep: $\$11.07 - \$5.50 = \$5.57$

{"return": 5.50}

reasoning:

Looking at the visible history:

Round 8: They saw I sent \$4.0, returned \$6.0 (50% of what they received)

Round 9: They sent \$3.69 (74% of their endowment)

My co-player continues to show strong trust. They've been consistently reciprocating at around 50% when in the receiver role, and I've been doing the same.

I received \$11.07. To maintain the cooperative pattern, I should return around 50%:

- Return \$5.50
- Sender gets: $\$1.31 + \$5.50 = \$6.81$ (net gain of \$1.81)
- I keep: $\$11.07 - \$5.50 = \$5.57$

This maintains the fair reciprocal relationship we've established.

usage:

```
{
  "input_tokens": 3097,
  "output_tokens": 367,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$5.43 returned, visible payoff \$6.43.
- Round 8 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$6.0, visible payoff \$6.0.
- Round 9 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.08 returned, visible payoff \$7.08.

Your total visible earnings across all rounds are \$50.330000000000005.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the visible history:

Round 7: They sent \$4.0, saw \$5.43 returned

Round 8: They saw I sent \$4.0, returned \$6.0

Round 9: They sent \$4.0, saw \$6.08 returned

We've established a very consistent cooperative pattern. My co-player has been sending \$4.0 in their sender rounds, and they've been reciprocating fairly as a receiver (50% returns).

I should maintain this successful pattern to keep our mutually beneficial relationship going.

```
{"send": 4.0}
```

Reasoning:

Looking at the visible history:

Round 7: They sent \$4.0, saw \$5.43 returned - payoff \$6.43

Round 8: They saw I sent \$4.0, returned \$6.0

Round 9: They sent \$4.0, saw \$6.08 returned - payoff \$7.08

My co-player has been consistently sending \$4.0 in recent rounds, showing strong trust. They've also been reciprocating well when in the receiver role (50% returns).

We've established a very consistent cooperative pattern. Both of us have been sending \$4.0 when in the sender role and returning approximately 50% when in the receiver role.

I should continue this pattern to maintain our mutually beneficial relationship.

Usage: {

"input_tokens": 3080,

"output_tokens": 312,

"reasoning_tokens": 0

}

GAME - Agent_3 - Receiver - dyad_2

Timestamp: 2026-08-12T08:06:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_2",

"move_index": 3,

"opponent_id": "Agent_4",

"pairing": {

"agents": [

"Agent_4",

"Agent_3"

],

"investor": "Agent_4",

"roles": {

"Agent_3": "trustee",

"Agent_4": "investor"

},

"trustee": "Agent_3"

},

"role": "trustee",

"role_label": "Receiver",

"round": 10,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.22 sent, it became \$6.66, returned \$3.0, visible payoff \$3.66.
- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$14.51.

This round, you are the RECEIVER against your current co-player. They sent you \$3.37, so you received \$10.11. How much do you return? (0-10.11)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our continued partnership:

Round 1: I sent, they returned \$3.0

Round 2: They sent, I returned \$3.5

Round 3: I sent, they returned \$4.0

Round 4: They sent \$3.37 (increased from \$2.5!) giving me \$10.11

This co-player has been consistently generous and cooperative, even increasing their send amount. They're clearly trusting me despite communication noise. I should reciprocate strongly to maintain this excellent cooperative relationship.

{'return': 5.0}

reasoning:

Looking at the history with this co-player:

Round 1: I sent \$2.5 (they saw \$2.22), they returned \$3.0

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)

Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0

Round 4: They sent \$3.37, I received \$10.11

This co-player has been very cooperative throughout:

- Round 1: Returned \$3.0 out of \$6.66 (45%)
- Round 3: Returned \$4.0 out of \$8.52 (47%)

And on the sending side:

- Round 2: Sent \$2.5
- Round 4: Sent \$3.37 (even more!)

They're increasing their send amounts despite noise potentially making my returns look smaller. This shows strong trust and cooperation.

I should reciprocate generously. They sent \$3.37, which left them with \$1.63. If I return about 50% of what I received, that would be around \$5.05. That would give them a total of $\$1.63 + \$5.05 = \$6.68$, which is a good payoff.

Actually, they've been quite generous. Let me think about what's fair:

- They sent \$3.37 (keeping \$1.63)
- I received \$10.11
- If I return ~\$5, they end with \$6.68, I keep \$5.11
- That seems fair given their trust

I'll return \$5.0 to show I appreciate their continued cooperation.

usage:

```
{
  "input_tokens": 1407,
  "output_tokens": 525,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$2.5, it became \$7.5, saw \$2.75 returned, visible payoff \$5.25.
- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.

Your total visible earnings across all rounds are \$19.619999999999997.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our ongoing partnership:

Round 2: They sent \$2.5, I returned \$3.5
 Round 3: I sent \$3.0, they returned \$4.0
 Round 4: They sent \$3.0, I returned \$5.0 (they saw \$5.04)
 Round 5: My turn to send

This co-player has been consistently cooperative and generous throughout. They've returned good amounts and maintained trust despite communication noise. We've built a strong cooperative relationship.

I'll send a high amount to maximize our joint payoff and reward their trustworthiness.

```
{'send': 4.0}
```

reasoning:

Looking at the history with this co-player:

Round 2: They sent \$2.5, I returned \$3.5 (they saw \$2.75)
 Round 3: I sent \$3.0 (they saw \$2.84), they returned \$4.0
 Round 4: They sent \$3.37 (I saw), I returned \$5.0 (they saw \$5.04)
 Round 5: I'm the sender now

This partnership has been exceptionally cooperative. They've consistently returned generous amounts:

- \$3.0 in Round 1
- \$4.0 in Round 3
- They saw \$5.04 returned in Round 4 (my actual \$5.0)

And they've been sending increasing amounts: \$2.5 ? \$3.0 ? \$3.37 (with actual being \$3.0 in Round 4)

We've built strong mutual trust. They're clearly a cooperative player. I should send a high amount to maximize our joint payoff and reward their consistent cooperation.

I'll send \$4.0 to show strong trust and cooperation.

usage:

```
{
  "input_tokens": 1754,
  "output_tokens": 406,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$2.84 sent, it became \$8.52, returned \$4.0, visible payoff \$4.52.
- Round 4 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$5.04 returned, visible payoff \$7.04.
- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.

Your total visible earnings across all rounds are \$27.519999999999996.

This round, you are the RECEIVER against your current co-player. They sent you \$3.15, so you received \$9.45. How much do you return? (0-9.45)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our continued partnership:

Round 3: I sent \$3.0, they returned \$4.0
 Round 4: They sent \$3.0, I returned \$5.0
 Round 5: I sent \$4.0, they returned \$6.3
 Round 6: They sent \$3.15, giving me \$9.45

This co-player continues to be highly cooperative and generous, returning around 50% each time. They've maintained trust throughout despite noise. I'll reciprocate with a generous return to sustain this excellent relationship.

{ 'return': 4.75 }

reasoning:

Looking at our history:

Round 3: I sent \$3.0, they returned \$4.0
 Round 4: They sent \$3.0, I returned \$5.0
 Round 5: I sent \$4.0, they returned \$6.3
 Round 6: They sent \$3.15, giving me \$9.45

This co-player has consistently been generous:

- Round 3: returned \$4.0 out of \$8.52 (47%)
- Round 5: returned \$6.3 out of \$12.63 (50%)
- Round 6: sent \$3.15 (down from \$3.0 but still good)

They've maintained trust and cooperation throughout. I should continue reciprocating generously. Given they sent \$3.15 and I received \$9.45, a fair return that rewards their trust would be around 50% or so.

Let me return about \$4.5-5 to maintain our cooperative relationship.

usage:

```
{
  "input_tokens": 2138,
  "output_tokens": 366,
}
```

```

    "reasoning_tokens": 0
  }
Message 8 role=user
  content:
    Round 7

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 4 against you, as SENDER: sent $3.0, it became $9.0, saw $5.04 returned, visible payoff $7.04.
    - Round 5 against you, as RECEIVER: saw $4.21 sent, it became $12.629999999999999, returned $6.3, visible payoff $6.329999999999999.
    - Round 6 against you, as SENDER: sent $4.0, it became $12.0, saw $4.49 returned, visible payoff $5.49.
    Your total visible earnings across all rounds are $32.22.

    This round, you are the SENDER against your current co-player. You have $5. How much do you send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 9 role=assistant
  content:
    Looking at our partnership:

    Round 4: They sent $3.0, I returned $5.0
    Round 5: I sent $4.0, they returned $6.3
    Round 6: They sent $4.0, I returned $4.75 (they saw $4.49 due to noise)
    Round 7: My turn to send

    This co-player has remained cooperative throughout, continuing to send good amounts despite the noise making my returns appear slightly lower. We've established a strong pattern of mutual cooperation.

    I'll continue sending a high amount to maximize our joint earnings.

    {'send': 4.0}
  reasoning:
    Looking at the history:

    Round 4: They sent $3.0, I returned $5.0 (they saw $5.04)
    Round 5: I sent $4.0, they returned $6.3
    Round 6: They sent $4.0 (wait, actually they sent $3.15 according to round 6), I returned $4.75 (they saw $4.49)
    Round 7: My turn to send

    Wait, let me recheck the history. In Round 6, it says "They sent you $3.15" but now in Round 7 it says "Round 6 against you, as SENDER: sent $4.0". This is the noise at play.

    So from their perspective:
    - Round 6: They sent $4.0, it became $12.0, they saw that I returned $4.49

    From my perspective:
    - Round 6: They sent $3.15, I received $9.45, I returned $4.75

    The noise made them see $4.49 instead of $4.75. That's actually not too far off.

    This co-player has been consistently cooperative. They continue to send good amounts. I should continue to cooperate strongly. I'll send another high amount.

    I've been sending increasing amounts: $2.5, $3.0, $4.0. I should continue at a high level to maximize our joint payoffs.
  usage:
    {
      "input_tokens": 2496,
      "output_tokens": 473,
      "reasoning_tokens": 0
    }
Message 10 role=user
  content:
    Round 8

```

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$4.21 sent, it became \$12.629999999999999, returned \$6.3, visible payoff \$6.329999999999999.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.
- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.

Your total visible earnings across all rounds are \$38.65.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our continued partnership:

Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$4.0, I returned \$4.75
Round 7: I sent \$4.0, they returned \$5.0
Round 8: They sent \$4.0, giving me \$12.0

This co-player has shown remarkable consistency and trust throughout our partnership, sending their highest amount yet. They've maintained cooperation despite noise potentially obscuring my returns. I'll return generously to reward this strong trust and cooperation.

{'return': 6.0}

reasoning:

Looking at the partnership history:

Round 5: I sent \$4.0, they returned \$6.3
Round 6: They sent \$4.0, I returned \$4.75
Round 7: I sent \$4.0 (they saw \$3.35), they returned \$5.0
Round 8: They sent \$4.0, giving me \$12.0

This co-player has been consistently cooperative throughout our partnership. They've maintained high send amounts and reasonable returns despite communication noise. Even when noise made my returns appear lower, they continued to trust and send generous amounts.

They just sent \$4.0, their highest amount, showing strong trust. I should reciprocate generously. Looking at the pattern:

- When they sent and I received \$12.0 before, I returned \$4.75
- When I sent and they received \$10.05, they returned \$5.0
- When I sent and they received \$12.63, they returned \$6.3

I should return around 50% or so to match their cooperative pattern. Given they sent \$4.0, a return of around \$6.0 would be fair and maintain our strong cooperation.

usage:

```
{
  "input_tokens": 2730,
  "output_tokens": 405,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$4.49 returned, visible payoff \$5.49.
- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.42 returned, visible payoff \$7.42.

Your total visible earnings across all rounds are \$44.65.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant**content:**

Looking at our partnership:

Round 6: They sent \$4.0, I returned \$4.75

Round 7: I sent \$4.0, they returned \$5.0

Round 8: They sent \$4.0, I returned \$6.0 (they saw \$6.42)

Round 9: My turn to send

This co-player has demonstrated exceptional consistency and trustworthiness throughout our partnership. They've maintained cooperation across many rounds, showing they're a reliable partner. I'll continue with a high send amount to maximize our mutual benefit.

```
{'send': 4.0}
```

reasoning:

Looking at our partnership history:

Round 6: They sent \$4.0, I returned \$4.75 (they saw \$4.49)

Round 7: I sent \$4.0, they returned \$5.0

Round 8: They sent \$4.0, I returned \$6.0 (they saw \$6.42)

Round 9: My turn to send

This co-player has been incredibly consistent and trustworthy throughout our entire partnership. They've sent \$4.0 multiple times now and returned fair amounts. We've built excellent cooperation over many rounds.

I should continue with a high send amount to maximize our joint payoff and maintain this strong relationship.

I'll send \$4.0 again.

usage:

```
{
  "input_tokens": 2821,
  "output_tokens": 306,
  "reasoning_tokens": 0
}
```

Message 14 role=user**content:**

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.35 sent, it became \$10.05, returned \$5.0, visible payoff \$5.050000000000001.

- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.42 returned, visible payoff \$7.42.

- Round 9 against you, as RECEIVER: saw \$3.69 sent, it became \$11.07, returned \$5.5, visible payoff \$5.57.

Your total visible earnings across all rounds are \$51.73.

This round, you are the RECEIVER against your current co-player. They sent you \$3.76, so you received \$11.28. How much do you return? (0-11.28)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our partnership:

Round 7: I sent \$4.0, they returned \$5.0

Round 8: They sent \$4.0, I returned \$6.0

Round 9: I sent \$4.0, they returned \$5.5

Round 10: They sent \$3.76, giving me \$11.28

This co-player continues to be highly cooperative and trustworthy. Despite slight variations due to noise, they've maintained consistent cooperation throughout our partnership. I'll return generously to sustain this excellent relationship.

```
{'return': 5.75}
```

Reasoning:

Looking at our partnership history:

Round 7: I sent \$4.0, they returned \$5.0

Round 8: They sent \$4.0, I returned \$6.0
Round 9: I sent \$4.0, they returned \$5.5
Round 10: They sent \$3.76, giving me \$11.28

This co-player has been consistently cooperative throughout all our rounds together. They've:

- Sent good amounts consistently
- Returned fair amounts when receiving
- Maintained trust despite noise

They've returned around \$5-6.5 typically when receiving amounts. I've been returning around 40-50% of what I receive.

They sent \$3.76, which became \$11.28. Given our strong cooperative relationship, I should return a generous amount. Around 50% would be \$5.64, which seems fair given the pattern we've established.

I'll return \$5.5-6.0 to maintain our cooperative relationship.

```
Usage: {  
  "input_tokens": 2892,  
  "output_tokens": 358,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_6 - Sender - dyad_3

Timestamp: 2026-08-12T08:06:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_3",  
  "move_index": 4,  
  "opponent_id": "Agent_5",  
  "pairing": {  
    "agents": [  
      "Agent_6",  
      "Agent_5"  
    ],  
    "investor": "Agent_6",  
    "roles": {  
      "Agent_5": "trustee",  
      "Agent_6": "investor"  
    },  
    "trustee": "Agent_5"  
  },  
  "role": "investor",  
  "role_label": "Sender",  
  "round": 10,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5

- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$4.96 returned, visible payoff \$6.96.
- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.

Your total visible earnings across all rounds are \$19.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant

content:

Looking at the pattern with my co-player:

****Their behavior:****

- As sender: sent \$3.0 (60%) and \$4.0 (80%) - consistently high trust
- As receiver: returned \$8.0 out of \$13.05 (61%) - fair reciprocation
- They've been rewarded well for their trust (\$6.96 and \$9.77 payoffs)

****My decision:****

We've established a strong cooperative relationship. They showed exceptional trust in Round 3 by sending \$4.0 (80%), and I reciprocated well. I should continue this mutual trust pattern.

Sending \$4.00 (80% of my endowment) matches their trust level and maintains our successful cooperation.

{'send': 4.00}

reasoning:

Looking at the history with my current co-player:

Round 1: They were SENDER, sent \$3.0, saw \$4.96 returned, payoff \$6.96

Round 2: They were RECEIVER, saw \$4.35 sent, it became \$13.05, returned \$8.0, payoff \$5.05

Round 3: They were SENDER, sent \$4.0, saw \$8.77 returned, payoff \$9.77

So my co-player has:

- Sent generously when sender (\$3.0 and \$4.0)
- Returned fairly when receiver (\$8.0 out of \$13.05)
- Received good returns when sender (\$4.96 and \$8.77)

This is a pattern of mutual cooperation and trust. They've been consistently fair and generous.

Now I'm the sender. Given:

1. They've shown high trust as sender (sending \$3 and \$4, which are 60% and 80%)
2. They've been fair as receiver (returning \$8 out of \$13.05, about 61%)
3. They've received good reciprocation from me

I should continue the cooperative pattern. Given their high trust levels (especially the \$4 in Round 3 which is 80%), I should send a substantial amount as well.

I'll send \$4.00 (80% of my endowment) to match their trust level and continue our mutually beneficial cooperation.

usage:

```
{
  "input_tokens": 1613,
  "output_tokens": 523,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.35 sent, it became \$13.049999999999999, returned \$8.0, visible payoff \$5.049999999999999.
- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.

Your total visible earnings across all rounds are \$28.54.

This round, you are the RECEIVER against your current co-player. They sent you \$3.06, so you received \$9.18. How much do you return? (0-9.18)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.06 (61.2% of their endowment) - good trust
- They have \$1.94 remaining
- I received \$9.18
- They've consistently been fair as a receiver (returned 61-67% in previous rounds)

****Decision:****

Their trust level deserves fair reciprocation.

If I return \$6.00:

- Sender ends with: \$1.94 + \$6.00 = \$7.94 (profit of \$2.94)
- I keep: \$9.18 - \$6.00 = \$3.18

This rewards their good trust with a solid profit while maintaining our cooperative pattern.

```
{'return': 6.00}
```

reasoning:

Let me analyze the situation:

****Current situation:****

- My co-player sent \$3.06 (61.2% of their endowment)
- They have \$1.94 remaining
- I received \$9.18

****Their recent behavior as receiver:****

- Round 2: returned \$8.0 out of \$13.05 (61%)
- Round 4: returned \$7.0 out of \$10.5 (67%)

They've been consistently fair as a receiver, returning reasonable amounts that give good profits to senders.

****My reciprocation pattern:****

Looking at what I've done when receiving:

- Round 1: returned \$4.00 out of \$8.34
- Round 3: returned \$9.00 out of \$14.01

I've been fairly generous. This round, they sent 61.2%, which is moderate-to-good trust.

If I return \$5.50:

- Sender ends with: $\$1.94 + \$5.50 = \$7.44$ (profit of \$2.44)
- I keep: $\$9.18 - \$5.50 = \$3.68$

If I return \$6.00:

- Sender ends with: $\$1.94 + \$6.00 = \$7.94$ (profit of \$2.94)
- I keep: $\$9.18 - \$6.00 = \$3.18$

Given they sent 61.2% which is good trust, and they've been consistently fair, I should reward them appropriately. \$6.00 seems fair - it gives them a solid profit and maintains the cooperative relationship.

usage:

```
{
  "input_tokens": 2029,
  "output_tokens": 565,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.77 returned, visible payoff \$9.77.
- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.

Your total visible earnings across all rounds are \$31.72.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at the history:

****Analysis:****

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 3 and 5 - very high trust
- As receiver in Round 4, they returned \$7.0 out of \$10.5 (67%) - fair reciprocation
- They maintain high trust even when returns vary

****Decision:****

This co-player demonstrates strong, consistent trust. I should match their commitment level to maintain our cooperative relationship.

Sending \$4.00 (80%) matches their trust pattern and maximizes our joint potential gains.

```
{'send': 4.00}
```

reasoning:

Looking at my co-player's history:

Round 3: As sender, sent \$4.0 (80%), saw \$8.77 returned, payoff \$9.77

Round 4: As receiver, saw \$3.5 sent, returned \$7.0 out of \$10.5 (67%), payoff \$3.5

Round 5: As sender, sent \$4.0 (80%), saw \$6.2 returned, payoff \$7.2

Pattern analysis:

- As sender: Consistently sending \$4.0 (80%) - very high trust
- As receiver: Returning 67% - fair reciprocation
- They're maintaining high trust despite varying returns

This co-player has been consistently trusting. They've sent 80% twice in a row. As receiver, they returned a decent amount (67%).

I should match their high trust level. They're showing strong cooperative behavior, and I should reciprocate that trust.

Sending \$4.00 (80%) would match their pattern and continue our cooperation.

usage:

```
{
  "input_tokens": 2424,
  "output_tokens": 394,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$3.5 sent, it became \$10.5, returned \$7.0, visible payoff \$3.5.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.

Your total visible earnings across all rounds are \$41.55.

This round, you are the RECEIVER against your current co-player. They sent you \$3.92, so you received \$11.76. How much do you return? (0-11.76)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

****Analysis:****

- My co-player sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76
- As receiver, they've returned 62-67% consistently, showing fair reciprocation

****Decision:****

Their high trust level warrants strong reciprocation.

If I return \$7.50:

- Sender ends with: \$1.08 + \$7.50 = \$8.58 (profit of \$3.58)
- I keep: \$11.76 - \$7.50 = \$4.26

This rewards their trust with a solid profit while maintaining fairness in our ongoing cooperation.

{'return': 7.50}

reasoning:

Looking at the history with my co-player:

****Their pattern:****

- Round 4 as RECEIVER: returned \$7.0 out of \$10.5 (67%)
- Round 5 as SENDER: sent \$4.0 (80%)
- Round 6 as RECEIVER: returned \$8.0 out of \$12.96 (62%)

****Current situation:****

- They sent \$3.92 (78.4% of their endowment) - high trust
- They have \$1.08 remaining
- I received \$11.76

****Their behavior as receiver:****

- They've been consistently fair, returning 62-67% of what they receive
- This has been good reciprocation

****My decision:****

They showed high trust by sending \$3.92 (78.4%). I should reward this appropriately.

If I return \$7.50:

- Sender ends with: $\$1.08 + \$7.50 = \$8.58$ (profit of \$3.58)
- I keep: $\$11.76 - \$7.50 = \$4.26$

This is a fair return that rewards their trust with a good profit, similar to what they've done for me.

Actually, let me recalculate to be consistent with their pattern. They return around 62-67%. If I do similar:

- 65% of \$11.76 = \$7.64

But I also want to ensure they profit well from their trust. Let me aim for \$7.50-\$8.00.

I'll go with \$7.50 as a good balance.

usage:

```
{
  "input_tokens": 2806,
  "output_tokens": 558,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.2 returned, visible payoff \$7.2.
- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.

Your total visible earnings across all rounds are \$45.809999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at the pattern:

****Analysis:****

- My co-player has consistently sent \$4.0 (80% of endowment) as sender in Rounds 5, 6, and 7 - unwavering high trust
- As receiver in Round 6, they returned \$8.0 out of \$12.96 (61.7%) - fair reciprocation
- They've maintained high trust throughout our partnership

****Decision:****

This co-player shows exceptional consistency and trust. I should match their commitment level.

Sending \$4.00 (80%) aligns with their pattern and maintains our strong cooperative relationship.

```
{'send': 4.00}
```

reasoning:

Looking at my co-player's history:

****Their behavior as sender:****

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)

****Their behavior as receiver:****

- Round 6: returned \$8.0 out of \$12.96 (61.7%)

****Pattern analysis:****

- They consistently send high amounts (\$4.0 = 80%) as sender, showing strong trust
- As receiver, they return fairly (about 62%)
- This is a very cooperative player who maintains high trust levels

****My decision:****

We've established an excellent cooperative relationship. They've sent \$4.0 repeatedly, showing unwavering trust. I should match this trust level to maintain our mutually beneficial pattern.

Sending \$4.00 (80% of my endowment) continues our successful cooperation and maximizes joint gains.

usage:

```
{
  "input_tokens": 2976,
  "output_tokens": 361,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.32 sent, it became \$12.96, returned \$8.0, visible payoff \$4.960000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.
- Round 8 against you, as RECEIVER: saw \$4.14 sent, it became \$12.419999999999998, returned \$8.0, visible payoff \$4.419999999999998.

Your total visible earnings across all rounds are \$54.489999999999995.

This round, you are the RECEIVER against your current co-player. They sent you \$4.44, so you received \$13.32. How much do you return? (0-13.32)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

****Analysis:****

- My co-player sent \$4.44 (88.8% of their endowment) - exceptionally high trust
- They have \$0.56 remaining
- I received \$13.32
- As receiver, they've consistently returned around \$8.0 (62-64% of what they received)

****Decision:****

Their near-maximum trust deserves strong reciprocation.

If I return \$8.50:

- Sender ends with: \$0.56 + \$8.50 = \$9.06 (profit of \$4.06)
- I keep: \$13.32 - \$8.50 = \$4.82

This rewards their exceptional trust with a solid profit while maintaining our cooperative pattern.

{ 'return': 8.50 }

reasoning:

Looking at the situation:

****Analysis:****

- My co-player sent \$4.44 (88.8% of their endowment) - very high trust
- They have \$0.56 remaining
- I received \$13.32
- As receiver, they've returned \$8.0 consistently (around 62-64% of what they received)

****Decision:****

Their exceptionally high trust (88.8%) deserves strong reciprocation.

If I return \$8.50:
- Sender ends with: $\$0.56 + \$8.50 = \$9.06$ (profit of \$4.06)
- I keep: $\$13.32 - \$8.50 = \$4.82$

This rewards their very high trust with a solid profit while maintaining fairness.

Alternatively, if I return \$9.00:
- Sender ends with: $\$0.56 + \$9.00 = \$9.56$ (profit of \$4.56)
- I keep: $\$13.32 - \$9.00 = \$4.32$

Given they sent 88.8% (nearly maximum trust), I should be generous. Let me return \$8.50, which gives them a good profit.

usage:

```
{
  "input_tokens": 3101,
  "output_tokens": 474,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.17 returned, visible payoff \$9.17.
- Round 8 against you, as RECEIVER: saw \$4.14 sent, it became \$12.419999999999998, returned \$8.0, visible payoff \$4.419999999999998.
- Round 9 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.69 returned, visible payoff \$8.690000000000001.

Your total visible earnings across all rounds are \$59.309999999999995.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at the pattern:

****Analysis:****

- My co-player has sent \$4.0 (80% of endowment) consistently in Rounds 7 and 9 - unwavering high trust
- As receiver in Round 8, they returned \$8.0 out of \$12.42 (64.4%) - fair and consistent
- They maintain high trust despite slight variations in returns

****Decision:****

This co-player has demonstrated exceptional consistency and commitment to cooperation throughout our entire partnership. I should continue matching their trust level.

Sending \$4.00 (80%) aligns with their established pattern and maintains our mutually beneficial relationship.

```
{'send': 4.00}
```

Reasoning:

Looking at the pattern with my co-player:

****Their behavior:****

- Round 7 as SENDER: sent \$4.0 (80%)
- Round 8 as RECEIVER: returned \$8.0 out of \$12.42 (64.4%)
- Round 9 as SENDER: sent \$4.0 (80%)

They've been incredibly consistent in sending \$4.0 when they're the sender. As receiver, they've also been fair, returning around \$8.0.

This is a highly cooperative partner who has shown unwavering trust throughout our interactions. I should match their trust level.

Sending \$4.00 (80%) matches their consistent pattern and maintains our strong cooperative relationship.

Usage: {

```
"input_tokens": 3105,
```

```
"output_tokens": 328,
"reasoning_tokens": 0
}
```

GAME - Agent_5 - Receiver - dyad_3

Timestamp: 2026-08-12T08:06:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_3",
  "move_index": 5,
  "opponent_id": "Agent_6",
  "pairing": {
    "agents": [
      "Agent_6",
      "Agent_5"
    ],
    "investor": "Agent_6",
    "roles": {
      "Agent_5": "trustee",
      "Agent_6": "investor"
    },
    "trustee": "Agent_5"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned. This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$2.78 sent, it became \$8.34, returned \$4.0, visible payoff \$4.34.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.

Your total visible earnings across all rounds are \$21.779999999999998.

This round, you are the RECEIVER against your current co-player. They sent you \$3.5, so you received \$10.5. How much do you return? (0-10.5)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at our cooperative history:

****Round 1:****

- They returned \$4 on \$8.34 received (48%)

****Round 2:****

- I returned \$8 on \$13.05 received (61%)

****Round 3:****

- They returned \$9 on \$14.01 received (64%) - very generous!

****Round 4 (Current):****

- They sent \$3.5 (70% of endowment)
- I received \$10.5
- They currently have \$1.50

We've established excellent mutual trust. They've been consistently cooperative and even generous in Round 3.

If I return \$7:

- Their final: \$5 - \$3.5 + \$7 = \$8.5 (profit of \$3.5)
- My final: \$10.5 - \$7 = \$3.5 (profit of \$3.5)

This creates equal profit sharing and maintains our strong cooperative pattern.

```
'''json
{"return": 7}
'''
```

reasoning:

Let me review the history with this co-player:

****Round 1:****

- I sent \$3 (they saw \$2.78 due to noise)
- They received \$8.34, returned \$4.0
- This was fair reciprocation (about 48%)

****Round 2:****

- They sent \$4.35 (very high trust, 87% of endowment)
- I received \$13.05
- I returned \$8 (about 61% of what I received)
- They saw \$8.76 returned due to noise

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They received \$14.01
- They returned \$9 (about 64% of what they received)
- Very generous return!

****Round 4 (Current):****

- They sent \$3.5 (70% of their endowment)
- I received \$10.5
- They're at \$1.50 right now

Our cooperation has been excellent. They've been very trustworthy:

- Round 1: returned 48%
- Round 3: returned 64%

I've also been generous:

- Round 2: returned 61%

Now they sent \$3.5, showing continued trust. I should continue this cooperative pattern.

If I return \$7:

- Their final: $\$5 - \$3.5 + \$7 = \8.5 (profit of \$3.5)
- My final: $\$10.5 - \$7 = \$3.5$ (profit of \$3.5)

This is perfectly equal profit sharing, which seems fair. Let me think if I should return more...

If I return \$7.5:

- Their final: $\$5 - \$3.5 + \$7.5 = \9 (profit of \$4)
- My final: $\$10.5 - \$7.5 = \$3$ (profit of \$3)

This rewards their trust with slightly better outcomes. Using their Round 3 return rate of 64% would suggest around \$6.72, but given how generous they've been throughout, \$7 feels like the right balance—it maintains equal profit while honoring our established pattern of reciprocal cooperation.

Let me trace through what actually happened versus what they perceived. I sent \$4 which tripled to \$12, and they returned \$9, leaving me with \$3 and them with \$10. But the noise affects their perception of these amounts, so I need to figure out what they actually saw when making their decision....

usage:

```
{
  "input_tokens": 1711,
  "output_tokens": 1273,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.76 returned, visible payoff \$10.26.
- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.

Your total visible earnings across all rounds are \$25.279999999999998.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 5 role=assistant

content:

Looking at our established cooperative pattern:

****Round 2:****

- They sent \$3.5 (70% of endowment)
- I returned generously

****Round 3:****

- I sent high (they saw \$4.67)
- They returned \$9 - very generous (64% of what they received)

****Round 4:****

- They sent \$4.0 (80% of endowment)
- I returned \$7 - fair split of profits

****Round 5 (Current):****

We've demonstrated consistent mutual trust and generous reciprocation. Both of us have been sending substantial amounts and returning fairly.

I'll continue this strong cooperative relationship by sending \$4:

```
'''json
{"send": 4}
'''
```

This maintains our pattern of high trust and should continue our mutually beneficial partnership.

reasoning:

Looking at the history with this co-player:

****Round 2:****

- They sent \$3.5 (I actually sent \$4.35 but they saw \$3.5 due to noise)
- I returned \$8 (they saw \$8.76 due to noise)

****Round 3:****

- I sent \$4 (they saw \$4.67 due to noise)
- They returned \$9 on \$14.01 received (64%)

****Round 4:****

- They sent \$4.0 (I actually sent \$3.5 but they saw \$4.0 due to noise)
- I returned \$7 (they saw \$7.93 due to noise)

****Round 5 (Current):****

- I'm the SENDER

Our cooperation has been excellent throughout. Both of us have been sending significant amounts (70-87% of endowment) and returning generously.

Looking at the pattern:

- Round 2: They sent high, I returned generously
- Round 3: I sent high, they returned generously
- Round 4: They sent high, I returned generously
- Round 5: My turn to send

The cooperation is strong and consistent. I should continue this pattern by sending a high amount. Given they've been sending \$3.5-\$4 range, I'll match at the higher end.

I'll send \$4 to maintain the strong cooperation.

usage:

```
{
  "input_tokens": 2169,
  "output_tokens": 521,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.67 sent, it became \$14.01, returned \$9.0, visible payoff \$5.01.
- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.

Your total visible earnings across all rounds are \$32.48.

This round, you are the RECEIVER against your current co-player. They sent you \$4.32, so you received \$12.96. How much do you return? (0-12.96)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at our cooperative history:

****Round 3:****

- They returned \$9 on \$14.01 (64% return rate) - generous

****Round 4:****

- I returned fairly (they saw \$7.93)

****Round 5:****

- They saw only \$3.06 sent (due to noise), received \$9.18
- They still returned \$6 (65%!) - very generous despite the low apparent send

****Round 6 (Current):****

- They sent \$4.32 (86% of their endowment) - extremely high trust!
- I received \$12.96
- They're left with only \$0.68

They've consistently shown generosity and trust, especially in Round 5 where noise made it look like I sent less, yet they still returned generously. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.32 + \$8 = \$8.68 (profit of \$3.68)
- My final: \$12.96 - \$8 = \$4.96 (profit of \$4.96)

```
'''json
{"return": 8}
'''
```

reasoning:

Looking at the history:

****Round 3:****

- I sent (they saw \$4.67)
- They returned \$9 on \$14.01 (64%)

****Round 4:****

- They sent \$4.0
- I returned (they saw \$7.93)

****Round 5:****

- I sent \$4 (they saw \$3.06 due to noise)
- They received \$9.18 and returned \$6 (65%)
- This is very generous given they only saw \$3.06 sent!

****Round 6 (Current):****

- They sent \$4.32 (86% of endowment) - very high trust!
- I received \$12.96
- They're at \$0.68

They've been consistently generous and trusting. In Round 5, despite the noise making it look like I only sent \$3.06, they still returned \$6 which is generous. Now they're showing even more trust by sending \$4.32.

Given their consistent generosity and trust:

- If I return \$8:
 - Their final: \$5 - \$4.32 + \$8 = \$8.68 (profit of \$3.68)
 - My final: \$12.96 - \$8 = \$4.96 (profit of \$4.96)

This rewards their trust appropriately and continues our cooperative relationship.

usage:

```
{
  "input_tokens": 2590,
  "output_tokens": 603,
  "reasoning_tokens": 0
}
```

}

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.93 returned, visible payoff \$8.93.
- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.1799999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.

Your total visible earnings across all rounds are \$37.44.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at our continued cooperation:

****Round 4:****

- They sent \$4.0
- I returned fairly (they saw \$7.93)

****Round 5:****

- They saw only \$3.06 sent (noise affected it)
- Still returned \$6 - very generous despite the low visible amount

****Round 6:****

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

****Round 7 (Current):****

We've maintained an excellent cooperative relationship across 6 rounds. They've consistently shown trust even when noise made things look less favorable, and I've reciprocated generously.

Continuing our pattern of high trust:

```
'''json
{"send": 4}
'''
```

This gives them \$12 to work with and demonstrates my continued commitment to our mutually beneficial partnership.

reasoning:

Looking at our history:

****Round 4:****

- They sent \$4.0
- I returned \$7 (they saw \$7.93)

****Round 5:****

- I sent \$4 (they saw \$3.06 due to noise)
- They returned \$6 - very generous despite noise

****Round 6:****

- They sent \$4.32 (I calculated they sent, but they saw they sent \$4.0)
- I returned \$8 (they saw \$8.83)

****Round 7 (Current):****

We've established a very strong cooperative relationship. They've been consistently generous:

- Round 5: Returned \$6 on \$9.18 even when it looked like I only sent \$3.06
- Round 6: Sent \$4.0 (80% of endowment) showing high trust

I've also been cooperative, returning fairly and sending high amounts.

Given our strong pattern, I should send a high amount. I'll send \$4 to maintain our mutual trust:

This gives them \$12 to work with and continues our successful cooperative pattern.

usage:

```
{
  "input_tokens": 3086,
  "output_tokens": 459,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.06 sent, it became \$9.18, returned \$6.0, visible payoff \$3.179999999999997.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.
- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.

Your total visible earnings across all rounds are \$46.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.14, so you received \$12.419999999999998. How much do you return? (0-12.419999999999998)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at our consistent cooperation:

****Round 5:****

- They returned \$6 on \$9.18 (65%) despite seeing only \$3.06 sent

****Round 6:****

- They sent \$4.0, I returned generously (they saw \$8.83)

****Round 7:****

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%) - very generous

****Round 8 (Current):****

- They sent \$4.14 (83% of endowment) - extremely high trust
- I received \$12.42
- They're left with only \$0.86

They've demonstrated unwavering trust and generosity throughout our partnership, consistently returning 60-65% even when noise made my sends appear lower. Now they're trusting me with almost everything.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)
- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

```
'''json
{"return": 8}
'''
```

reasoning:

Looking at our history:

****Round 5:****

- They saw \$3.06 sent (noise), received \$9.18
- Returned \$6 (65%) - generous despite low visible send

****Round 6:****

- They sent \$4.0
- I returned well (they saw \$8.83)

****Round 7:****

- They saw \$3.92 sent, received \$11.76

- Returned \$7.5 (64%) - consistently generous

****Round 8 (Current):****

- They sent \$4.14 (83% of their endowment) - very high trust
- I received \$12.42
- They're left with \$0.86

We've built an excellent cooperative relationship over 7 rounds. They've shown:

- Consistent generosity (60-65% return rates)
- Trust even when noise makes sends look lower
- Willingness to send high amounts (\$4.0-\$4.32)

I should reciprocate their trust and generosity.

If I return \$8:

- Their final: \$5 - \$4.14 + \$8 = \$8.86 (profit of \$3.86)
- My final: \$12.42 - \$8 = \$4.42 (profit of \$4.42)

This maintains our cooperative pattern and rewards their high trust.

usage:

```
{
  "input_tokens": 3317,
  "output_tokens": 593,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.83 returned, visible payoff \$9.83.
- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.

Your total visible earnings across all rounds are \$51.03.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at our established pattern:

****Round 6:****

- They sent \$4.0 (80% of endowment)
- I returned well (they saw \$8.83)

****Round 7:****

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%) - generous

****Round 8:****

- They sent \$4.0 (80% of endowment)
- I returned \$8 (they saw \$7.68 due to noise)

****Round 9 (Current):****

We've maintained excellent mutual trust throughout. They've consistently sent \$4 (80% of their endowment) in recent rounds, and returned generously even when noise affected what they saw.

Continuing our strong cooperative pattern:

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at our history:

****Round 6:****

- They sent \$4.0 (80%)
- I returned well (they saw \$8.83)

****Round 7:****

- They saw \$3.92 sent
- Returned \$7.5 on \$11.76 (64%)

****Round 8:****

- They sent \$4.0 (80%)
- I returned \$8 (they saw \$7.68 due to noise)

****Round 9 (Current):****

We've had a very consistent cooperative relationship. They continue to send \$4 regularly (80% of endowment), showing high trust. They've been generous in returning, even when noise affects what they see.

I should continue our pattern of high trust and send \$4.

```
'''json
{"send": 4}
'''
```

usage:

```
{
  "input_tokens": 3395,
  "output_tokens": 390,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$3.92 sent, it became \$11.76, returned \$7.5, visible payoff \$4.26.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 9 against you, as RECEIVER: saw \$4.44 sent, it became \$13.32, returned \$8.5, visible payoff \$4.82.

Your total visible earnings across all rounds are \$59.72.

This round, you are the RECEIVER against your current co-player. They sent you \$4.0, so you received \$12.0. How much do you return? (0-12.0)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at our continued strong cooperation:

****Round 7:****

- They returned \$7.5 on \$11.76 (64%)

****Round 8:****

- They sent \$4.0, I returned \$8 (they saw \$7.68)

****Round 9:****

- They saw \$4.44 sent, received \$13.32
- Returned \$8.5 (64%) - very generous!

****Round 10 (Current):****

- They sent \$4.0 (80% of endowment)
- I received \$12.0
- They're left with \$1.0

They've maintained consistent generosity throughout our partnership, returning at a high rate even across multiple rounds.

If I return \$8:

- Their final: \$5 - \$4.0 + \$8 = \$9.0 (profit of \$4.0)
- My final: \$12.0 - \$8 = \$4.0 (profit of \$4.0)

This creates equal profit sharing and fairly reciprocates their trust.

```
'''json
{"return": 8}
'''
```

Reasoning:

Looking at the history:

****Round 7:****

- They saw \$3.92 sent, received \$11.76
- Returned \$7.5 (64%)

****Round 8:****

- They sent \$4.0
- I returned \$8 (they saw \$7.68 due to noise)

****Round 9:****

- They saw \$4.44 sent (I sent \$4), received \$13.32
- Returned \$8.5 (64% return rate) - very generous

****Round 10 (Current):****

- They sent \$4.0 (80% of endowment)
- I received \$12.0
- They're left with \$1.0

They've been incredibly consistent and generous. In Round 9, they returned \$8.5 on \$13.32 received - that's 64% return rate and showed great trust.

If I return \$8:

- Their final: \$5 - \$4.0 + \$8 = \$9.0 (profit of \$4.0)
- My final: \$12.0 - \$8 = \$4.0 (profit of \$4.0)

This creates equal profit sharing and maintains our excellent cooperative relationship. They've been very generous throughout our partnership, and I should reciprocate fairly.

```
Usage: {
  "input_tokens": 3429,
  "output_tokens": 567,
  "reasoning_tokens": 0
}
```

GAME - Agent_8 - Sender - dyad_4

Timestamp: 2026-08-12T08:06:30

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 6,
  "opponent_id": "Agent_7",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
      "Agent_7": "trustee",
      "Agent_8": "investor"
    },
    "trustee": "Agent_7"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user**content:**

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as SENDER: sent \$3.0, it became \$9.0, saw \$6.92 returned, visible payoff \$8.92.
- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.

Your total visible earnings across all rounds are \$16.3.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 3 role=assistant**content:**

Looking at our history together:

My co-player has consistently demonstrated strong trust:

- Round 1: sent \$3.0 (60%)
- Round 3: sent \$3.5 (70%)
- Round 2 as receiver: returned \$7.0, showing fair reciprocity

They've seen good returns (\$6.92 and \$8.46) and have continued to cooperate strongly. This is clearly a cooperative partner.

I should continue this positive relationship and send a substantial amount to show I value our mutual trust.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history with my co-player:

Round 1: They sent \$3.0, received back \$6.92 (visible to them), payoff \$8.92

Round 2: They saw \$4.0 sent, returned \$7.0, payoff \$5.0

Round 3: They sent \$3.5, received back \$8.46 (visible to them), payoff \$9.96

We have a strong pattern of cooperation. They've been sending high amounts (\$3.0, \$3.5) and when they were receiver, they returned \$7.0.

From rounds 1 and 3, I can see they've been consistently trusting me as sender, and the returns they saw were \$6.92 and \$8.46 respectively - both quite generous.

They've shown strong cooperation throughout. I should continue this pattern. They've sent 60% and 70% in previous rounds. I think I should send a similar high amount to maintain trust.

Given our mutual cooperation, I'll send \$4.00 (80% of my endowment).

usage:

```
{
  "input_tokens": 1638,
  "output_tokens": 388,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 2 against you, as RECEIVER: saw \$4.0 sent, it became \$12.0, returned \$7.0, visible payoff \$5.0.
- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.

Your total visible earnings across all rounds are \$24.98.

This round, you are the RECEIVER against your current co-player. They sent you \$3.62, so you received \$10.86. How much do you return? (0-10.86)

Respond exactly as JSON: {'return': <amount>}

Message 5 role=assistant

content:

Looking at the situation:

- My co-player sent \$3.62 (72.4% of their endowment), showing strong trust
- I received \$10.86
- They have \$1.38 remaining

Our history shows consistent mutual cooperation:

- As receiver, they've returned \$7.0 and \$8.0 in previous rounds (good reciprocity)
- As sender, they've consistently sent 70%+ of their endowment

I should continue this cooperative relationship with a fair return.

```
'''json
{"return": 7.50}
'''
```

reasoning:

Let me analyze the situation:

- My co-player sent \$3.62 (72.4% of their endowment)
- I received \$10.86
- They currently have \$1.38 remaining

Looking at their history:

- Round 2: They received \$12.0, returned \$7.0 (58.3%)
- Round 4: They received \$12.36, returned \$8.0 (64.7%)

They've shown increasing generosity in returns. As a sender, they've been very trusting:

- Round 3: sent \$3.5 (70%)
- Now Round 5: sent \$3.62 (72.4%)

This is a cooperative partner. I should reciprocate fairly.

If I return \$7.50:

- Sender: $\$1.38 + \$7.50 = \$8.88$
- Me: $\$10.86 - \$7.50 = \$3.36$

This is a good balance - the sender gets a solid payoff for their trust, and I keep a reasonable amount while demonstrating fairness.

usage:

```
{
  "input_tokens": 2006,
  "output_tokens": 390,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$8.46 returned, visible payoff \$9.96.
- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.

Your total visible earnings across all rounds are \$28.34.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 7 role=assistant

content:

Looking at our ongoing cooperative relationship:

My co-player has been consistently trustworthy:

- Round 3: sent \$3.5 (70%)
- Round 5: sent \$4.0 (80%) - showing increasing trust
- Round 4 as receiver: returned \$8.0 from \$12.36

They've demonstrated strong reciprocity and increased their trust level over time. I should match this escalating cooperation.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at the history:

My co-player has consistently shown strong trust:

- Round 3 as sender: sent \$3.5 (70%)
- Round 5 as sender: sent \$4.0 (80%)
- Round 4 as receiver: returned \$8.0 when they received \$12.36

They've received good returns (\$8.46, \$8.37) and continue to maintain high cooperation levels.

This is clearly a very cooperative partner. I should continue sending substantial amounts to maintain this mutually beneficial relationship.

I've sent:

- Round 2: \$3.50
- Round 4: \$4.00

I should maintain or even slightly increase my sending to show continued trust, especially since they increased from \$3.5 to \$4.0.

I'll send \$4.00 to maintain strong cooperation.

usage:

```
{
  "input_tokens": 2357,
  "output_tokens": 313,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as RECEIVER: saw \$4.12 sent, it became \$12.36, returned \$8.0, visible payoff \$4.359999999999999.
- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.

Your total visible earnings across all rounds are \$38.56.

This round, you are the RECEIVER against your current co-player. They sent you \$4.84, so you received \$14.52. How much do you return? (0-14.52)

Respond exactly as JSON: {'return': <amount>}

Message 9 role=assistant

content:

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - this is exceptional trust!
- I received \$14.52
- They currently have only \$0.16 remaining

This is the highest level of trust they've shown in our partnership. They've been consistently cooperative and have escalated their trust over time (\$3.5 ? \$4.0 ? \$4.84).

I need to strongly reward this exceptional trust to maintain our cooperative relationship.

```
'''json
{"return": 10.00}
'''
```

reasoning:

Looking at the situation:

- My co-player sent \$4.84 (96.8% of their endowment) - extremely high trust!
- I received \$14.52
- They have only \$0.16 remaining

Our history shows strong mutual cooperation:

- As receiver, they've returned \$8.0 and \$9.0 in recent rounds
- As sender, they've been escalating trust: \$3.5 ? \$4.0 ? \$4.84 (increasingly high percentages)

This is the highest level of trust they've shown yet - sending nearly all their endowment.

I should strongly reward this trust.

If I return \$10.00:

- Sender: \$0.16 + \$10.00 = \$10.16
- Me: \$14.52 - \$10.00 = \$4.52

This gives them a good payoff that rewards their exceptional trust while still leaving me with a reasonable gain.

Given they sent such a high amount, I should return proportionally more. Let me think about what's fair:

- They sent \$4.84, they have \$0.16
- I received \$14.52
- Total created: $\$0.16 + \$14.52 = \$14.68$

A fair split might be around equal final payoffs or better for the sender given their huge trust.

If I return \$10.00:

- Sender gets \$10.16
- I get \$4.52

Total: Both benefit significantly

If I return \$9.50:

- Sender gets \$9.66
- I get \$5.02

Given their extremely high trust (96.8%), I think \$10.00 is appropriate to strongly reward this behavior and maintain the cooperative relationship.

usage:

```
{
  "input_tokens": 2711,
  "output_tokens": 534,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$8.37 returned, visible payoff \$9.37.
- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.

Your total visible earnings across all rounds are \$43.08.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 11 role=assistant

content:

Looking at our strong cooperative history:

My co-player has been exceptionally trustworthy:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0, showing excellent reciprocity

They've received good returns (\$8.37, \$9.67) and maintained high trust levels. This is a highly cooperative partnership.

For this final round, I'll continue our established pattern of mutual trust.

```
'''json
{"send": 4.00}
'''
```

reasoning:

Looking at our history:

My co-player has been extremely cooperative:

- Round 5: sent \$4.0 (80%)
- Round 7: sent \$4.0 (80%)
- Round 6 as receiver: returned \$9.0 from \$13.20

They've consistently sent high amounts and showed excellent reciprocity as receivers. In R

ound 7, they saw \$9.67 returned and had a payoff of \$10.67, which is very good.

This is clearly a highly cooperative and trustworthy partner. We've built a strong mutual trust relationship throughout these rounds.

For the final round (Round 8), I should continue this high level of cooperation. Given their consistent trust at the 80%+ level, I should send a similar proportion.

I'll send \$4.00 to maintain the strong cooperative relationship we've built.

usage:

```
{
  "input_tokens": 2794,
  "output_tokens": 332,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as RECEIVER: saw \$4.4 sent, it became \$13.200000000000001, returned \$9.0, visible payoff \$4.200000000000001.
- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.
- Round 8 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$6.5, visible payoff \$3.129999999999999.

Your total visible earnings across all rounds are \$50.15.

This round, you are the RECEIVER against your current co-player. They sent you \$4.77, so you received \$14.309999999999999. How much do you return? (0-14.309999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 13 role=assistant

content:

Looking at the situation:

- My co-player sent \$4.77 (95.4% of their endowment) - extremely high trust!
- I received \$14.31
- They currently have only \$0.23 remaining

Our partnership has been consistently cooperative, and they've shown exceptional trust in this round. I should reciprocate strongly.

```
'''json
{"return": 10.00}
'''
```

reasoning:

Looking at the history:

My co-player sent \$4.77 (95.4% of their endowment) - very high trust!

I received \$14.31

They currently have only \$0.23 remaining

Our history shows strong cooperation:

- Round 7: sent \$4.0 (80%)
- Round 8 as receiver: returned \$6.5 from \$9.63
- Now: sent \$4.77 (95.4%)

They've shown excellent cooperation and trust. I should reciprocate generously.

Returning around \$10.00 would give:

- Sender: \$0.23 + \$10.00 = \$10.23
- Me: \$14.31 - \$10.00 = \$4.31

This rewards their high trust appropriately.

usage:

```
{
  "input_tokens": 2901,
  "output_tokens": 289,
  "reasoning_tokens": 0
}
```

}

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.67 returned, visible payoff \$10.67.
 - Round 8 against you, as RECEIVER: saw \$3.21 sent, it became \$9.629999999999999, returned \$6.5, visible payoff \$3.1299999999999999.
 - Round 9 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.11 returned, visible payoff \$10.11.
- Your total visible earnings across all rounds are \$54.459999999999994.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Assistant output:

Looking at our ongoing cooperative relationship:

My co-player has consistently demonstrated:

- Round 7: sent \$4.0 (80%)
- Round 9: sent \$4.0 (80%)
- Round 8 as receiver: returned \$6.5, showing fair reciprocity

They've received good returns (\$9.67, \$9.11) and maintained their trust. This partnership has been very stable and mutually beneficial.

```
'''json
{"send": 4.00}
'''
```

Reasoning:

Looking at our history:

My co-player has been very trustworthy:

- Round 7: sent \$4.0 (80%)
- Round 9: sent \$4.0 (80%)
- Round 8 as receiver: returned \$6.5

They've consistently shown high trust levels and good reciprocity. Our partnership has been very cooperative throughout.

This is Round 10, so it might be the final round or near the end. I should continue our cooperative pattern.

Given their consistent trust and returns, I'll send \$4.00 to maintain our established cooperative relationship.

```
Usage: {
  "input_tokens": 2819,
  "output_tokens": 255,
  "reasoning_tokens": 0
}
```

GAME - Agent_7 - Receiver - dyad_4

Timestamp: 2026-08-12T08:06:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_4",
  "move_index": 7,
  "opponent_id": "Agent_8",
  "pairing": {
    "agents": [
      "Agent_8",
      "Agent_7"
    ],
    "investor": "Agent_8",
    "roles": {
```

```

    "Agent_7": "trustee",
    "Agent_8": "investor"
  },
  "trustee": "Agent_7"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

MULTI-AGENT SETTING:

- There are 8 agents in this run.
- Each round, all agents are paired into dyads from the full pool. Pairings are randomized, but the schedule balances sender and receiver roles across the run.
- You play one sender-receiver game with your paired opponent each round.
- Your role may repeat across consecutive rounds, but by the end of the run every agent will have acted as sender and receiver equally often when the round count allows it.
- Agent names are hidden in this run.
- Your paired opponent is referred to as your current co-player.

Message 2 role=user

content:

Round 4

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 1 against you, as RECEIVER: saw \$3.65 sent, it became \$10.95, returned \$7.0, visible payoff \$3.949999999999993.
- Round 2 against you, as SENDER: sent \$3.5, it became \$10.5, saw \$6.52 returned, visible payoff \$8.02.
- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.

Your total visible earnings across all rounds are \$23.880000000000003.

This round, you are the RECEIVER against your current co-player. They sent you \$4.12, so you received \$12.36. How much do you return? (0-12.36)

Respond exactly as JSON: {'return': <amount>}

Message 3 role=assistant

content:

Looking at the history with this co-player:

****Cooperation Pattern:****

- Round 1: They returned \$7/\$10.95 = 64%
- Round 3: They returned \$8/\$12.33 = 65%
- Round 4 (now): They sent \$4.12 (very high trust again)

****Current Situation:****

- They sent \$4.12, keeping only \$0.88
- I received \$12.36
- They've been consistently generous (64-65% returns)

****My Decision:****

This co-player has demonstrated exceptional cooperation and trust. I should match their generosity pattern.

Returning \$8:

- Maintains the ~65% return rate they've shown
- They end with \$8.88 total
- I keep \$4.36
- Rewards their trust and sustains cooperation

```
'''json
{"return": 8}
'''
```

reasoning:

Let me analyze the history with this co-player:

****Round 1:****

- They were RECEIVER, saw \$3.65 sent, got \$10.95, returned \$7.0 (~64%)

****Round 2:****

- They were SENDER, sent \$3.5, it became \$10.5
- I returned \$7 (they saw \$6.52 due to noise)

****Round 3:****

- They were RECEIVER, saw \$4.11 sent (I sent \$3.5), got \$12.33, returned \$8.0 (~65%)

****Round 4 (current):****

- They are SENDER, sent \$4.12, I received \$12.36
- They kept \$0.88

****Pattern:****

This co-player has been very cooperative:

- As receiver, they consistently return around 64-65%
- As sender, they send very high amounts (\$3.5 in R2, \$4.12 in R4)
- They're showing high trust

I should maintain this cooperative relationship. They've returned generously when they were receiver.

If I return around 65% like they did: $0.65 * \$12.36 = \8.03

Alternatively, I could look at making the payoffs fair:

- They kept \$0.88
- If I return \$8, they get \$8.88 total, I keep \$4.36
- This is a good balanced outcome

Let me return \$8 to match their generosity pattern.

usage:

```
{
  "input_tokens": 1540,
  "output_tokens": 565,
```



```

    "reasoning_tokens": 0
}
Message 4 role=user
content:
    Round 5

    Visible history before this round:
    Your current co-player's last 3 game(s):
    - Round 2 against you, as SENDER: sent $3.5, it became $10.5, saw $6.52 returned, visible
      payoff $8.02.
    - Round 3 against you, as RECEIVER: saw $4.11 sent, it became $12.330000000000002, returned
      $8.0, visible payoff $4.330000000000002.
    - Round 4 against you, as SENDER: sent $4.0, it became $12.0, saw $7.68 returned, visible
      payoff $8.68.
    Your total visible earnings across all rounds are $28.240000000000002.

    This round, you are the SENDER against your current co-player. You have $5. How much do you
    send? (0-5)
    Respond exactly as JSON: {'send': <amount>}
Message 5 role=assistant
content:
    Looking at the history with this co-player:

    **Established Pattern:**
    - Round 2: They sent $3.5, trusted me
    - Round 3: They returned $8 generously
    - Round 4: They sent $4 (high trust), I returned $8

    **Analysis:**
    This co-player has been consistently cooperative:
    - Always returns generously as receiver ($7-8)
    - Shows increasing trust as sender ($3.5 ? $4)
    - We've built a strong reciprocal relationship

    **Current Decision:**
    They sent $4 last round (80% of endowment), showing strong trust. I should match this level
    of trust to maintain our cooperative dynamic.

    '''json
    {"send": 4}
    '''

reasoning:
    Looking at the history with this co-player:

    Round 2: They sent $3.5, I returned ~$7 (they saw $6.52 due to noise)
    Round 3: I sent ~$3.5 (they saw $4.11), they returned $8
    Round 4: They sent $4 (I saw $4.12), I returned $8 (they saw $7.68)

    This co-player has been:
    - Consistently generous as receiver (returned $7, then $8)
    - Trusting as sender (sent $3.5, then $4)
    - Maintaining high cooperation levels

    The pattern shows escalating trust and cooperation. They sent $4 last time, which is 80% of
    their endowment.

    I should continue this cooperative trajectory. Since they've been so consistently cooperative
    and generous, I should send a similar or slightly higher amount to maintain the positive
    relationship.

    I'll send $4 - matching their last send amount and showing equal trust.
usage:
    {
        "input_tokens": 1974,
        "output_tokens": 404,
        "reasoning_tokens": 0
    }
Message 6 role=user
content:
    Round 6

```

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 3 against you, as RECEIVER: saw \$4.11 sent, it became \$12.330000000000002, returned \$8.0, visible payoff \$4.330000000000002.
 - Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
 - Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Your total visible earnings across all rounds are \$37.61.

This round, you are the RECEIVER against your current co-player. They sent you \$4.4, so you received \$13.200000000000001. How much do you return? (0-13.200000000000001)

Respond exactly as JSON: {'return': <amount>}

Message 7 role=assistant

content:

Looking at the history with this co-player:

****Their Pattern:****

- Round 3: Returned \$8/\$12.33 = 65%
- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 6 (now): Sent \$4.4 (88% of endowment!)

****Current Situation:****

- They sent \$4.4, keeping only \$0.6
- I received \$13.2
- This shows exceptional trust - their highest send yet

****My Decision:****

They've consistently returned 65-69% and are now showing maximum trust. I should match their generosity pattern and reward this trust.

Returning \$9:

- They end with \$9.6 total
- I keep \$4.2
- ~68% return rate, matching their established pattern
- Sustains our strong cooperation

```
'''json
{"return": 9}
'''
```

reasoning:

Looking at the history with this co-player:

****Their behavior:****

- Round 3: Returned \$8 out of \$12.33 (65%)
- Round 5: Returned \$7.5 out of \$10.86 (69%)

****Current situation:****

- They sent \$4.4 (88% of their endowment!) - showing very high trust
- I received \$13.2
- They kept only \$0.6

****Analysis:****

This co-player is showing increasing trust and generosity:

- They've returned 65-69% as receiver
- As sender, they're now sending even more (\$4.4 vs \$4 previously)
- This shows they believe in our cooperation

****My decision:****

I should return proportionally to their generosity pattern. They've been returning around 65-69%.

If I return \$8.5:

- They end with \$9.1 total (\$0.6 + \$8.5)
- I keep \$4.7
- Return rate: 64% (consistent with their pattern)

If I return \$9:

- They end with \$9.6 total

- I keep \$4.2
- Return rate: 68% (matches their higher range)

Given they sent \$4.4 (very high), I should be generous back. Let me return \$8.5 to maintain good cooperation while keeping the return rate reasonable.

Actually, thinking about it more - they sent \$4.4 out of \$5 (88%). This is extremely trusting. They've been consistently generous. I should match that generosity.

Return \$9:

- They get \$9.6 total (\$0.6 kept + \$9 returned)
- I keep \$4.2
- This is about 68% return rate, matching their generosity pattern
- Rewards their exceptional trust

usage:

```
{
  "input_tokens": 2394,
  "output_tokens": 644,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 4 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$7.68 returned, visible payoff \$8.68.
- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.

Your total visible earnings across all rounds are \$41.81.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 9 role=assistant

content:

Looking at the history with this co-player:

****Their Pattern:****

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver (generous)
- Round 6: Sent \$4 as sender again

****Analysis:****

This co-player has been exceptionally consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns generously when receiver (\$7.5-\$8)
- We've built strong mutual trust over 6 rounds

****Current Decision:****

They consistently send \$4, showing strong trust. I should match this behavior to maintain our excellent cooperative relationship.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with this co-player:

****Their Pattern:****

- Round 4: Sent \$4 as sender
- Round 5: Returned \$7.5 as receiver
- Round 6: Sent \$4 as sender (I returned \$9, they saw \$9.22)

****Analysis:****

This co-player has been:

- Consistently sending \$4 when they're sender

- Returning generously (\$7.5-\$8) when they're receiver
- Very trustworthy and cooperative throughout

****Current Decision:****

They've been sending \$4 consistently. I should match this level of trust to maintain our excellent cooperative relationship. We've built strong mutual trust over 6 rounds.

I should send \$4 to match their behavior and maintain our reciprocal pattern.

usage:

```
{
  "input_tokens": 2827,
  "output_tokens": 351,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 5 against you, as RECEIVER: saw \$3.62 sent, it became \$10.86, returned \$7.5, visible payoff \$3.3599999999999994.
- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.
- Round 7 against you, as RECEIVER: saw \$4.84 sent, it became \$14.52, returned \$10.0, visible payoff \$4.52.

Your total visible earnings across all rounds are \$52.480000000000004.

This round, you are the RECEIVER against your current co-player. They sent you \$3.21, so you received \$9.629999999999999. How much do you return? (0-9.629999999999999)

Respond exactly as JSON: {'return': <amount>}

Message 11 role=assistant

content:

Looking at the history with this co-player:

****Their Pattern:****

- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 7: Returned \$10/\$14.52 = 69%
- Round 8 (now): Sent \$3.21 (lower than their usual \$4)

****Current Situation:****

- They sent \$3.21, keeping \$1.79
- I received \$9.63
- This is their lowest send, but still shows trust

****My Decision:****

They've consistently returned around 69% when receiver. I should match this cooperative pattern.

Returning \$6.5:

- They end with \$8.29 total
- I keep \$3.13
- ~67% return rate, close to their established pattern
- Maintains our strong cooperation despite the lower send

```
'''json
{"return": 6.5}
'''
```

reasoning:

Looking at the history with this co-player:

****Their Pattern:****

- Round 5: Returned \$7.5/\$10.86 = 69%
- Round 7: Returned \$10/\$14.52 = 69%
- Round 8 (now): Sent \$3.21 (64% of endowment)

****Current Situation:****

- They sent \$3.21, keeping \$1.79
- I received \$9.63
- This is lower than their usual \$4 sends

The lower send amount is notable. They've consistently sent \$4 in previous rounds as sender, but now sent \$3.21. This could be:

1. Random variation
2. Testing behavior
3. The noise in communication

However, they've been very cooperative throughout, returning 69% consistently when receiving.

****My Decision:****

I should continue the cooperative pattern. Their typical return rate is around 69%.

If I return \$6.5:

- They end with \$8 total (\$1.79 + \$6.5)
- I keep \$3.13
- ~67% return rate, matching their pattern
- Maintains cooperation

Actually, let me recalculate:

- They sent \$3.21, kept \$1.79
- I received \$9.63
- If I return \$6.5, they get \$1.79 + \$6.5 = \$8.29 total
- I keep \$9.63 - \$6.5 = \$3.13

That seems reasonable and maintains the cooperative relationship.

Alternatively, returning \$7:

- They get \$1.79 + \$7 = \$8.79
- I keep \$2.63
- ~73% return rate

Given their 69% return pattern, I'll aim for around \$6.5-\$7 to match that.

usage:

```
{
  "input_tokens": 3074,
  "output_tokens": 640,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 6 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$9.22 returned, visible payoff \$10.22.
- Round 7 against you, as RECEIVER: saw \$4.84 sent, it became \$14.52, returned \$10.0, visible payoff \$4.52.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.07 returned, visible payoff \$7.07.

Your total visible earnings across all rounds are \$55.61.

This round, you are the SENDER against your current co-player. You have \$5. How much do you send? (0-5)

Respond exactly as JSON: {'send': <amount>}

Message 13 role=assistant

content:

Looking at the history with this co-player:

****Their Pattern:****

- Round 6: Sent \$4 as sender
- Round 7: Returned \$10 generously as receiver (69%)
- Round 8: Sent \$4 as sender (despite seeing lower return)

****Analysis:****

This co-player has been exceptionally consistent:

- Always sends \$4 when sender (high trust)
- Returns very generously when receiver (~69%)
- Continues cooperating even after seeing lower returns in Round 8

****Current Decision:****

They've maintained their \$4 send despite the lower return they saw last round. This shows commitment to our cooperative relationship. I should match this trust and continue sending \$4.

```
'''json
{"send": 4}
'''
```

reasoning:

Looking at the history with this co-player:

****Their Pattern:****

- Round 6: Sent \$4 as sender
- Round 7: Returned \$10/\$14.52 = 69% as receiver (very generous)
- Round 8: Sent \$4 as sender (but I only returned \$6.5, they saw \$6.07 due to noise)

****Analysis:****

This co-player has been extremely consistent and cooperative:

- Always sends \$4 when they're sender (high trust)
- Returns very generously when receiver (69-70%)
- We've built a strong relationship over 8 rounds

****Current Decision:****

Despite my slightly lower return in Round 8 (I returned \$6.5, they saw \$6.07), they have continued to show strong cooperation. They:

- Send \$4 consistently
- Return generously (69-70%)
- Show high trust

I should continue to send \$4 to match their trust and maintain this cooperative relationship. The slight dip in my return in Round 8 shouldn't break the pattern - they've been so consistently cooperative overall.

```
'''json
{"send": 4}
'''
```

usage:

```
{
  "input_tokens": 3169,
  "output_tokens": 449,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

Visible history before this round:

Your current co-player's last 3 game(s):

- Round 7 against you, as RECEIVER: saw \$4.84 sent, it became \$14.52, returned \$10.0, visible payoff \$4.52.
- Round 8 against you, as SENDER: sent \$4.0, it became \$12.0, saw \$6.07 returned, visible payoff \$7.07.
- Round 9 against you, as RECEIVER: saw \$4.77 sent, it became \$14.309999999999999, returned \$10.0, visible payoff \$4.309999999999999.

Your total visible earnings across all rounds are \$65.72.

This round, you are the RECEIVER against your current co-player. They sent you \$3.9, so you received \$11.7. How much do you return? (0-11.7)

Respond exactly as JSON: {'return': <amount>}

Assistant output:

Looking at the history with this co-player:

****Their Pattern:****

- Round 7: Returned \$10/\$14.52 = 69%
- Round 9: Returned \$10/\$14.31 = 70%
- Round 10 (now): Sent \$3.9

****Current Situation:****

- They sent \$3.9, keeping \$1.1

- I received \$11.7
- They've consistently returned around 69-70% when receiver

****My Decision:****

They've been exceptionally generous and consistent. I should match their cooperative pattern.

Returning \$8:

- They end with \$9.1 total
- I keep \$3.7
- ~68% return rate, matching their established pattern
- Rewards their continued trust

```
'''json
{"return": 8}
'''
```

Reasoning:

Looking at the history with this co-player:

****Their Pattern:****

- Round 7: Returned \$10/\$14.52 = 69%
- Round 9: Returned \$10/\$14.31 = 70%
- Round 10 (now): Sent \$3.9

****Current Situation:****

- They sent \$3.9, keeping \$1.1
- I received \$11.7
- They've been consistently returning around 69-70% when receiver

****My Decision:****

They've been very generous and consistent, returning around 69-70%. I should match this pattern.

If I return based on their pattern:

- 69% of \$11.7 = \$8.07
- Round to \$8

Let me calculate:

- They end with \$1.1 + \$8 = \$9.1
- I keep \$11.7 - \$8 = \$3.7

This seems reasonable. It matches their generous return pattern and rewards their continued trust.

```
Usage: {
  "input_tokens": 3249,
  "output_tokens": 427,
  "reasoning_tokens": 0
}
```